PROBLEMS SHEET 4, PART 1 • OFFICIAL SOLUTIONS

Department of Mathematics • UC3M Aerospace Engineering

Chapter 4 Problem Set, Part 1: Linear Systems, Elimination and Real Eigenvalues

Complete, step-by-step solutions to the eleven exercises of the first Chapter 4 sheet: higher-order equations as systems, superposition, Wronskian and general solution of linear systems, elimination with initial conditions, general solutions and phase portraits from eigen-data, decoupling of four matrices with real eigenvalues, the trajectories of an uncoupled system and a zero eigenvalue.

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EXERCISE 1

An n-th Order Equation as a First-Order System and Its Uniqueness

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Reduction to a System • Existence and Uniqueness

STATEMENT

By choosing an appropriate collection of new variables $x_1, \dots, x_n$, rewrite the nth order differential equation

$$\frac{d^n x}{dt^n} = f\left(\frac{d^{n-1}x}{dt^{n-1}}, \dots, \frac{dx}{dt}, x, t\right)$$

as a set of n coupled linear first order equations. Find the conditions on the function $f(x_n, \dots, x_1, t)$ for the original differential equation to have a unique solution.

SETUP AND HYPOTHESES

Unknown: $x(t)$, n times differentiable on an interval I. Data: $f(x_n, \dots, x_1, t)$ defined on a region $\Omega \subset \mathbb{R}^{n+1}$. The initial value problem prescribes the n numbers $x(t_0), \dot{x}(t_0), \dots, x^{(n-1)}(t_0)$. The tool for uniqueness is the existence and uniqueness theorem for first-order systems $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, t)$ (Robinson Thm 25.2, Topic 4 Sec. 1).

FORMULATION: THE NEW VARIABLES

Introduce one variable for the function and for each derivative below the highest order (Robinson Sec. 25.1):

$$x_1 = x, \quad x_2 = \frac{dx}{dt}, \quad x_3 = \frac{d^2x}{dt^2}, \quad \dots, \quad x_n = \frac{d^{n-1}x}{dt^{n-1}}, \qquad \text{i.e. } x_k = x^{(k-1)} \ (k = 1, \dots, n)$$

METHOD OF THE CLASS NOTES [NOTES, P. 9]

The important point of the notes: an ODE of order $n \geq 2$ is a system of n first-order ODEs; for $\ddot{x} = f(\dot{x}, x, t)$ they set $x_1 = x$, $x_2 = \dot{x}$, so that $\dot{x}_1 = x_2$ and $\dot{x}_2 = f(x_2, x_1, t)$. The variables above are the same choice for order n. Uniqueness then follows from the condition the notes state for systems: $\mathbf{f}$ and its Jacobian $D\mathbf{f}$ continuous near $(\mathbf{x}_0, t_0)$. Only the last component f is not trivial, so the condition falls on f and its partial derivatives $\partial f / \partial x_k$.

DERIVATION

The system

By construction $\dot{x}_k = x^{(k)} = x_{k+1}$ for $k = 1, \dots, n-1$, and $\dot{x}_n = x^{(n)} = f(x_n, \dots, x_1, t)$ by the original equation:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ &\vdots \\ \dot{x}_{n-1} &= x_n \\ \dot{x}_n &= f(x_n, x_{n-1}, \dots, x_1, t) \end{aligned} \qquad \Longleftrightarrow \qquad \dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, t) = \begin{pmatrix} x_2 \\ x_3 \\ \vdots \\ x_n \\ f(x_n, \dots, x_1, t) \end{pmatrix}$$

The first $n-1$ equations are linear in the unknowns (they say $\dot{x}_k = x_{k+1}$); the last one is linear exactly when f is a linear function of $x_1, \dots, x_n$ (as for a linear ODE, where the whole system is linear, $\dot{\mathbf{x}} = A(t)\mathbf{x} + \mathbf{b}(t)$). The first-order system form is valid for any f.

Equivalence

(a) If x solves the nth-order equation, then $\mathbf{x} = (x, \dot{x}, \dots, x^{(n-1)})^T$ solves the system. (b) Conversely, if $(x_1, \dots, x_n)$ solves the system, then $\dot{x}_1 = x_2, \ddot{x}_1 = \dot{x}_2 = x_3$, ..., $x_1^{(k)} = x_{k+1}$ for $k \le n-1$, and $x_1^{(n)} = \dot{x}_n = f(x_n, \dots, x_1, t) = f(x_1^{(n-1)}, \dots, \dot{x}_1, x_1, t)$: the function $x = x_1$ solves the original equation. The initial condition $\mathbf{x}(t_0) = (x_1^0, \dots, x_n^0)$ is exactly $x(t_0) = x_1^0, \dot{x}(t_0) = x_2^0$, ..., $x^{(n-1)}(t_0) = x_n^0$.

Conditions for a unique solution

Theorem 25.2 (Robinson p. 250) requires $\mathbf{F}$ and the matrix of partial derivatives $D\mathbf{F}$ to be continuous on an open set containing the initial point. Compute it row by row: for $F_k = x_{k+1}$ ($k < n$) the partial derivatives are 0 or 1, and for $F_n = f$ they are $\partial f / \partial x_j$:

$$D\mathbf{F} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ \dfrac{\partial f}{\partial x_1} & \dfrac{\partial f}{\partial x_2} & \dfrac{\partial f}{\partial x_3} & \cdots & \dfrac{\partial f}{\partial x_n} \end{pmatrix}$$

The first $n-1$ rows are constant, hence continuous. Therefore all the conditions reduce to the last row:

CONDITION FOR UNIQUENESS

$f(x_n, \dots, x_1, t)$ and its partial derivatives $\partial f / \partial x_j$, $j = 1, \dots, n$, must be continuous in an open region containing the initial point $(x(t_0), \dots, x^{(n-1)}(t_0), t_0)$. Then the IVP has exactly one solution on some interval around t_0 (local existence and uniqueness), and two solutions with the same $x(t_0), \dots, x^{(n-1)}(t_0)$ coincide wherever both exist.

INTERPRETATION AND CHECKS

- For $n = 2$ the system is $\dot{x}_1 = x_2, \dot{x}_2 = f(x_2, x_1, t)$ and the condition is Theorem 11.1 of Topic 3 (continuity of $f, \partial f/\partial x_1, \partial f/\partial x_2$); for $n = 1$ it is Picard's theorem.
- No condition on $\partial f / \partial t$ is needed: the theorem asks continuity in t but no differentiability in t.
- Dependence on the n derivatives: continuity of the partial derivatives $\partial f / \partial x_j$ gives a local Lipschitz bound in $(x_1, \dots, x_n)$, which is what excludes branching solutions (compare Exercise 2.14 of Chapter 2, where the Lipschitz condition fails).
- For a linear equation $x^{(n)} + a_{n-1}(t) x^{(n-1)} + \cdots + a_0(t) x = g(t)$, $f = g - \sum a_k x_{k+1}$ and $\partial f / \partial x_{k+1} = -a_k(t)$: the conditions reduce to continuity of the a_k and g. For linear equations the solution in fact exists on the whole interval of continuity of the coefficients (a standard result for linear systems, not in Robinson Thm 25.2, which is local; see the continuation argument in Exercise 4).
- Check with the mass-spring-damper (Topic 4 Sec. 1.1): $n = 2$, $f = -\lambda x_2 - \omega^2 x_1$ has constant partial derivatives $-\omega^2$ and $-\lambda$: unique solutions for all data.

EXERCISE 2

Superposition for Homogeneous Linear Systems

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Superposition • Linear Combination of Solutions

STATEMENT

Show that if a homogeneous linear system

$$\begin{cases} \dfrac{dx}{dt} = a_1(t)x + b_1(t)y, \\ \dfrac{dy}{dt} = a_2(t)x + b_2(t)y \end{cases} \tag{1}$$

has two solutions

$$\begin{cases} x = x_1(t) \\ y = y_1(t) \end{cases} \quad \text{and} \quad \begin{cases} x = x_2(t) \\ y = y_2(t) \end{cases} \tag{2}$$

then any linear combination

$$\begin{cases} x = c_1x_1(t) + c_2x_2(t) \\ y = c_1y_1(t) + c_2y_2(t) \end{cases} \tag{3}$$

with constants $c_1, c_2 \in \mathbb{R}$ is a solution as well.

SETUP

Hypotheses: a_i, b_i are functions of t (not necessarily constant) defined on an interval I, and (x_1, y_1), (x_2, y_2) are differentiable on I with

$$\dot{x}_j = a_1 x_j + b_1 y_j, \qquad \dot{y}_j = a_2 x_j + b_2 y_j \qquad (j = 1, 2) \tag{H}$$

FORMULATION

To show: with $x = c_1 x_1 + c_2 x_2$ and $y = c_1 y_1 + c_2 y_2$ the two equations of (1) hold. Tool: differentiation is linear, $\frac{d}{dt}(c_1 f + c_2 g) = c_1 \dot{f} + c_2 \dot{g}$ for constants c_1, c_2. In vector notation: $\mathbf{x} = c_1 \mathbf{x}_1 + c_2 \mathbf{x}_2$ solves $\dot{\mathbf{x}} = A(t)\mathbf{x}$ because the right-hand side is linear in $\mathbf{x}$ (Robinson Eq. 27.3).

METHOD OF THE CLASS NOTES [NOTES, P. 9]

The notes state superposition for $\dot{\mathbf{x}} = A(t)\mathbf{x}$: combinations of solutions are solutions. This exercise is that statement written out in components, with $A(t) = \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}$.

DERIVATION

First equation. Differentiate $x = c_1 x_1 + c_2 x_2$ and use (H):

$$\dot{x} = c_1 \dot{x}_1 + c_2 \dot{x}_2 = c_1 (a_1 x_1 + b_1 y_1) + c_2 (a_1 x_2 + b_1 y_2)$$

Regroup the terms with a_1 and with b_1:

$$\dot{x} = a_1 (c_1 x_1 + c_2 x_2) + b_1 (c_1 y_1 + c_2 y_2) = a_1\, x + b_1\, y$$

Second equation. In the same way

$$\dot{y} = c_1 \dot{y}_1 + c_2 \dot{y}_2 = c_1 (a_2 x_1 + b_2 y_1) + c_2 (a_2 x_2 + b_2 y_2) = a_2 (c_1 x_1 + c_2 x_2) + b_2 (c_1 y_1 + c_2 y_2) = a_2\, x + b_2\, y$$

Both equations of (1) are satisfied by (3). ■

Matrix form

$$\dot{\mathbf{x}} = c_1\dot{\mathbf{x}}_1 + c_2\dot{\mathbf{x}}_2 = c_1A\mathbf{x}_1 + c_2A\mathbf{x}_2 = A(c_1\mathbf{x}_1 + c_2\mathbf{x}_2) = A\mathbf{x}, \qquad A(t) = \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}$$

INTERPRETATION AND CHECKS

- Linearity and homogeneity are both used: homogeneity guarantees that the right-hand side of (1) vanishes when $x = y = 0$ (a forcing term $f(t)$ would be multiplied by $c_1 + c_2$ instead of by 1, and the combination would not be a solution unless $c_1 + c_2 = 1$ and so on).
- Constant coefficients were not needed: only linearity of the derivative and of $(x, y) \mapsto (a_1x + b_1y, a_2x + b_2y)$.
- The set of solutions is therefore a vector space; Exercises 3 and 4 show that it has dimension 2.
- Quick example: $\dot{x} = y, \dot{y} = -x$ has $(\cos t, -\sin t)$ and $(\sin t, \cos t)$ as solutions, and so $(c_1\cos t + c_2\sin t,\ -c_1\sin t + c_2\cos t)$ does.

EXERCISE 3

The Wronskian of a Linear System: Liouville's Formula and the Dichotomy

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Wronskian • Liouville • Dichotomy

STATEMENT

Show that given a homogeneous linear system (1) with continuous coefficients in $[a, b]$, then the Wronskian $W(t)$ of two solutions (2) satisfies the first order ODE

$$\frac{dW}{dt} = \left(a_1(t) + b_2(t)\right) W.$$

Deduce that this Wronskian is either identically 0 or nowhere zero in $[a, b]$.

SETUP

System (1) of Exercise 2 with a_1, b_1, a_2, b_2 continuous on $[a, b]$. For two solutions (x_1, y_1) and (x_2, y_2) the Wronskian is the determinant of the matrix whose columns are the solutions (Topic 4 Sec. 3):

$$W(t) = \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} = x_1 y_2 - x_2 y_1$$

FORMULATION

Differentiate W with the product rule, substitute the equations (H) of Exercise 2 for each derivative, and recognise W. The resulting scalar linear ODE is solved by an integrating factor.

CLASS NOTES AND COMPLEMENT [NOTES, P. 9]

The notes state that $\mathbf{x}_1, \mathbf{x}_2$ are independent if and only if $W = \det(x_{ij}) \neq 0$ for all $t \in I$; the words for all t are what this exercise proves. The equation $W' = (\operatorname{tr} A)\, W$ and its solution $W(t) = W(t_0) e^{\int_{t_0}^{t} \operatorname{tr} A}$ (Liouville's formula) are a complement (not in the class notes). For the system of a second-order equation $\operatorname{tr} A = -p$, and it becomes the $W' = -pW$ of Topic 3 (Exercise 6).

DERIVATION

Step 1: the product rule

$$\dot{W} = \dot{x}_1 y_2 + x_1 \dot{y}_2 - \dot{x}_2 y_1 - x_2 \dot{y}_1$$

Step 2: substitute the system

$$\dot{W} = (a_1 x_1 + b_1 y_1) y_2 + x_1 (a_2 x_2 + b_2 y_2) - (a_1 x_2 + b_1 y_2) y_1 - x_2 (a_2 x_1 + b_2 y_1)$$

Expand and collect by coefficient:

$$\dot{W} = a_1 (x_1 y_2 - x_2 y_1) + b_2 (x_1 y_2 - x_2 y_1) + b_1 (y_1 y_2 - y_2 y_1) + a_2 (x_1 x_2 - x_2 x_1)$$

The terms in b_1 and a_2 vanish identically ($y_1 y_2 - y_2 y_1 = 0$ and $x_1 x_2 - x_2 x_1 = 0$), and the remaining brackets are W:

$$\boxed{\dot{W} = (a_1 + b_2)\, W} \qquad (= (\operatorname{tr} A)\, W)$$

Step 3: solve the scalar equation

Let $\tau(t) = \int_{t_0}^{t} [a_1(s) + b_2(s)]\, ds$, which exists and is differentiable because $a_1 + b_2$ is continuous, with $\tau'(t) = a_1(t) + b_2(t)$ (fundamental theorem of calculus). The chain rule gives $\frac{d}{dt} e^{-\tau(t)} = -\tau'(t) e^{-\tau(t)}$, so by the product rule $\frac{d}{dt}\left(e^{-\tau} W\right) = e^{-\tau} \dot{W} - \tau' e^{-\tau} W = e^{-\tau} (\dot{W} - \tau' W) = e^{-\tau} \left(\dot{W} - (a_1 + b_2) W\right) = 0$, so $e^{-\tau} W$ is constant on $[a, b]$. Barrow's rule, $\left[e^{-\tau(s)} W(s)\right]_{t_0}^{t} = 0$, gives $e^{-\tau(t)} W(t) = e^{-\tau(t_0)} W(t_0) = W(t_0)$ (as $\tau(t_0) = 0$):

$$W(t) = W(t_0)\exp\left(\int_{t_0}^{t} [a_1(s) + b_2(s)]\,ds\right), \qquad t \in [a,b]$$

Step 4: the dichotomy

The exponential is finite and strictly positive, so $W(t) = 0$ exactly when $W(t_0) = 0$. Since $t_0 \in [a,b]$ was arbitrary: either $W(t_0) = 0$ for one (hence every) t_0, and then $W \equiv 0$ on $[a,b]$; or $W(t_0) \neq 0$ for one (hence every) t_0, and then W does not vanish anywhere on $[a,b]$. There is no third possibility. ■

INTERPRETATION AND CHECKS

- Constant coefficients: $W(t) = W(0)e^{(a_1+b_2)t} = W(0)e^{(\operatorname{tr} A)t}$ (Topic 4 Sec. 3, Liouville). Check with Example 28.1: eigenvalues 2 and -3 have trace -1 and $W \propto e^{2t}e^{-3t} = e^{-t}$.
- For the system of a second-order equation $\ddot{x} + p\dot{x} + qx = 0$, $a_1 = 0$ and $b_2 = -p$ and the result is Abel's identity $\dot{W} = -pW$ (Exercise 6).
- Geometric meaning: $|W|$ is the area of the parallelogram spanned by the two solution vectors; $\operatorname{tr} A$ is the divergence of the flow, which rates the growth of areas. If $\operatorname{tr} A = 0$ areas are conserved (W constant).
- This dichotomy is what makes the Wronskian a test for independence: it is enough to evaluate it at a single instant (Exercise 4).

EXERCISE 4

A Non-Vanishing Wronskian Gives the General Solution

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Fundamental Set • General Solution • Uniqueness

STATEMENT

> Show that if two solutions (2) of (1) have a Wronskian that does not vanish in $[a, b]$, then (3) gives the general solution of the system.

SETUP AND HYPOTHESES

System (1) with coefficients continuous on $[a, b]$ (as in Exercise 3). (x_1, y_1) and (x_2, y_2) are solutions with $W(t) = x_1 y_2 - x_2 y_1 \neq 0$ for all $t \in [a, b]$. To show: for every solution (x, y) of (1) there are constants c_1, c_2 with $x = c_1 x_1 + c_2 x_2$ and $y = c_1 y_1 + c_2 y_2$ on $[a, b]$. (By Exercise 2 every combination is a solution, so this shows that (3) describes all of them.)

FORMULATION

Tools: (a) uniqueness for linear systems with continuous coefficients: two solutions with the same value at one instant $t_0 \in [a, b]$ coincide on all of $[a, b]$. Robinson Thm 25.2 gives this locally (near t_0); the extension to $[a, b]$ is a continuation argument: the set S of instants where the two solutions agree contains t_0, is closed (the solutions are continuous) and is open in $[a, b]$ (at each point of S the local uniqueness applies, since the coefficients are continuous), and an interval is connected, so $S = [a, b]$; (b) Cramer's rule: a 2×2 linear system with non-zero determinant has exactly one solution. Plan: match the given solution at one instant with a combination, then invoke uniqueness.

> METHOD OF THE CLASS NOTES [NOTES, P. 9]
>
> The notes: for $n = 2$ a fundamental set is a pair of solutions with $W(\mathbf{x}_1, \mathbf{x}_2) \neq 0$, and every solution is a combination of it. This exercise proves that statement. Matching the data at t_0 is the 2×2 system with determinant $W(t_0)$, as in the second-order example of the notes (p. 5), and uniqueness does the rest.

DERIVATION

Step 1: choose the constants at one instant

Let (x, y) be any solution and fix $t_0 \in [a, b]$. We look for c_1, c_2 such that the combination equals (x, y) at t_0:

$$\begin{pmatrix} x_1(t_0) & x_2(t_0) \\ y_1(t_0) & y_2(t_0) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} x(t_0) \\ y(t_0) \end{pmatrix}$$

The determinant of the coefficient matrix is $W(t_0) \neq 0$ by hypothesis, so there is exactly one solution (Cramer):

$$c_1 = \frac{x(t_0)y_2(t_0) - x_2(t_0)y(t_0)}{W(t_0)}, \qquad c_2 = \frac{x_1(t_0)y(t_0) - x(t_0)y_1(t_0)}{W(t_0)}$$

Step 2: uniqueness

Define $\tilde{x} = c_1 x_1 + c_2 x_2$, $\tilde{y} = c_1 y_1 + c_2 y_2$. By Exercise 2 $(\tilde{x}, \tilde{y})$ solves (1), and by Step 1 $(\tilde{x}(t_0), \tilde{y}(t_0)) = (x(t_0), y(t_0))$. Both (x, y) and $(\tilde{x}, \tilde{y})$ solve the same IVP at t_0, so by uniqueness $(x, y) = (\tilde{x}, \tilde{y})$ on $[a, b]$. ■

Where Exercise 3 enters

If the Wronskian were known not to vanish at only some t_0, the argument would work at that t_0; Exercise 3 shows that a non-zero value at one point already implies $W \neq 0$ everywhere, so one test is enough. Also, two solutions with $W \equiv 0$ cannot form a general solution: then the matrix above is singular at each instant, and not every initial vector can be matched.

INTERPRETATION AND CHECKS

- The solution space is a 2-dimensional vector space with basis $\{(x_1, y_1), (x_2, y_2)\}$ (a **fundamental set**; $\Phi = [\mathbf{x}_1\ \mathbf{x}_2]$ is a fundamental matrix and $\mathbf{c} = \Phi(t_0)^{-1}\mathbf{x}(t_0)$, Topic 4 Sec. 3).
- Example: for $\dot{x} = x + y$, $\dot{y} = 4x - 2y$ the solutions $e^{2t}(1, 1)$ and $e^{-3t}(1, -4)$ have $W = e^{2t}e^{-3t}(-4 - 1) = -5e^{-t} \neq 0$, hence form a fundamental set (Topic 4 Example 28.1).
- Check: if $x_2 = kx_1$ and $y_2 = ky_1$ (parallel solutions) then $W = kx_1y_1 - kx_1y_1 = 0$ and the argument rightly fails.
- The same proof works for $n \times n$ systems with $\det \Phi(t_0) \neq 0$.

EXERCISE 5

General Solution of an Inhomogeneous System

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Inhomogeneous System • Particular Solution

STATEMENT

Suppose that we have an inhomogeneous system

$$\begin{cases} \dfrac{dx}{dt} = a_1(t)x + b_1(t)y + f_1(t), \\ \dfrac{dy}{dt} = a_2(t)x + b_2(t)y + f_2(t) \end{cases} \tag{4}$$

and one particular solution

$$\begin{cases} x = x_p(t) \\ y = y_p(t). \end{cases} \tag{5}$$

Show that, if $x_1(t)$ and $x_2(t)$ are linearly independent solutions of the homogeneous system, then the general solution of (4) is

$$\begin{cases} x = c_1x_1(t) + c_2x_2(t) + x_p(t) \\ y = c_1y_1(t) + c_2y_2(t) + y_p(t) \end{cases}$$

SETUP AND HYPOTHESES

Coefficients and forcings f_1, f_2 continuous on $[a,b]$. The homogeneous system is (1) of Exercise 2. $\mathbf{x}_1 = (x_1, y_1)$ and $\mathbf{x}_2 = (x_2, y_2)$ are linearly independent solutions of (1) (the statement writes "$x_1(t)$ and $x_2(t)$ " for the two solution vectors). Their Wronskian is non-zero on $[a,b]$: by Exercise 3 it is either $\equiv 0$ or never zero, and if $W(t_0) = 0$ there would be $(c_1, c_2) \neq (0,0)$ with $c_1\mathbf{x}_1(t_0) + c_2\mathbf{x}_2(t_0) = \mathbf{0}$; the combination $c_1\mathbf{x}_1 + c_2\mathbf{x}_2$ solves (1) with zero data at t_0, so it is $\equiv \mathbf{0}$ by uniqueness (Exercise 4, Step 2), contradicting independence. $\mathbf{x}_p = (x_p, y_p)$ solves (4).

FORMULATION

Two claims: (A) every function of the displayed form solves (4); (B) every solution of (4) has that form. The key idea (Robinson Sec. 14.1 for scalar equations, Topic 4 Sec. 3.1) is that the difference of two solutions of (4) solves the **homogeneous** system.

METHOD OF THE CLASS NOTES [NOTES, P. 9]

The last theorem of the notes for linear systems: $\mathbf{x} = \mathbf{x}_h + \mathbf{x}_p$. The proof is the one of the scalar case (Topic 3, notes p. 6): the difference of two solutions of the inhomogeneous system solves the homogeneous one.

DERIVATION

(A) The family solves (4)

Write $\mathbf{x} = c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \mathbf{x}_p$ and $\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}$ with $A = \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}$, $\mathbf{f} = (f_1, f_2)^T$. Using $\dot{\mathbf{x}}_j = A\mathbf{x}_j$ and $\dot{\mathbf{x}}_p = A\mathbf{x}_p + \mathbf{f}$:

$$\dot{\mathbf{x}} = c_1 A\mathbf{x}_1 + c_2 A\mathbf{x}_2 + A\mathbf{x}_p + \mathbf{f} = A(c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \mathbf{x}_p) + \mathbf{f} = A\mathbf{x} + \mathbf{f}$$

(B) Every solution has this form

Let $\mathbf{x} = (x, y)$ be any solution of (4) and put $\mathbf{z} = \mathbf{x} - \mathbf{x}_p$. Then

$$\dot{\mathbf{z}} = \dot{\mathbf{x}} - \dot{\mathbf{x}}_p = (A\mathbf{x} + \mathbf{f}) - (A\mathbf{x}_p + \mathbf{f}) = A(\mathbf{x} - \mathbf{x}_p) = A\mathbf{z}$$

so $\mathbf{z}$ solves the homogeneous system (1). Since $\mathbf{x}_1, \mathbf{x}_2$ are independent solutions of (1) (non-zero Wronskian), Exercise 4 gives constants c_1, c_2 with $\mathbf{z} = c_1\mathbf{x}_1 + c_2\mathbf{x}_2$. Hence $\mathbf{x} = c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \mathbf{x}_p$. ■

INTERPRETATION AND CHECKS

- The structure "general = complementary + particular" is the same as for scalar linear equations (Topic 3 Sec. 5 lemma). The two constants are fixed by an initial condition $\mathbf{x}(t_0) = \mathbf{x}_0$: $\mathbf{c} = \Phi(t_0)^{-1}\,[\mathbf{x}_0 - \mathbf{x}_p(t_0)]$.
- Example (Topic 4 Example 26.1): $\dot{x} = x + y$, $\dot{y} = 4x - 2y + 4e^{-2t}$ has the particular solution $(-e^{-2t}, 3e^{-2t})$: check $\dot{x}_p = 2e^{-2t} = x_p + y_p = -e^{-2t} + 3e^{-2t}$ and $\dot{y}_p = -6e^{-2t} = 4(-e^{-2t}) - 2(3e^{-2t}) + 4e^{-2t} = -6e^{-2t}$.
- A particular solution can be found by elimination (Exercise 7), undetermined coefficients, or variation of parameters $\mathbf{x}_p = \Phi \int \Phi^{-1}\mathbf{f}$ (Topic 4 Sec. 3.1).
- Different particular solutions differ by a homogeneous solution, which is absorbed in c_1, c_2.

EXERCISE 6

A Second-Order Equation as a System: the Two Wronskians Coincide

Original statement · probls_ch4-i_2627_v2.pdf, p. 2

Second Order to System • Wronskian • Abel

STATEMENT

Suppose that we have a second order linear ODE $x''(t) + p(t)x'(t) + q(t)x(t) = 0$, and two solutions $x_1(t)$ and $x_2(t)$. Write it as a linear system of two first order ODEs. Check that if

$$\begin{cases} x = x_1(t) \\ y = y_1(t) \end{cases} \quad \text{and} \quad \begin{cases} x = x_2(t) \\ y = y_2(t) \end{cases}$$

are the corresponding solutions of the system, then the Wronskian as defined in this chapter coincides with the definition that we had before for second order ODEs.

SETUP

p, q continuous on an interval I. Two definitions to compare: the scalar Wronskian $W_2[x_1, x_2] = x_1 x_2' - x_2 x_1'$ (Topic 3 Sec. 2.1) and the Wronskian of vector solutions $W_{sys} = \det[\mathbf{x}_1\ \mathbf{x}_2]$ (Topic 4 Sec. 3).

FORMULATION: THE SYSTEM

Introduce $y = x'$ (Robinson Sec. 25.1: $x_1 \to x$, $x_2 \to x'$). Then $x' = y$ and, from the ODE, $y' = x'' = -p\,x' - q\,x = -q\,x - p\,y$:

$$\begin{cases} x' = 0 \cdot x + 1 \cdot y \\ y' = -q(t)\,x - p(t)\,y \end{cases} \qquad \dot{\mathbf{x}} = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix} \mathbf{x}$$

This has the form (1) with $a_1 = 0, b_1 = 1, a_2 = -q, b_2 = -p$ (continuous). A solution $x_j(t)$ of the ODE gives the solution $(x_j, y_j) = (x_j, x_j')$ of the system, and conversely the first component of a system solution solves the ODE.

METHOD OF THE CLASS NOTES [NOTES, P. 9]

This is the important point of the notes with $x_1 = x$ and $x_2 = \dot{x}$ (here y), applied to $x'' + px' + qx = 0$: $A = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix}$. The Wronskian of the system, $\det \begin{pmatrix} x_1 & x_2 \\ y_1 & y_2 \end{pmatrix} = x_1 y_2 - x_2 y_1$ with $y_j = x_j'$, is then the Wronskian $x_1 x_2' - x_2 x_1'$ of the notes for second-order equations (p. 5).

DERIVATION

The corresponding solutions of the system are $\mathbf{x}_1 = (x_1, x_1')^T$ and $\mathbf{x}_2 = (x_2, x_2')^T$, so $y_j = x_j'$. Their system Wronskian is

$$W_{sys} = \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} = x_1 y_2 - x_2 y_1 = x_1 x_2' - x_2 x_1' = W_2[x_1, x_2]$$

The two definitions coincide identically. ■

Consistency with Exercise 3

Exercise 3 applied to this system ($a_1 + b_2 = 0 + (-p) = -p$) gives $\dot{W} = -pW$, i.e. $W(t) = W(t_0) \exp\left(-\int_{t_0}^{t} p\right)$, which is Abel's identity of Topic 3 (Robinson Exercise 11.2). Direct check from the scalar definition: $W_2' = x_1' x_2' + x_1 x_2'' - x_2' x_1' - x_2 x_1'' = x_1 x_2'' - x_2 x_1'' = x_1(-p x_2' - q x_2) - x_2(-p x_1' - q x_1) = -p(x_1 x_2' - x_2 x_1') = -p W_2$, the same equation.

INTERPRETATION AND CHECKS

- Therefore all the theory of systems (non-vanishing Wronskian $\Leftrightarrow$ independence, general solution, dichotomy) contains the scalar results of Topic 3 as the special case $A = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix}$; the notation $\mathbf{x}(t_0) = (x(t_0), \dot{x}(t_0))$ for the initial data is the same pair as in Theorem 11.1.
- Example: $\ddot{x} + x = 0$, $x_1 = \cos t$, $x_2 = \sin t$: $W_2 = \cos^2 t + \sin^2 t = 1$ and $W_{sys} = \begin{vmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{vmatrix} = 1$; $p = 0$ so W is constant, as Abel/Liouville predict.
- The matrix $\begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix}$ has trace $-p$ and determinant q; for constants it gives the characteristic polynomial $\mu^2 + p\mu + q$ of Topic 3.

EXERCISE 7

Seven Linear Systems Solved by Elimination

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Elimination Method • Seven IVPs • Undetermined Coefficients

STATEMENT

Find the general solutions of the following differential equations by converting them into a single second-order equation. Also find the solution that satisfies the given initial conditions.

- (i) $\dot{x} = 4x - y, \quad \dot{y} = 2x + y + t^2, \quad x(0) = 0,\ y(0) = 1;$
- (ii) $\dot{x} = x - 4y + \cos 2t, \quad \dot{y} = x + y, \quad x(0) = 1,\ y(0) = 1;$
- (iii) $\dot{x} = 2x + 2y, \quad \dot{y} = 6x + 3y + e^t, \quad x(0) = 0,\ \dot{x}(0) = 1;$
- (iv) $\dot{x} = 5x - 4y + e^{3t}, \quad \dot{y} = x + y, \quad x(0) = 1,\ y(0) = -1;$
- (v) $\dot{x} = 2x + 5y, \quad \dot{y} = -2x + \cos 3t, \quad x(0) = 2,\ y(0) = -1;$
- (vi) $\dot{x} = x + y + e^{-t}, \quad \dot{y} = 4x - 2y + e^{2t}, \quad x(0) = 1,\ y(0) = -1;$
- (vii) $\dot{x} = 8x + 14y, \quad \dot{y} = 7x + y, \quad x(0) = y(0) = 1.$

SETUP AND CLASSIFICATION

Each system is linear with constant coefficients, of the form $\dot{x} = ax + by + f(t), \dot{y} = cx + dy + g(t)$ (Robinson Eq. 26.1), homogeneous when $f = g = 0$. In every item $b \neq 0$, so the first equation can be solved for y. Hypotheses: coefficients and forcings continuous on $\mathbb{R}$, so the IVP has a unique solution, which exists on all of $\mathbb{R}$ (linear system with continuous coefficients; Thm 25.2 gives local existence and uniqueness and the continuation argument of Exercise 4 extends it). Item (iii) gives $x(0)$ and $\dot{x}(0)$; the value of $y(0)$ then follows from the first equation.

METHOD: ELIMINATION (TOPIC 4 SEC. 2.1)

Solve the first equation for $y = (\dot{x} - ax - f)/b$ (26.2), differentiate, and substitute in the second equation. The result is one second-order equation for x (26.3):

$$\ddot{x} - (a+d)\dot{x} + (ad - bc)x = \dot{f} + bg - df$$

Its coefficients are the trace and determinant of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, so its characteristic polynomial is the one of the eigenvalue method. It is solved by the methods of Topic 3 (complementary function plus undetermined coefficients). Then y is recovered from (26.2) and the constants from the data.

METHOD OF THE CLASS NOTES [NOTES, P. 9]

Method 1 of the notes: for $\dot{x} = ax + by$, $\dot{y} = cx + dy$ with $b \neq 0$ (true in all seven items), $y = (\dot{x} - ax)/b$ and $\dot{y} = (\ddot{x} - a\dot{x})/b$; substituting and multiplying by b gives $\ddot{x} - (a+d)\dot{x} + (ad - bc)x = 0$, the homogeneous part of the equation above. The forcing terms are a complement (not in the class notes); the second-order equation is then solved as in Topic 3, $x = x_H + x_N$ with the educated guess.

Discrepancy with the notes: they box $\ddot{x} - (a+b)\dot{x} + (ad - bc)x = 0$, but the steps give $a + d$. Check with $x' = 2x - y$, $y' = -x + 3y$: elimination gives $\ddot{x} - 5\dot{x} + 5x = 0$, and $a + d = 5$ while $a + b = 1$.

DERIVATION OF THE SEVEN SOLUTIONS

(i) $\dot{x} = 4x - y, \quad \dot{y} = 2x + y + t^2, \quad x(0) = 0,\ y(0) = 1$

The system is linear with constant coefficients, in the form $\dot{x} = ax + by + f(t)$, $\dot{y} = cx + dy + g(t)$ (Robinson Eq. 26.1) with

$$a = 4, \quad b = -1, \quad c = 2, \quad d = 1, \quad f(t) = 0, \quad g(t) = t^2$$

Elimination of y

Since $b = -1 \neq 0$, the first equation can be solved for y (Eq. 26.2):

$$y = \frac{\dot{x} - ax - f}{b} = \frac{1}{-1}\left(\dot{x} - (4)x - (0)\right)$$

Differentiate with respect to t (everything is differentiable):

$$\dot{y} = \frac{\ddot{x} - a\dot{x} - \dot{f}}{b} = \frac{1}{-1}\left(\ddot{x} - (4)\dot{x} - (0)\right)$$

Substitute both expressions into the second equation, $\dot{y} = cx + dy + g$, and multiply by b:

$$\ddot{x} - a\dot{x} - \dot{f} = bc\,x + d\left(\dot{x} - ax - f\right) + bg$$

Collect the terms in $\ddot{x}$, $\dot{x}$ and x (Eq. 26.3):

$$\ddot{x} - (a+d)\dot{x} + (ad - bc)x = \dot{f} + bg - df$$

Numerically, $a + d = 5$, $ad - bc = (4)(1) - (-1)(2) = 6$ and $F(t) := \dot{f} + bg - df = (0) + (-1)(t^2) - (1)(0) = -t^2$, so

$$\ddot{x} - 5\dot{x} + 6x = -t^2$$

Initial data. The condition on y is converted into one on $\dot{x}$ with the first equation at $t = 0$: $\dot{x}(0) = a\,x(0) + b\,y(0) + f(0) = (4)(0) + (-1)(1) + (0) = -1$.

The homogeneous equation $\ddot{x} - 5\dot{x} + 6x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 - 5k + 6 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (-5)^2 - 4(1)(6) = 1$.

Factorising, $P(k) = (k-3)\,(k-2)$, so the roots are

$$k = 2, \qquad k = 3$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{2t}, e^{3t}, so the complementary function is

$$x_h(t) = c_1 e^{2t} + c_2 e^{3t}$$

The forcing $-t^2$ is a polynomial of degree 2; a polynomial is e^{0t} times a polynomial, so the exponent to test is $s = 0$: $P(0) = 6$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 + K_2 t + K_3 t^2$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= K_2 + 2K_3 t \\ x_p''(t) &= 2K_3 \end{aligned}$$

Substitute into $L[x_p] = x_p'' - 5x_p' + 6x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 6K_3\, t^2 + (6K_2 - 10K_3)\, t + (6K_1 - 5K_2 + 2K_3)$$

This must equal the forcing $F(t) = -t^2$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in F:

$$\begin{aligned} t^2: &\quad 6K_3 = -1 \\ t: &\quad 6K_2 - 10K_3 = 0 \\ 1: &\quad 6K_1 - 5K_2 + 2K_3 = 0 \end{aligned}$$

The linear system is

$$\begin{aligned} 6K_3 &= -1 \quad (1) \\ 6K_2 - 10K_3 &= 0 \quad (2) \\ 6K_1 - 5K_2 + 2K_3 &= 0 \quad (3) \end{aligned}$$

$6K_3 = -1$ contains one unknown only, so $K_3 = -\frac{1}{6}$.

Substitute in $6K_2 - 10K_3 = 0$: $6K_2 + \frac{5}{3} = 0$.

Substitute in $6K_1 - 5K_2 + 2K_3 = 0$: $6K_1 - 5K_2 - \frac{1}{3} = 0$.

$6K_2 + \frac{5}{3} = 0$ contains one unknown only, so $K_2 = -\frac{5}{18}$.

Substitute in $6K_1 - 5K_2 - \frac{1}{3} = 0$: $6K_1 + \frac{19}{18} = 0$.

$6K_1 + \frac{19}{18} = 0$ contains one unknown only, so $K_1 = -\frac{19}{108}$.

Hence $K_1 = -\frac{19}{108}, K_2 = -\frac{5}{18}, K_3 = -\frac{1}{6}$.

Therefore

$$x_p(t) = -\frac{t^2}{6} - \frac{5t}{18} - \frac{19}{108}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{2t} + c_2 e^{3t} - \frac{t^2}{6} - \frac{5t}{18} - \frac{19}{108}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = 2c_1 e^{2t} + 3c_2 e^{3t} - \frac{t}{3} - \frac{5}{18}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 + c_2 - \frac{19}{108} = 0$$
$$x'(0) = 2c_1 + 3c_2 - \frac{5}{18} = -1$$

The linear system is

$$c_1 + c_2 - \frac{19}{108} = 0 \quad (1)$$
$$2c_1 + 3c_2 - \frac{5}{18} = -1 \quad (2)$$

Solve $c_1 + c_2 - \frac{19}{108} = 0$ for c_1: $c_1 = \frac{19}{108} - c_2$.

Substitute in $2c_1 + 3c_2 - \frac{5}{18} = -1$: $c_2 + \frac{2}{27} = -1$.

$c_2 + \frac{2}{27} = -1$ contains one unknown only, so $c_2 = -\frac{29}{27}$.

Back-substitution: $c_1 = \frac{19}{108} - c_2 = \frac{5}{4}$.

Hence $c_1 = \frac{5}{4}, c_2 = -\frac{29}{27}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = -\frac{t^2}{6} - \frac{5t}{18} - \frac{29e^{3t}}{27} + \frac{5e^{2t}}{4} - \frac{19}{108}}$$

Recovering y

From Eq. 26.2, $y = (\dot{x} - ax - f)/b$. With the solution found,

$$\dot{x}(t) = -\frac{t}{3} - \frac{29e^{3t}}{9} + \frac{5e^{2t}}{2} - \frac{5}{18}$$

$$y(t) = \frac{1}{-1}\left(\dot{x} - (4)x - (0)\right) = -\frac{2t^2}{3} - \frac{7t}{9} - \frac{29e^{3t}}{27} + \frac{5e^{2t}}{2} - \frac{23}{54}$$

Check in the original system

$$\begin{aligned} \dot{x} &= -\frac{t}{3} - \frac{29e^{3t}}{9} + \frac{5e^{2t}}{2} - \frac{5}{18} \\ ax + by + f &= -\frac{t}{3} - \frac{29e^{3t}}{9} + \frac{5e^{2t}}{2} - \frac{5}{18} \\ \dot{y} &= -\frac{4t}{3} - \frac{29e^{3t}}{9} + 5e^{2t} - \frac{7}{9} \\ cx + dy + g &= -\frac{4t}{3} - \frac{29e^{3t}}{9} + 5e^{2t} - \frac{7}{9} \end{aligned}$$

Both pairs of expressions coincide, and $x(0) = 0, y(0) = 1$ reproduce the initial data.

(ii) $\dot{x} = x - 4y + \cos 2t, \quad \dot{y} = x + y, \quad x(0) = 1,\ y(0) = 1$

The system is linear with constant coefficients, in the form $\dot{x} = ax + by + f(t), \dot{y} = cx + dy + g(t)$ (Robinson Eq. 26.1) with

$$a = 1, \quad b = -4, \quad c = 1, \quad d = 1, \quad f(t) = \cos 2t, \quad g(t) = 0$$

Elimination of y

Since $b = -4 \neq 0$, the first equation can be solved for y (Eq. 26.2):

$$y = \frac{\dot{x} - ax - f}{b} = \frac{1}{-4}\left(\dot{x} - (1)x - (\cos 2t)\right)$$

Differentiate with respect to t (everything is differentiable):

$$\dot{y} = \frac{\ddot{x} - a\dot{x} - \dot{f}}{b} = \frac{1}{-4}\left(\ddot{x} - (1)\dot{x} - (-2\sin 2t)\right)$$

Substitute both expressions into the second equation, $\dot{y} = cx + dy + g$, and multiply by b:

$$\ddot{x} - a\dot{x} - \dot{f} = bc\,x + d\left(\dot{x} - ax - f\right) + bg$$

Collect the terms in $\ddot{x}$, $\dot{x}$ and x (Eq. 26.3):

$$\ddot{x} - (a + d)\dot{x} + (ad - bc)x = \dot{f} + bg - df$$

Numerically, $a + d = 2$, $ad - bc = (1)(1) - (-4)(1) = 5$ and $F(t) := \dot{f} + bg - df = (-2\sin 2t) + (-4)(0) - (1)(\cos 2t) = -2\sin 2t - \cos 2t$, so

$$\ddot{x} - 2\dot{x} + 5x = -2\sin 2t - \cos 2t$$

Initial data. The condition on y is converted into one on $\dot{x}$ with the first equation at $t = 0$: $\dot{x}(0) = a\,x(0) + b\,y(0) + f(0) = (1)(1) + (-4)(1) + (1) = -2$.

The homogeneous equation $\ddot{x} - 2\dot{x} + 5x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 - 2k + 5 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (-2)^2 - 4(1)(5) = -16$.

Factorising, $P(k) = (k - 1 - 2i)\,(k - 1 + 2i)$, so the roots are

$$k = 1 - 2i, \qquad k = 1 + 2i$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $e^t \cos 2t$, $e^t \sin 2t$, so the complementary function is

$$x_h(t) = c_1 e^t \cos 2t + c_2 e^t \sin 2t$$

The forcing $-2\sin 2t - \cos 2t$ is a sinusoidal term: it is the real or imaginary part of $e^{(2i)t}$, so we test $s = 2i$ in the characteristic polynomial: $P(2i) = 1 - 4i$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 \cos 2t + K_2 \sin 2t$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= -2K_1 \sin 2t + 2K_2 \cos 2t \\ x_p''(t) &= -4K_1 \cos 2t - 4K_2 \sin 2t \end{aligned}$$

Substitute into $L[x_p] = x_p'' - 2x_p' + 5x_p$ and collect the terms that multiply the same function:

$$L[x_p] = (K_1 - 4K_2)\,\cos 2t + (4K_1 + K_2)\,\sin 2t$$

This must equal the forcing $F(t) = -2\sin 2t - \cos 2t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in F:

$$\begin{aligned} \cos 2t: &\quad K_1 - 4K_2 = -1 \\ \sin 2t: &\quad 4K_1 + K_2 = -2 \end{aligned}$$

The linear system is

$$K_1 - 4K_2 = -1 \quad (1)$$
$$4K_1 + K_2 = -2 \quad (2)$$

Solve $K_1 - 4K_2 = -1$ for K_1: $K_1 = 4K_2 - 1$.

Substitute in $4K_1 + K_2 = -2$: $17K_2 - 4 = -2$.

$17K_2 - 4 = -2$ contains one unknown only, so $K_2 = \frac{2}{17}$.

Back-substitution: $K_1 = 4K_2 - 1 = -\frac{9}{17}$.

Hence $K_1 = -\frac{9}{17}, K_2 = \frac{2}{17}$.

Therefore

$$x_p(t) = \frac{2\sin 2t}{17} - \frac{9\cos 2t}{17}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^t \cos 2t + c_2 e^t \sin 2t + \frac{2\sin 2t}{17} - \frac{9\cos 2t}{17}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -2c_1 e^t \sin 2t + c_1 e^t \cos 2t + c_2 e^t \sin 2t + 2c_2 e^t \cos 2t + \frac{18\sin 2t}{17} + \frac{4\cos 2t}{17}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 - \frac{9}{17} = 1$$
$$x'(0) = c_1 + 2c_2 + \frac{4}{17} = -2$$

The linear system is

$$c_1 - \frac{9}{17} = 1 \quad (1)$$
$$c_1 + 2c_2 + \frac{4}{17} = -2 \quad (2)$$

$c_1 - \frac{9}{17} = 1$ contains one unknown only, so $c_1 = \frac{26}{17}$.

Substitute in $c_1 + 2c_2 + \frac{4}{17} = -2$: $2c_2 + \frac{30}{17} = -2$.

$2c_2 + \frac{30}{17} = -2$ contains one unknown only, so $c_2 = -\frac{32}{17}$.

Hence $c_1 = \frac{26}{17}$, $c_2 = -\frac{32}{17}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = -\frac{32e^t \sin 2t}{17} + \frac{26e^t \cos 2t}{17} + \frac{2\sin 2t}{17} - \frac{9\cos 2t}{17}}$$

Recovering y

From Eq. 26.2, $y = (\dot{x} - ax - f)/b$. With the solution found,

$$\dot{x}(t) = -\frac{84e^t \sin 2t}{17} - \frac{38e^t \cos 2t}{17} + \frac{18\sin 2t}{17} + \frac{4\cos 2t}{17}$$

$$y(t) = \frac{1}{-4}\left(\dot{x} - (1)x - (\cos 2t)\right) = \frac{13e^t \sin 2t}{17} + \frac{16e^t \cos 2t}{17} - \frac{4\sin 2t}{17} + \frac{\cos 2t}{17}$$

Check in the original system

$$\begin{aligned} \dot{x} &= -\frac{84e^t \sin 2t}{17} - \frac{38e^t \cos 2t}{17} + \frac{18\sin 2t}{17} + \frac{4\cos 2t}{17} \\ ax + by + f &= -\frac{84e^t \sin 2t}{17} - \frac{38e^t \cos 2t}{17} + \frac{18\sin 2t}{17} + \frac{4\cos 2t}{17} \\ \dot{y} &= -\frac{19e^t \sin 2t}{17} + \frac{42e^t \cos 2t}{17} - \frac{2\sin 2t}{17} - \frac{8\cos 2t}{17} \\ cx + dy + g &= -\frac{19e^t \sin 2t}{17} + \frac{42e^t \cos 2t}{17} - \frac{2\sin 2t}{17} - \frac{8\cos 2t}{17} \end{aligned}$$

Both pairs of expressions coincide, and $x(0) = 1, y(0) = 1$ reproduce the initial data.

(iii) $\dot{x} = 2x + 2y, \quad \dot{y} = 6x + 3y + e^t, \quad x(0) = 0,\ \dot{x}(0) = 1$

The system is linear with constant coefficients, in the form $\dot{x} = ax + by + f(t), \dot{y} = cx + dy + g(t)$ (Robinson Eq. 26.1) with

$$a = 2, \quad b = 2, \quad c = 6, \quad d = 3, \quad f(t) = 0, \quad g(t) = e^t$$

Elimination of y

Since $b = 2 \neq 0$, the first equation can be solved for y (Eq. 26.2):

$$y = \frac{\dot{x} - a x - f}{b} = \frac{1}{2}\left(\dot{x} - (2)x - (0)\right)$$

Differentiate with respect to t (everything is differentiable):

$$\dot{y} = \frac{\ddot{x} - a\dot{x} - \dot{f}}{b} = \frac{1}{2}\left(\ddot{x} - (2)\dot{x} - (0)\right)$$

Substitute both expressions into the second equation, $\dot{y} = cx + dy + g$, and multiply by b:

$$\ddot{x} - a\dot{x} - \dot{f} = bc\,x + d\left(\dot{x} - a x - f\right) + bg$$

Collect the terms in $\ddot{x}$, $\dot{x}$ and x (Eq. 26.3):

$$\ddot{x} - (a + d)\dot{x} + (ad - bc)x = \dot{f} + bg - df$$

Numerically, $a + d = 5$, $ad - bc = (2)(3) - (2)(6) = -6$ and $F(t) := \dot{f} + bg - df = (0) + (2)(e^{t}) - (3)(0) = 2e^{t}$, so

$$\ddot{x} - 5\dot{x} - 6x = 2e^{t}$$

Initial data. Here $x(0)$ and $\dot{x}(0)$ are given; the value of $y(0)$ follows from the first equation at $t = 0$: $y(0) = \dfrac{\dot{x}(0) - a\,x(0) - f(0)}{b} = \dfrac{1 - (2)(0) - (0)}{2} = \tfrac{1}{2}$.

The homogeneous equation $\ddot{x} - 5\dot{x} - 6x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^{2} - 5k - 6 = 0$$

Its discriminant is $\Delta = b^{2} - 4ac = (-5)^{2} - 4(1)(-6) = 49$.

Factorising, $P(k) = (k - 6)\,(k + 1)$, so the roots are

$$k = -1, \qquad k = 6$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-t}, e^{6t}, so the complementary function is

$$x_h(t) = c_1 e^{-t} + c_2 e^{6t}$$

The forcing $2e^{t}$ is an exponential with exponent $s = 1$; we test it in the characteristic polynomial: $P(1) = -10$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 e^{t}$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= K_1 e^{t} \\ x_p''(t) &= K_1 e^{t} \end{aligned}$$

Substitute into $L[x_p] = x_p'' - 5x_p' - 6x_p$ and collect the terms that multiply the same function:

$$L[x_p] = -10K_1\, e^{t}$$

This must equal the forcing $F(t) = 2e^{t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in F:

$$e^{t}: \quad -10K_1 = 2$$

The linear system is

$$-10K_1 = 2 \quad (1)$$

$-10K_1 = 2$ contains one unknown only, so $K_1 = -\frac{1}{5}$.

Hence $K_1 = -\frac{1}{5}$.

Therefore

$$x_p(t) = -\frac{e^t}{5}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{-t} + c_2 e^{6t} - \frac{e^t}{5}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -c_1 e^{-t} + 6c_2 e^{6t} - \frac{e^t}{5}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 + c_2 - \frac{1}{5} = 0$$
$$x'(0) = -c_1 + 6c_2 - \frac{1}{5} = 1$$

The linear system is

$$c_1 + c_2 - \frac{1}{5} = 0 \quad (1)$$
$$-c_1 + 6c_2 - \frac{1}{5} = 1 \quad (2)$$

Solve $c_1 + c_2 - \frac{1}{5} = 0$ for c_1: $c_1 = \frac{1}{5} - c_2$.

Substitute in $-c_1 + 6c_2 - \frac{1}{5} = 1$: $7c_2 - \frac{2}{5} = 1$.

$7c_2 - \frac{2}{5} = 1$ contains one unknown only, so $c_2 = \frac{1}{5}$.

Back-substitution: $c_1 = \frac{1}{5} - c_2 = 0$.

Hence $c_1 = 0$, $c_2 = \frac{1}{5}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{e^{6t}}{5} - \frac{e^{t}}{5}}$$

Recovering y

From Eq. 26.2, $y = (\dot{x} - ax - f)/b$. With the solution found,

$$\dot{x}(t) = \frac{6e^{6t}}{5} - \frac{e^{t}}{5}$$

$$y(t) = \frac{1}{2}\left(\dot{x} - (2)x - (0)\right) = \frac{2e^{6t}}{5} + \frac{e^{t}}{10}$$

Check in the original system

$$\begin{aligned} \dot{x} &= \frac{6e^{6t}}{5} - \frac{e^{t}}{5} \\ ax + by + f &= \frac{6e^{6t}}{5} - \frac{e^{t}}{5} \\ \dot{y} &= \frac{12e^{6t}}{5} + \frac{e^{t}}{10} \\ cx + dy + g &= \frac{12e^{6t}}{5} + \frac{e^{t}}{10} \end{aligned}$$

Both pairs of expressions coincide, and $x(0) = 0$, $y(0) = \tfrac{1}{2}$ reproduce the initial data.

(iv) $\dot{x} = 5x - 4y + e^{3t}, \quad \dot{y} = x + y, \quad x(0) = 1,\ y(0) = -1$

The system is linear with constant coefficients, in the form $\dot{x} = ax + by + f(t)$, $\dot{y} = cx + dy + g(t)$ (Robinson Eq. 26.1) with

$$a = 5, \quad b = -4, \quad c = 1, \quad d = 1, \quad f(t) = e^{3t}, \quad g(t) = 0$$

Elimination of y

Since $b = -4 \neq 0$, the first equation can be solved for y (Eq. 26.2):

$$y = \frac{\dot{x} - ax - f}{b} = \frac{1}{-4}\left(\dot{x} - (5)x - (e^{3t})\right)$$

Differentiate with respect to t (everything is differentiable):

$$\dot{y} = \frac{\ddot{x} - a\dot{x} - \dot{f}}{b} = \frac{1}{-4}\left(\ddot{x} - (5)\dot{x} - (3e^{3t})\right)$$

Substitute both expressions into the second equation, $\dot{y} = cx + dy + g$, and multiply by b:

$$\ddot{x} - a\dot{x} - \dot{f} = bc\,x + d\,(\dot{x} - ax - f) + bg$$

Collect the terms in $\ddot{x}$, $\dot{x}$ and x (Eq. 26.3):

$$\ddot{x} - (a + d)\dot{x} + (ad - bc)x = \dot{f} + bg - df$$

Numerically, $a + d = 6$, $ad - bc = (5)(1) - (-4)(1) = 9$ and $F(t) := \dot{f} + bg - df = (3e^{3t}) + (-4)(0) - (1)(e^{3t}) = 2e^{3t}$, so

$$\ddot{x} - 6\dot{x} + 9x = 2e^{3t}$$

Initial data. The condition on y is converted into one on $\dot{x}$ with the first equation at $t = 0$: $\dot{x}(0) = a\,x(0) + b\,y(0) + f(0) = (5)(1) + (-4)(-1) + (1) = 10$.

The homogeneous equation $\ddot{x} - 6\dot{x} + 9x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 - 6k + 9 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (-6)^2 - 4(1)(9) = 0$.

Factorising, $P(k) = (k - 3)^2$, so the roots are

$$k = 3 \text{ (multiplicity } 2)$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{3t}, te^{3t}, so the complementary function is

$$x_h(t) = c_1 e^{3t} + c_2 t e^{3t}$$

The forcing $2e^{3t}$ is an exponential with exponent $s = 3$; we test it in the characteristic polynomial: $P(3) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 2, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by t^2.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 t^2 e^{3t}$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= 3K_1 t^2 e^{3t} + 2K_1 t e^{3t} \\ x_p''(t) &= 9K_1 t^2 e^{3t} + 12K_1 t e^{3t} + 2K_1 e^{3t} \end{aligned}$$

Substitute into $L[x_p] = x_p'' - 6x_p' + 9x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 2K_1\, e^{3t}$$

This must equal the forcing $F(t) = 2e^{3t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in F:

$$e^{3t}: \quad 2K_1 = 2$$

The linear system is

$$2K_1 = 2 \quad (1)$$

$2K_1 = 2$ contains one unknown only, so $K_1 = 1$.

Hence $K_1 = 1$.

Therefore

$$x_p(t) = t^2 e^{3t}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{3t} + c_2 t e^{3t} + t^2 e^{3t}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = 3c_1 e^{3t} + 3c_2 t e^{3t} + c_2 e^{3t} + 3t^2 e^{3t} + 2te^{3t}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 = 1$$
$$x'(0) = 3c_1 + c_2 = 10$$

The linear system is

$$c_1 = 1 \quad (1)$$
$$3c_1 + c_2 = 10 \quad (2)$$

$c_1 = 1$ contains one unknown only, so $c_1 = 1$.

Substitute in $3c_1 + c_2 = 10$: $c_2 + 3 = 10$.

$c_2 + 3 = 10$ contains one unknown only, so $c_2 = 7$.

Hence $c_1 = 1, c_2 = 7$.

Substituting the constants into the general solution,

$$\boxed{x(t) = t^2 e^{3t} + 7te^{3t} + e^{3t}}$$

Recovering y

From Eq. 26.2, $y = (\dot{x} - ax - f)/b$. With the solution found,

$$\dot{x}(t) = 3t^2 e^{3t} + 23te^{3t} + 10e^{3t}$$

$$y(t) = \frac{1}{-4}\left(\dot{x} - (5)x - (e^{3t})\right) = \frac{t^2 e^{3t}}{2} + 3te^{3t} - e^{3t}$$

Check in the original system

$$\begin{aligned} \dot{x} &= 3t^2 e^{3t} + 23te^{3t} + 10e^{3t} \\ ax + by + f &= 3t^2 e^{3t} + 23te^{3t} + 10e^{3t} \\ \dot{y} &= \frac{3t^2 e^{3t}}{2} + 10te^{3t} \\ cx + dy + g &= \frac{3t^2 e^{3t}}{2} + 10te^{3t} \end{aligned}$$

Both pairs of expressions coincide, and $x(0) = 1, y(0) = -1$ reproduce the initial data.

(v) $\dot{x} = 2x + 5y, \quad \dot{y} = -2x + \cos 3t, \quad x(0) = 2,\ y(0) = -1$

The system is linear with constant coefficients, in the form $\dot{x} = ax + by + f(t), \dot{y} = cx + dy + g(t)$ (Robinson Eq. 26.1) with

$$a = 2, \quad b = 5, \quad c = -2, \quad d = 0, \quad f(t) = 0, \quad g(t) = \cos 3t$$

Elimination of y

Since $b = 5 \neq 0$, the first equation can be solved for y (Eq. 26.2):

$$y = \frac{\dot{x} - ax - f}{b} = \frac{1}{5}\left(\dot{x} - (2)x - (0)\right)$$

Differentiate with respect to t (everything is differentiable):

$$\dot y = \frac{\ddot x - a\dot x - \dot f}{b} = \frac{1}{5}\left(\ddot x - (2)\dot x - (0)\right)$$

Substitute both expressions into the second equation, $\dot y = cx + dy + g$, and multiply by b:

$$\ddot x - a\dot x - \dot f = bc\,x + d\left(\dot x - ax - f\right) + bg$$

Collect the terms in $\ddot x$, $\dot x$ and x (Eq. 26.3):

$$\ddot x - (a+d)\dot x + (ad - bc)x = \dot f + bg - df$$

Numerically, $a + d = 2$, $ad - bc = (2)(0) - (5)(-2) = 10$ and $F(t) := \dot f + bg - df = (0) + (5)(\cos 3t) - (0)(0) = 5\cos 3t$, so

$$\ddot x - 2\dot x + 10x = 5\cos 3t$$

Initial data. The condition on y is converted into one on $\dot x$ with the first equation at $t = 0$: $\dot x(0) = a\,x(0) + b\,y(0) + f(0) = (2)(2) + (5)(-1) + (0) = -1$.

The homogeneous equation $\ddot x - 2\dot x + 10x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 - 2k + 10 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (-2)^2 - 4(1)(10) = -36$.

Factorising, $P(k) = (k - 1 - 3i)\,(k - 1 + 3i)$, so the roots are

$$k = 1 - 3i, \qquad k = 1 + 3i$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $e^{t}\cos 3t$, $e^{t}\sin 3t$, so the complementary function is

$$x_h(t) = c_1 e^{t}\cos 3t + c_2 e^{t}\sin 3t$$

The forcing $5\cos 3t$ is a sinusoidal term: it is the real or imaginary part of $e^{(3i)t}$, so we test $s = 3i$ in the characteristic polynomial: $P(3i) = 1 - 6i$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 \cos 3t + K_2 \sin 3t$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= -3K_1 \sin 3t + 3K_2 \cos 3t \\ x_p''(t) &= -9K_1 \cos 3t - 9K_2 \sin 3t \end{aligned}$$

Substitute into $L[x_p] = x_p'' - 2x_p' + 10x_p$ and collect the terms that multiply the same function:

$$L[x_p] = (K_1 - 6K_2)\, \cos 3t + (6K_1 + K_2)\, \sin 3t$$

This must equal the forcing $F(t) = 5\cos 3t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in F:

$$\begin{aligned} \cos 3t: &\quad K_1 - 6K_2 = 5 \\ \sin 3t: &\quad 6K_1 + K_2 = 0 \end{aligned}$$

The linear system is

$$\begin{aligned} K_1 - 6K_2 &= 5 \quad (1) \\ 6K_1 + K_2 &= 0 \quad (2) \end{aligned}$$

Solve $K_1 - 6K_2 = 5$ for K_1: $K_1 = 6K_2 + 5$.

Substitute in $6K_1 + K_2 = 0$: $37K_2 + 30 = 0$.

$37K_2 + 30 = 0$ contains one unknown only, so $K_2 = -\frac{30}{37}$.

Back-substitution: $K_1 = 6K_2 + 5 = \frac{5}{37}$.

Hence $K_1 = \frac{5}{37}, K_2 = -\frac{30}{37}$.

Therefore

$$x_p(t) = -\frac{30\sin 3t}{37} + \frac{5\cos 3t}{37}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^t \cos 3t + c_2 e^t \sin 3t - \frac{30\sin 3t}{37} + \frac{5\cos 3t}{37}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -3c_1 e^t \sin 3t + c_1 e^t \cos 3t + c_2 e^t \sin 3t + 3c_2 e^t \cos 3t - \frac{15\sin 3t}{37} - \frac{90\cos 3t}{37}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$\begin{aligned} x(0) &= c_1 + \frac{5}{37} = 2 \\ x'(0) &= c_1 + 3c_2 - \frac{90}{37} = -1 \end{aligned}$$

The linear system is

$$\begin{aligned} c_1 + \frac{5}{37} &= 2 \quad (1) \\ c_1 + 3c_2 - \frac{90}{37} &= -1 \quad (2) \end{aligned}$$

$c_1 + \frac{5}{37} = 2$ contains one unknown only, so $c_1 = \frac{69}{37}$.

Substitute in $c_1 + 3c_2 - \frac{90}{37} = -1$: $3c_2 - \frac{21}{37} = -1$.

$3c_2 - \frac{21}{37} = -1$ contains one unknown only, so $c_2 = -\frac{16}{111}$.

Hence $c_1 = \frac{69}{37}$, $c_2 = -\frac{16}{111}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = -\frac{16e^t \sin 3t}{111} + \frac{69e^t \cos 3t}{37} - \frac{30\sin 3t}{37} + \frac{5\cos 3t}{37}}$$

Recovering y

From Eq. 26.2, $y = (\dot{x} - ax - f)/b$. With the solution found,

$$\dot{x}(t) = -\frac{637e^{t}\sin 3t}{111} + \frac{53e^{t}\cos 3t}{37} - \frac{15\sin 3t}{37} - \frac{90\cos 3t}{37}$$

$$y(t) = \frac{1}{5}\left(\dot{x} - (2)x - (0)\right) = -\frac{121e^{t}\sin 3t}{111} - \frac{17e^{t}\cos 3t}{37} + \frac{9\sin 3t}{37} - \frac{20\cos 3t}{37}$$

Check in the original system

$$\begin{aligned}
\dot{x} &= -\frac{637e^{t}\sin 3t}{111} + \frac{53e^{t}\cos 3t}{37} - \frac{15\sin 3t}{37} - \frac{90\cos 3t}{37} \\
ax + by + f &= -\frac{637e^{t}\sin 3t}{111} + \frac{53e^{t}\cos 3t}{37} - \frac{15\sin 3t}{37} - \frac{90\cos 3t}{37} \\
\dot{y} &= \frac{32e^{t}\sin 3t}{111} - \frac{138e^{t}\cos 3t}{37} + \frac{60\sin 3t}{37} + \frac{27\cos 3t}{37} \\
cx + dy + g &= \frac{32e^{t}\sin 3t}{111} - \frac{138e^{t}\cos 3t}{37} + \frac{60\sin 3t}{37} + \frac{27\cos 3t}{37}
\end{aligned}$$

Both pairs of expressions coincide, and $x(0) = 2, y(0) = -1$ reproduce the initial data.

(vi) $\dot{x} = x + y + e^{-t}, \quad \dot{y} = 4x - 2y + e^{2t}, \quad x(0) = 1,\ y(0) = -1$

The system is linear with constant coefficients, in the form $\dot{x} = ax + by + f(t), \dot{y} = cx + dy + g(t)$ (Robinson Eq. 26.1) with

$$a = 1, \quad b = 1, \quad c = 4, \quad d = -2, \quad f(t) = e^{-t}, \quad g(t) = e^{2t}$$

Elimination of y

Since $b = 1 \neq 0$, the first equation can be solved for y (Eq. 26.2):

$$y = \frac{\dot{x} - ax - f}{b} = \frac{1}{1}\left(\dot{x} - (1)x - (e^{-t})\right)$$

Differentiate with respect to t (everything is differentiable):

$$\dot y = \frac{\ddot x - a\dot x - \dot f}{b} = \frac{1}{1}\left(\ddot x - (1)\dot x - (-e^{-t})\right)$$

Substitute both expressions into the second equation, $\dot y = cx + dy + g$, and multiply by b:

$$\ddot x - a\dot x - \dot f = bc\,x + d\left(\dot x - ax - f\right) + bg$$

Collect the terms in $\ddot x$, $\dot x$ and x (Eq. 26.3):

$$\ddot x - (a+d)\dot x + (ad - bc)x = \dot f + bg - df$$

Numerically, $a + d = -1$, $ad - bc = (1)(-2) - (1)(4) = -6$ and $F(t) := \dot f + bg - df = (-e^{-t}) + (1)(e^{2t}) - (-2)(e^{-t}) = e^{2t} + e^{-t}$, so

$$\ddot x + \dot x - 6x = e^{2t} + e^{-t}$$

Initial data. The condition on y is converted into one on $\dot x$ with the first equation at $t = 0$: $\dot x(0) = a\,x(0) + b\,y(0) + f(0) = (1)(1) + (1)(-1) + (1) = 1$.

The homogeneous equation $\ddot x + \dot x - 6x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + k - 6 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (1)^2 - 4(1)(-6) = 25$.

Factorising, $P(k) = (k - 2)\,(k + 3)$, so the roots are

$$k = -3, \qquad k = 2$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-3t}, e^{2t}, so the complementary function is

$$x_h(t) = c_1 e^{-3t} + c_2 e^{2t}$$

The forcing is a sum of terms. By linearity (Robinson Eq. 14.7) a particular integral is the sum of particular integrals of the separate terms, each found in turn.

Particular integral for the term e^{-t}

The forcing e^{-t} is an exponential with exponent $s = -1$; we test it in the characteristic polynomial: $P(-1) = -6$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 e^{-t}$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$x_p'(t) = -K_1 e^{-t}$$
$$x_p''(t) = K_1 e^{-t}$$

Substitute into $L[x_p] = x_p'' + x_p' - 6x_p$ and collect the terms that multiply the same function:

$$L[x_p] = -6K_1\, e^{-t}$$

This must equal the forcing $F(t) = e^{-t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in F:

$$e^{-t}: \quad -6K_1 = 1$$

The linear system is

$$-6K_1 = 1 \quad (1)$$

$-6K_1 = 1$ contains one unknown only, so $K_1 = -\frac{1}{6}$.

Hence $K_1 = -\frac{1}{6}$.

Therefore

$$x_p(t) = -\frac{e^{-t}}{6}$$

Particular integral for the term e^{2t}

The forcing e^{2t} is an exponential with exponent $s = 2$; we test it in the characteristic polynomial: $P(2) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 1, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by t^1.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 t e^{2t}$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$x_p'(t) = 2K_1 t e^{2t} + K_1 e^{2t}$$
$$x_p''(t) = 4K_1 t e^{2t} + 4K_1 e^{2t}$$

Substitute into $L[x_p] = x_p'' + x_p' - 6x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 5K_1\, e^{2t}$$

This must equal the forcing $F(t) = e^{2t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in F:

$$e^{2t}: \quad 5K_1 = 1$$

The linear system is

$$5K_1 = 1 \quad (1)$$

$5K_1 = 1$ contains one unknown only, so $K_1 = \frac{1}{5}$.

Hence $K_1 = \frac{1}{5}$.

Therefore

$$x_p(t) = \frac{te^{2t}}{5}$$

Adding the particular integrals,

$$x_p(t) = \frac{te^{2t}}{5} - \frac{e^{-t}}{6}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{-3t} + c_2 e^{2t} + \frac{te^{2t}}{5} - \frac{e^{-t}}{6}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -3c_1 e^{-3t} + 2c_2 e^{2t} + \frac{2te^{2t}}{5} + \frac{e^{2t}}{5} + \frac{e^{-t}}{6}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 + c_2 - \frac{1}{6} = 1$$
$$x'(0) = -3c_1 + 2c_2 + \frac{11}{30} = 1$$

The linear system is

$$c_1 + c_2 - \frac{1}{6} = 1 \quad (1)$$
$$-3c_1 + 2c_2 + \frac{11}{30} = 1 \quad (2)$$

Solve $c_1 + c_2 - \frac{1}{6} = 1$ for c_1: $c_1 = \frac{7}{6} - c_2$.

Substitute in $-3c_1 + 2c_2 + \frac{11}{30} = 1$: $5c_2 - \frac{47}{15} = 1$.

$5c_2 - \frac{47}{15} = 1$ contains one unknown only, so $c_2 = \frac{62}{75}$.

Back-substitution: $c_1 = \frac{7}{6} - c_2 = \frac{17}{50}$.

Hence $c_1 = \frac{17}{50}, c_2 = \frac{62}{75}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{te^{2t}}{5} + \frac{62e^{2t}}{75} - \frac{e^{-t}}{6} + \frac{17e^{-3t}}{50}}$$

Recovering y

From Eq. 26.2, $y = (\dot{x} - ax - f)/b$. With the solution found,

$$\dot{x}(t) = \frac{2te^{2t}}{5} + \frac{139e^{2t}}{75} + \frac{e^{-t}}{6} - \frac{51e^{-3t}}{50}$$

$$y(t) = \frac{1}{1}\left(\dot{x} - (1)x - (e^{-t})\right) = \frac{te^{2t}}{5} + \frac{77e^{2t}}{75} - \frac{2e^{-t}}{3} - \frac{34e^{-3t}}{25}$$

Check in the original system

$$\begin{aligned} \dot{x} &= \frac{2te^{2t}}{5} + \frac{139e^{2t}}{75} + \frac{e^{-t}}{6} - \frac{51e^{-3t}}{50} \\ ax + by + f &= \frac{2te^{2t}}{5} + \frac{139e^{2t}}{75} + \frac{e^{-t}}{6} - \frac{51e^{-3t}}{50} \\ \dot{y} &= \frac{2te^{2t}}{5} + \frac{169e^{2t}}{75} + \frac{2e^{-t}}{3} + \frac{102e^{-3t}}{25} \\ cx + dy + g &= \frac{2te^{2t}}{5} + \frac{169e^{2t}}{75} + \frac{2e^{-t}}{3} + \frac{102e^{-3t}}{25} \end{aligned}$$

Both pairs of expressions coincide, and $x(0) = 1, y(0) = -1$ reproduce the initial data.

(vii) $\dot{x} = 8x + 14y, \quad \dot{y} = 7x + y, \quad x(0) = y(0) = 1$

The system is linear with constant coefficients, in the form $\dot{x} = ax + by + f(t), \dot{y} = cx + dy + g(t)$ (Robinson Eq. 26.1) with

$$a = 8, \quad b = 14, \quad c = 7, \quad d = 1, \quad f(t) = 0, \quad g(t) = 0$$

Elimination of y

Since $b = 14 \neq 0$, the first equation can be solved for y (Eq. 26.2):

$$y = \frac{\dot{x} - ax - f}{b} = \frac{1}{14}\left(\dot{x} - (8)x - (0)\right)$$

Differentiate with respect to t (everything is differentiable):

$$\dot{y} = \frac{\ddot{x} - a\dot{x} - \dot{f}}{b} = \frac{1}{14}\left(\ddot{x} - (8)\dot{x} - (0)\right)$$

Substitute both expressions into the second equation, $\dot{y} = cx + dy + g$, and multiply by b:

$$\ddot{x} - a\dot{x} - \dot{f} = bc\,x + d\left(\dot{x} - ax - f\right) + bg$$

Collect the terms in $\ddot{x}$, $\dot{x}$ and x (Eq. 26.3):

$$\ddot{x} - (a+d)\dot{x} + (ad - bc)x = \dot{f} + bg - df$$

Numerically, $a + d = 9$, $ad - bc = (8)(1) - (14)(7) = -90$ and $F(t) := \dot{f} + bg - df = (0) + (14)(0) - (1)(0) = 0$, so

$$\ddot{x} - 9\dot{x} - 90x = 0$$

Initial data. The condition on y is converted into one on $\dot{x}$ with the first equation at $t = 0$: $\dot{x}(0) = a\,x(0) + b\,y(0) + f(0) = (8)(1) + (14)(1) + (0) = 22$.

The homogeneous equation $\ddot{x} - 9\dot{x} - 90x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 - 9k - 90 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (-9)^2 - 4(1)(-90) = 441$.

Factorising, $P(k) = (k - 15)\,(k + 6)$, so the roots are

$$k = -6, \qquad k = 15$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-6t}, e^{15t}, so the complementary function is

$$x_h(t) = c_1 e^{-6t} + c_2 e^{15t}$$

The equation is homogeneous, so there is no particular integral and the general solution is the complementary function:

$$x(t) = x_h(t) = c_1 e^{-6t} + c_2 e^{15t}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -6c_1 e^{-6t} + 15c_2 e^{15t}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$\begin{aligned} x(0) &= c_1 + c_2 = 1 \\ x'(0) &= -6c_1 + 15c_2 = 22 \end{aligned}$$

The linear system is

$$\begin{aligned} c_1 + c_2 &= 1 \quad (1) \\ -6c_1 + 15c_2 &= 22 \quad (2) \end{aligned}$$

Solve $c_1 + c_2 = 1$ for c_1: $c_1 = 1 - c_2$.

Substitute in $-6c_1 + 15c_2 = 22$: $21c_2 - 6 = 22$.

$21c_2 - 6 = 22$ contains one unknown only, so $c_2 = \frac{4}{3}$.

Back-substitution: $c_1 = 1 - c_2 = -\frac{1}{3}$.

Hence $c_1 = -\frac{1}{3}, c_2 = \frac{4}{3}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{4e^{15t}}{3} - \frac{e^{-6t}}{3}}$$

Recovering y

From Eq. 26.2, $y = (\dot{x} - ax - f)/b$. With the solution found,

$$\dot{x}(t) = 20e^{15t} + 2e^{-6t}$$

$$y(t) = \frac{1}{14}\left(\dot{x} - (8)x - (0)\right) = \frac{2e^{15t}}{3} + \frac{e^{-6t}}{3}$$

Check in the original system

$$\begin{aligned} \dot{x} &= 20e^{15t} + 2e^{-6t} \\ ax + by + f &= 20e^{15t} + 2e^{-6t} \\ \dot{y} &= 10e^{15t} - 2e^{-6t} \\ cx + dy + g &= 10e^{15t} - 2e^{-6t} \end{aligned}$$

Both pairs of expressions coincide, and $x(0) = 1, y(0) = 1$ reproduce the initial data.

INTERPRETATION AND CHECKS

Item	$x(t)$	$y(t)$
(i)	$-\frac{t^2}{6} - \frac{5t}{18} - \frac{29e^{3t}}{27} + \frac{5e^{2t}}{4} - \frac{19}{108}$	$-\frac{2t^2}{3} - \frac{7t}{9} - \frac{29e^{3t}}{27} + \frac{5e^{2t}}{2} - \frac{23}{54}$
(ii)	$-\frac{32e^t\sin 2t}{17} + \frac{26e^t\cos 2t}{17} + \frac{2\sin 2t}{17} - \frac{9\cos 2t}{17}$	$\frac{13e^t\sin 2t}{17} + \frac{16e^t\cos 2t}{17} - \frac{4\sin 2t}{17} + \frac{\cos 2t}{17}$
(iii)	$\frac{e^{6t}}{5} - \frac{e^t}{5}$	$\frac{2e^{6t}}{5} + \frac{e^t}{10}$
(iv)	$t^2e^{3t} + 7te^{3t} + e^{3t}$	$\frac{t^2e^{3t}}{2} + 3te^{3t} - e^{3t}$
(v)	$-\frac{16e^t\sin 3t}{111} + \frac{69e^t\cos 3t}{37} - \frac{30\sin 3t}{37} + \frac{5\cos 3t}{37}$	$-\frac{121e^t\sin 3t}{111} - \frac{17e^t\cos 3t}{37} + \frac{9\sin 3t}{37} - \frac{20\cos 3t}{37}$
(vi)	$\frac{te^{2t}}{5} + \frac{62e^{2t}}{75} - \frac{e^{-t}}{6} + \frac{17e^{-3t}}{50}$	$\frac{te^{2t}}{5} + \frac{77e^{2t}}{75} - \frac{2e^{-t}}{3} - \frac{34e^{-3t}}{25}$
(vii)	$\frac{4e^{15t}}{3} - \frac{e^{-6t}}{3}$	$\frac{2e^{15t}}{3} + \frac{e^{-6t}}{3}$

- Every pair (x, y) was substituted in **both** original equations and in the initial data (shown in each item and re-checked symbolically in `scripts/verify_topic3_topic4.py`).
- Characteristic exponents seen: (i) $2, 3$ (real); (ii) $1 \pm 2i$ (spiral) with the forcing $\cos 2t$ at a different frequency; (iii) $6, -1$ (saddle; the forcing e^t has exponent 1, which is **not** a root, so no extra factor t); (iv) 3 double (improper-node type), and the forcing e^{3t} resonates with the double root, giving t^2e^{3t}; (v) $1 \pm 3i$; (vi) $2, -3$ with the forcing exponent 2 resonant (term te^{2t}) and -1 not; (vii) $15, -6$ (homogeneous saddle).

- Cross-check of the homogeneous parts with Topic 4: (iv) and (v) have the same matrices as Examples 8.2 and 7.1; (vii) is the matrix of Exercise 9 (i).
- Dimensions: with t in seconds the matrix entries have units s^{-1} and the forcings those of $\dot{x}$; the elimination keeps this consistent ($a+d$ in s^{-1}, $ad-bc$ in s^{-2}).

EXERCISE 8

General Solution and Phase Portrait from Eigenvalues and Eigenvectors

Original statement · probls_ch4-i_2627_v2.pdf, p. 2

Eigen-data • Nodes and Saddles • Eight Portraits

STATEMENT

Write down the general solution and draw the phase portrait for the equation $\dot{\mathbf{x}} = A\mathbf{x}$, when the eigenvalues and eigenvectors of A are as given in each case. You should take particular care with stable (or unstable) nodes to ensure that the trajectories approach (or move away from) the origin tangent to the correct eigenvector.

- (i) $\lambda_1 = 1, \mathbf{v}_1 = (1, 1)$ and $\lambda_2 = 2, \mathbf{v}_2 = (1, -1)$;
- (ii) $\lambda_1 = 1, \mathbf{v}_1 = (1, 0)$ and $\lambda_2 = -2, \mathbf{v}_2 = (1, 1)$;
- (iii) $\lambda_1 = -2, \mathbf{v}_1 = (1, 2)$ and $\lambda_2 = -3, \mathbf{v}_2 = (2, -3)$;
- (iv) $\lambda_1 = 3, \mathbf{v}_1 = (2, 3)$ and $\lambda_2 = -5, \mathbf{v}_2 = (0, 1)$;
- (v) $\lambda_1 = 3, \mathbf{v}_1 = (1, 2)$ and $\lambda_2 = 1, \mathbf{v}_2 = (1, -3)$;
- (vi) $\lambda_1 = 2, \mathbf{v}_1 = (0, 1)$ and $\lambda_2 = -3, \mathbf{v}_2 = (1, 5)$;
- (vii) $\lambda_1 = 1, \mathbf{v}_1 = (1, 1)$ and $\lambda_2 = 2, \mathbf{v}_2 = (2, 1)$;
- (viii) $\lambda_1 = -3, \mathbf{v}_1 = (1, 3)$ and $\lambda_2 = -1, \mathbf{v}_2 = (-3, 2)$;

SETUP AND CLASSIFICATION

In every item the two eigenvalues are real and distinct, with linearly independent eigenvectors (so $\det[\mathbf{v}_1\ \mathbf{v}_2] \neq 0$, checked in each item). The general solution is then $\mathbf{x} = c_1 e^{\lambda_1 t}\mathbf{v}_1 + c_2 e^{\lambda_2 t}\mathbf{v}_2$ (Robinson Eq. 28.1) and the portrait is determined by the signs of λ_1, λ_2 (Topic 4 Sec. 6 and 9):

Signs	Origin	Behaviour
$\lambda_1, \lambda_2 < 0$	stable node	all orbits tend to $\mathbf{0}$, tangent to the eigenvector of the smaller $\lvert\lambda\rvert$
$\lambda_1, \lambda_2 > 0$	unstable node	all orbits leave $\mathbf{0}$, tangent to the eigenvector of the smaller $\lvert\lambda\rvert$
$\lambda_1\lambda_2 < 0$	saddle	only the orbits on the stable eigenline reach $\mathbf{0}$

Item	λ_1, λ_2	Type	Slow direction (smaller $\lvert\lambda\rvert$)
(i)	$1, 2$	unstable node	$\mathbf{v}_1 = (1, 1)$
(ii)	$1, -2$	saddle	not applicable
(iii)	$-2, -3$	stable node	$\mathbf{v}_1 = (1, 2)$
(iv)	$3, -5$	saddle	not applicable
(v)	$3, 1$	unstable node	$\mathbf{v}_2 = (1, -3)$
(vi)	$2, -3$	saddle	not applicable
(vii)	$1, 2$	unstable node	$\mathbf{v}_1 = (1, 1)$
(viii)	$-3, -1$	stable node	$\mathbf{v}_2 = (-3, 2)$

FORMULATION: HOW EACH ITEM IS TREATED

For each item: (1) the general solution; (2) the matrix $A = Q\Lambda Q^{-1}$ reconstructed from the data, used to draw and to check; (3) the canonical system $\dot{\mathbf{z}} = \Lambda\mathbf{z}$ with $\mathbf{z} = Q^{-1}\mathbf{x}$ and its solution; (4) the portrait in the $\mathbf{z}$-plane and its image under $\mathbf{x} = Q\mathbf{z}$ in the (x, y)-plane, with the arrows and the tangent direction decided by the rules above.

METHOD OF THE CLASS NOTES [NOTES, P. 9]

Case 1 of the notes: $\mathbf{x} = C_1 e^{\lambda_1 t}\mathbf{v}_1 + C_2 e^{\lambda_2 t}\mathbf{v}_2$. With $P = (\mathbf{v}_1, \mathbf{v}_2)$ (the Q of this page), $\mathbf{z} = P^{-1}\mathbf{x}$ gives $\dot{\mathbf{z}} = D\mathbf{z}$ and the orbits $z_2 = \tilde{C} z_1^{\lambda_2/\lambda_1}$. The table of the notes reads the portrait from $r = \lambda_2/\lambda_1$: eigenvalues of one sign give a node, tangent at the origin to the axis of the eigenvalue of smaller modulus (z_1-axis when $r > 1$, z_2-axis when $0 < r < 1$); opposite signs give a saddle. Back in $\mathbf{x}$ the z-axes become the eigenvector lines.

DERIVATION OF THE EIGHT PORTRAITS

(i) $\lambda_1 = 1$, $\mathbf{v}_1 = (1, 1)^T$; $\lambda_2 = 2$, $\mathbf{v}_2 = (1, -1)^T$

The statement gives the eigen-data, not the matrix. Since $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$ for $j = 1, 2$ means $AQ = Q\Lambda$ with $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, the matrix is $A = Q\Lambda Q^{-1}$ (it is needed to draw the portrait and to check the data):

$$Q = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \Lambda = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad \det Q = -2, \qquad Q^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

$$
\begin{aligned}
(Q\Lambda)_{11} &= 1\cdot 1 + 1\cdot 0 = 1\\
(Q\Lambda)_{12} &= 1\cdot 0 + 1\cdot 2 = 2\\
(Q\Lambda)_{21} &= 1\cdot 1 + (-1)\cdot 0 = 1\\
(Q\Lambda)_{22} &= 1\cdot 0 + (-1)\cdot 2 = -2
\end{aligned}
$$

$$
Q\Lambda = \begin{pmatrix} 1 & 2 \\ 1 & -2 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= 1\cdot \frac{1}{2} + 2\cdot \frac{1}{2} = \frac{3}{2}\\
(A)_{12} &= 1\cdot \frac{1}{2} + 2\cdot (-\frac{1}{2}) = -\frac{1}{2}\\
(A)_{21} &= 1\cdot \frac{1}{2} + (-2)\cdot \frac{1}{2} = -\frac{1}{2}\\
(A)_{22} &= 1\cdot \frac{1}{2} + (-2)\cdot (-\frac{1}{2}) = \frac{3}{2}
\end{aligned}
$$

$$
A = Q\Lambda Q^{-1} = \begin{pmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{pmatrix}
$$

Change of coordinates

The eigenvectors of distinct eigenvalues are linearly independent (Topic 4, Appendix B argument), so they form a basis. Put them as columns, $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, and define the new coordinates by $\mathbf{z} = Q^{-1}\mathbf{x}$, i.e. $\mathbf{x} = Q\mathbf{z}$:

$$
Q = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}
$$

$\det Q = (1)(-1) - (1)(1) = -2$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$
Q^{-1} = \frac{1}{-2}\begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}
$$

Since Q is constant, $\dot{\mathbf{z}} = Q^{-1}\dot{\mathbf{x}} = Q^{-1}A\mathbf{x} = (Q^{-1}AQ)\mathbf{z}$. Compute AQ one column at a time, with $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$:

$$A\mathbf{v}_1 = \begin{pmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1\mathbf{v}_1, \qquad A\mathbf{v}_2 = \begin{pmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix} = 2\mathbf{v}_2$$

$$AQ = [A\mathbf{v}_1 \mid A\mathbf{v}_2] = \begin{pmatrix} 1 & 2 \\ 1 & -2 \end{pmatrix}$$

Now multiply by Q^{-1} on the left, entry by entry:

$$\begin{aligned} (Q^{-1}AQ)_{11} &= \frac{1}{2}\cdot 1 + \frac{1}{2}\cdot 1 = 1 \\ (Q^{-1}AQ)_{12} &= \frac{1}{2}\cdot 2 + \frac{1}{2}\cdot (-2) = 0 \\ (Q^{-1}AQ)_{21} &= \frac{1}{2}\cdot 1 + (-\frac{1}{2})\cdot 1 = 0 \\ (Q^{-1}AQ)_{22} &= \frac{1}{2}\cdot 2 + (-\frac{1}{2})\cdot (-2) = 2 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = \Lambda = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

The transformation has decoupled the system (the matrix is diagonal), as claimed.

Solution in the canonical plane

$$\dot{z}_1 = z_1, \qquad \dot{z}_2 = 2z_2$$

Each equation is separable. If $z_i \neq 0$, $\frac{dz_i}{z_i} = \lambda_i\,dt$ and integrating, $\ln|z_i| = \lambda_i t + C_i$, so $z_i = c_i e^{\lambda_i t}$ with $c_i = \pm e^{C_i}$ (the solution $z_i \equiv 0$ corresponds to $c_i = 0$). Equivalently, $\frac{d}{dt}(e^{-\lambda_i t} z_i) = 0$.

$$z_1(t) = c_1 e^{1t}, \qquad z_2(t) = c_2 e^{2t}$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{v}_1 + z_2\mathbf{v}_2$, so the general solution is

$$\mathbf{x}(t) = c_1 e^{1t}\begin{pmatrix}1\\1\end{pmatrix} + c_2 e^{2t}\begin{pmatrix}1\\-1\end{pmatrix}$$

$$x(t) = c_1 e^{t} + c_2 e^{2t}, \qquad y(t) = c_1 e^{t} - c_2 e^{2t}$$

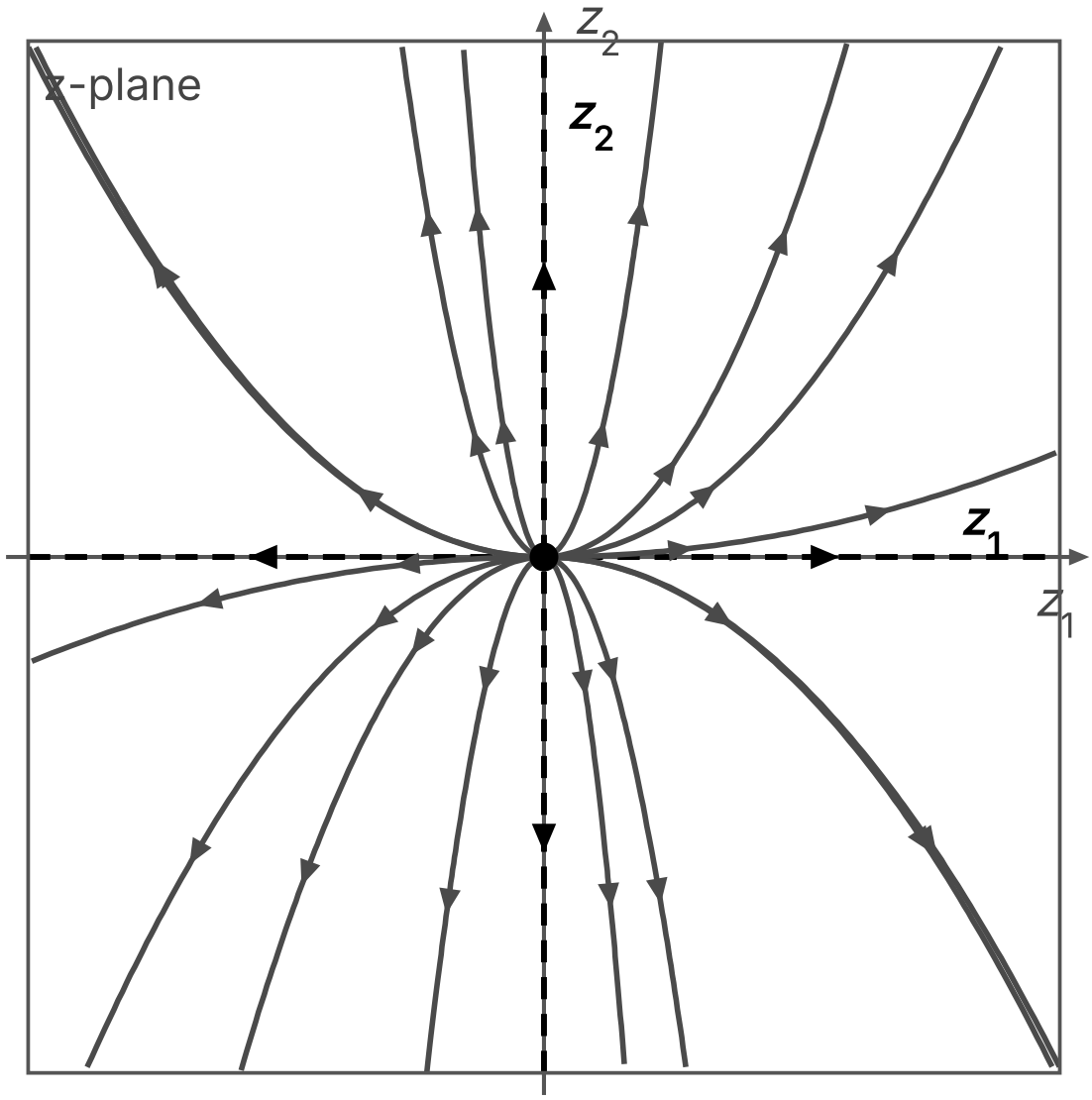
Phase portrait

$\lambda_1 = 1$ and $\lambda_2 = 2$ are both positive: the origin is a **unstable node** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

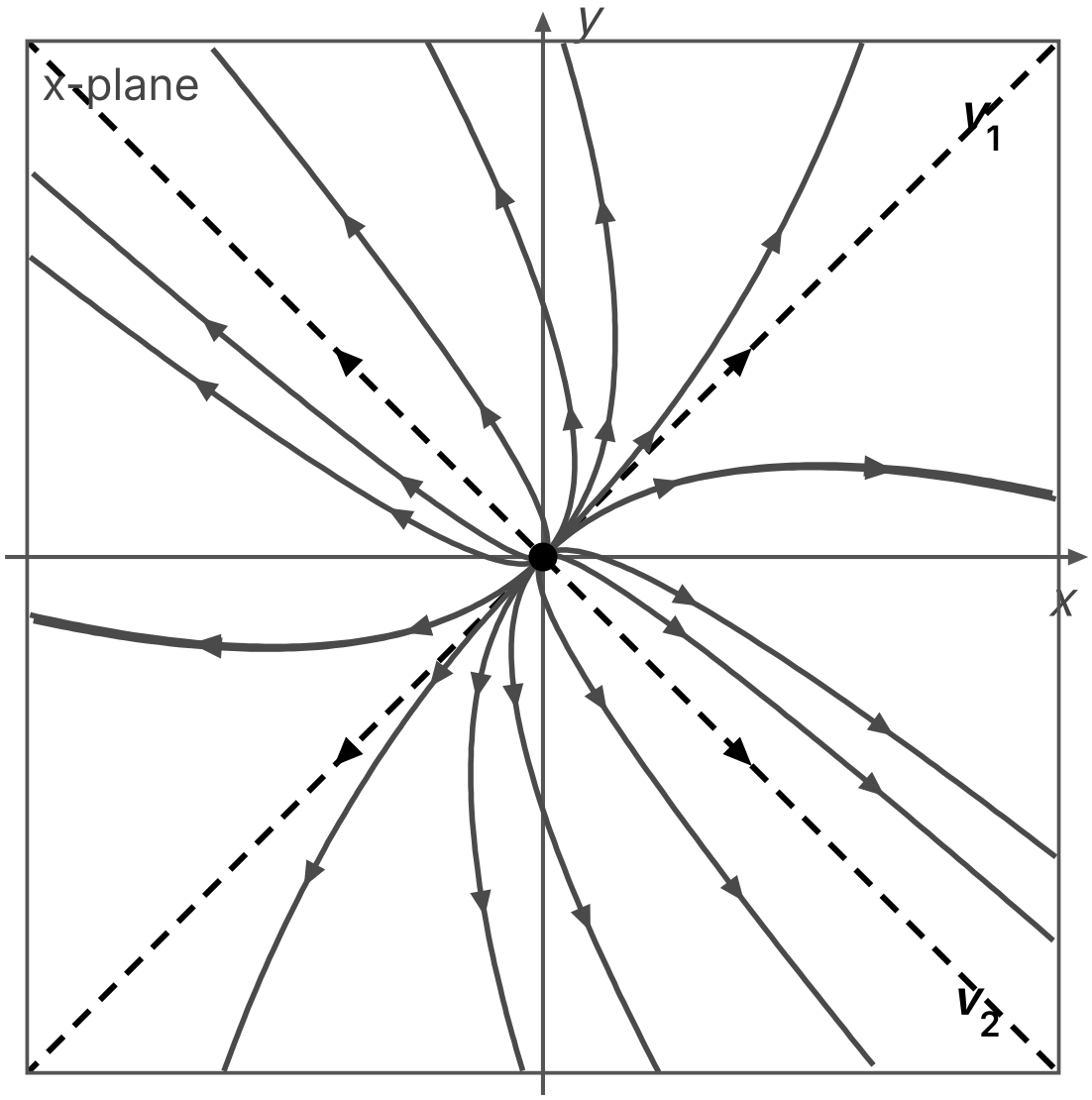
In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{1t}$ and $z_2 = c_2 e^{2t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^t = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2|\,(|z_1|/|c_1|)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K\,|z_1|^{\lambda_2/\lambda_1} = K\,|z_1|^2$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

The exponent $\lambda_2/\lambda_1 = 2$ is positive, so the orbits are parabola-like curves through the origin. At the origin they are tangent to the eigenvector of the eigenvalue of **smaller modulus**, $|\lambda| = 1$, i.e. to the line through $(1, 1)$ (they leave from there (the limit $t \to -\infty$)), and far from the origin they become parallel to the other eigenvector (the fast direction). The arrows point away from the origin on both eigenlines.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (1, 1)$ and the z_2-axis to the line through $\mathbf{v}_2 = (1, -1)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = 1\mathbf{v}_1$ points outward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = 2\mathbf{v}_2$ points outward.



Canonical $\mathbf{z}$-plane (unstable node): invariant axes and orbits $z_2 = K|z_1|^2$.



Original (x, y)-plane (unstable node): dashed lines are the eigenlines $\mathbf{v}_1 = (1, 1)$ and $\mathbf{v}_2 = (1, -1)$.

(ii) $\lambda_1 = 1,\ \mathbf{v}_1 = (1, 0)^T;\ \lambda_2 = -2,\ \mathbf{v}_2 = (1, 1)^T$

The statement gives the eigen-data, not the matrix. Since $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$ for $j = 1, 2$ means $AQ = Q\Lambda$ with $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, the matrix is $A = Q\Lambda Q^{-1}$ (it is needed to draw the portrait and to check the data):

$$Q = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \Lambda = \begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}, \qquad \det Q = 1, \qquad Q^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

$$
\begin{aligned}
(Q\Lambda)_{11} &= 1\cdot 1 + 1\cdot 0 = 1\\
(Q\Lambda)_{12} &= 1\cdot 0 + 1\cdot (-2) = -2\\
(Q\Lambda)_{21} &= 0\cdot 1 + 1\cdot 0 = 0\\
(Q\Lambda)_{22} &= 0\cdot 0 + 1\cdot (-2) = -2
\end{aligned}
$$

$$
Q\Lambda = \begin{pmatrix} 1 & -2 \\ 0 & -2 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= 1\cdot 1 + (-2)\cdot 0 = 1\\
(A)_{12} &= 1\cdot (-1) + (-2)\cdot 1 = -3\\
(A)_{21} &= 0\cdot 1 + (-2)\cdot 0 = 0\\
(A)_{22} &= 0\cdot (-1) + (-2)\cdot 1 = -2
\end{aligned}
$$

$$
A = Q\Lambda Q^{-1} = \begin{pmatrix} 1 & -3 \\ 0 & -2 \end{pmatrix}
$$

Change of coordinates

The eigenvectors of distinct eigenvalues are linearly independent (Topic 4, Appendix B argument), so they form a basis. Put them as columns, $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, and define the new coordinates by $\mathbf{z} = Q^{-1}\mathbf{x}$, i.e. $\mathbf{x} = Q\mathbf{z}$:

$$
Q = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}
$$

$\det Q = (1)(1) - (1)(0) = 1$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$
Q^{-1} = \frac{1}{1}\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}
$$

Since Q is constant, $\dot{\mathbf{z}} = Q^{-1}\dot{\mathbf{x}} = Q^{-1}A\mathbf{x} = (Q^{-1}AQ)\mathbf{z}$. Compute AQ one column at a time, with $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$:

$$A\mathbf{v}_1 = \begin{pmatrix} 1 & -3 \\ 0 & -2 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1\mathbf{v}_1, \qquad A\mathbf{v}_2 = \begin{pmatrix} 1 & -3 \\ 0 & -2 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -2 \end{pmatrix} = (-2)\mathbf{v}_2$$

$$AQ = [A\mathbf{v}_1 \mid A\mathbf{v}_2] = \begin{pmatrix} 1 & -2 \\ 0 & -2 \end{pmatrix}$$

Now multiply by Q^{-1} on the left, entry by entry:

$$\begin{aligned} (Q^{-1}AQ)_{11} &= 1\cdot 1 + (-1)\cdot 0 = 1 \\ (Q^{-1}AQ)_{12} &= 1\cdot(-2) + (-1)\cdot(-2) = 0 \\ (Q^{-1}AQ)_{21} &= 0\cdot 1 + 1\cdot 0 = 0 \\ (Q^{-1}AQ)_{22} &= 0\cdot(-2) + 1\cdot(-2) = -2 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix} = \Lambda = \begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}$$

The transformation has decoupled the system (the matrix is diagonal), as claimed.

Solution in the canonical plane

$$\dot{z}_1 = z_1, \qquad \dot{z}_2 = -2z_2$$

Each equation is separable. If $z_i \neq 0$, $\frac{dz_i}{z_i} = \lambda_i\,dt$ and integrating, $\ln|z_i| = \lambda_i t + C_i$, so $z_i = c_i e^{\lambda_i t}$ with $c_i = \pm e^{C_i}$ (the solution $z_i \equiv 0$ corresponds to $c_i = 0$). Equivalently, $\frac{d}{dt}(e^{-\lambda_i t} z_i) = 0$.

$$z_1(t) = c_1 e^{1t}, \qquad z_2(t) = c_2 e^{-2t}$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{v}_1 + z_2\mathbf{v}_2$, so the general solution is

$$\mathbf{x}(t) = c_1 e^{1t}\begin{pmatrix}1\\0\end{pmatrix} + c_2 e^{-2t}\begin{pmatrix}1\\1\end{pmatrix}$$

$$x(t) = c_1 e^{t} + c_2 e^{-2t}, \qquad y(t) = c_2 e^{-2t}$$

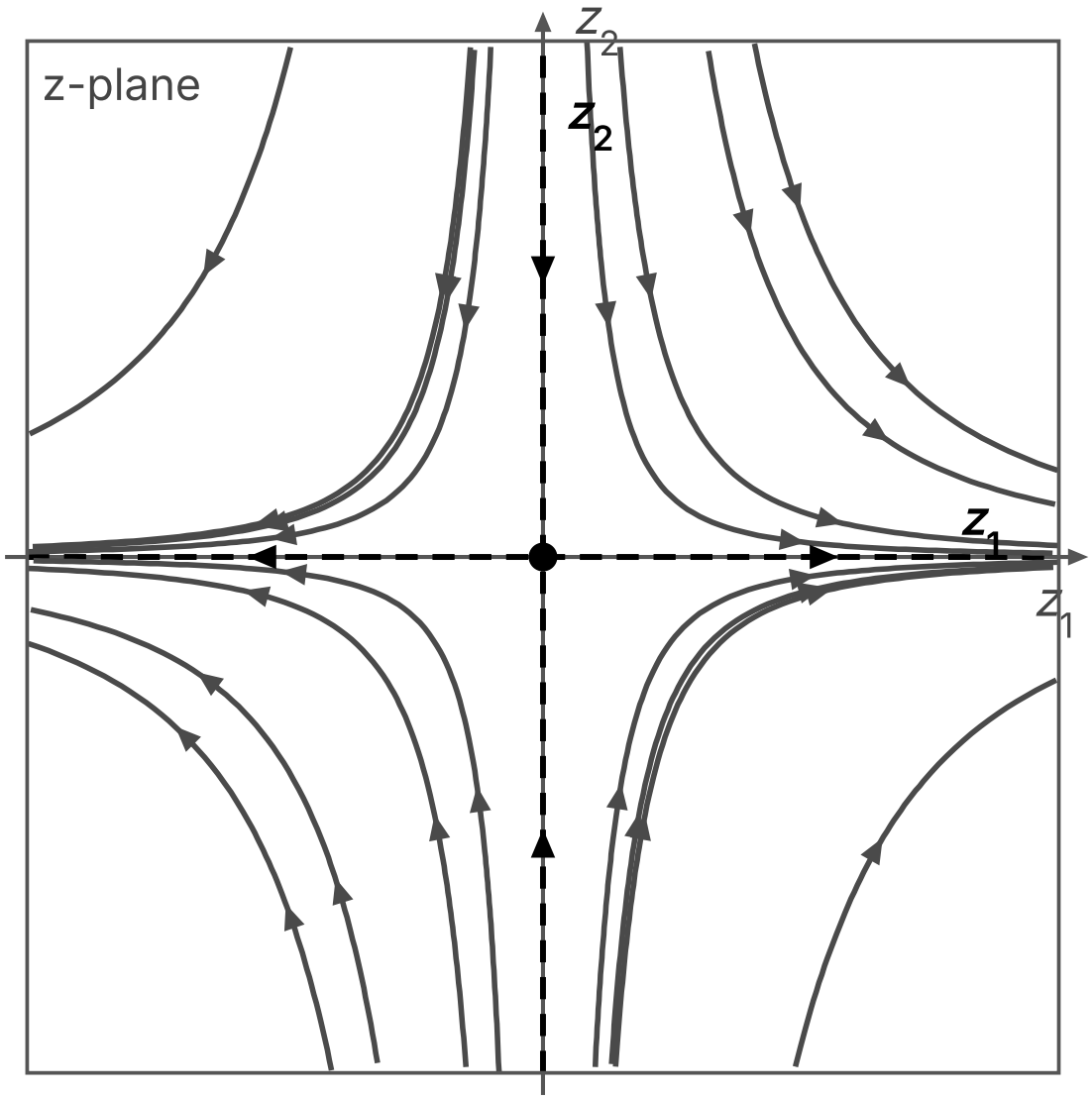
Phase portrait

$\lambda_1 = 1$ and $\lambda_2 = -2$ are of opposite signs: the origin is a **saddle** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

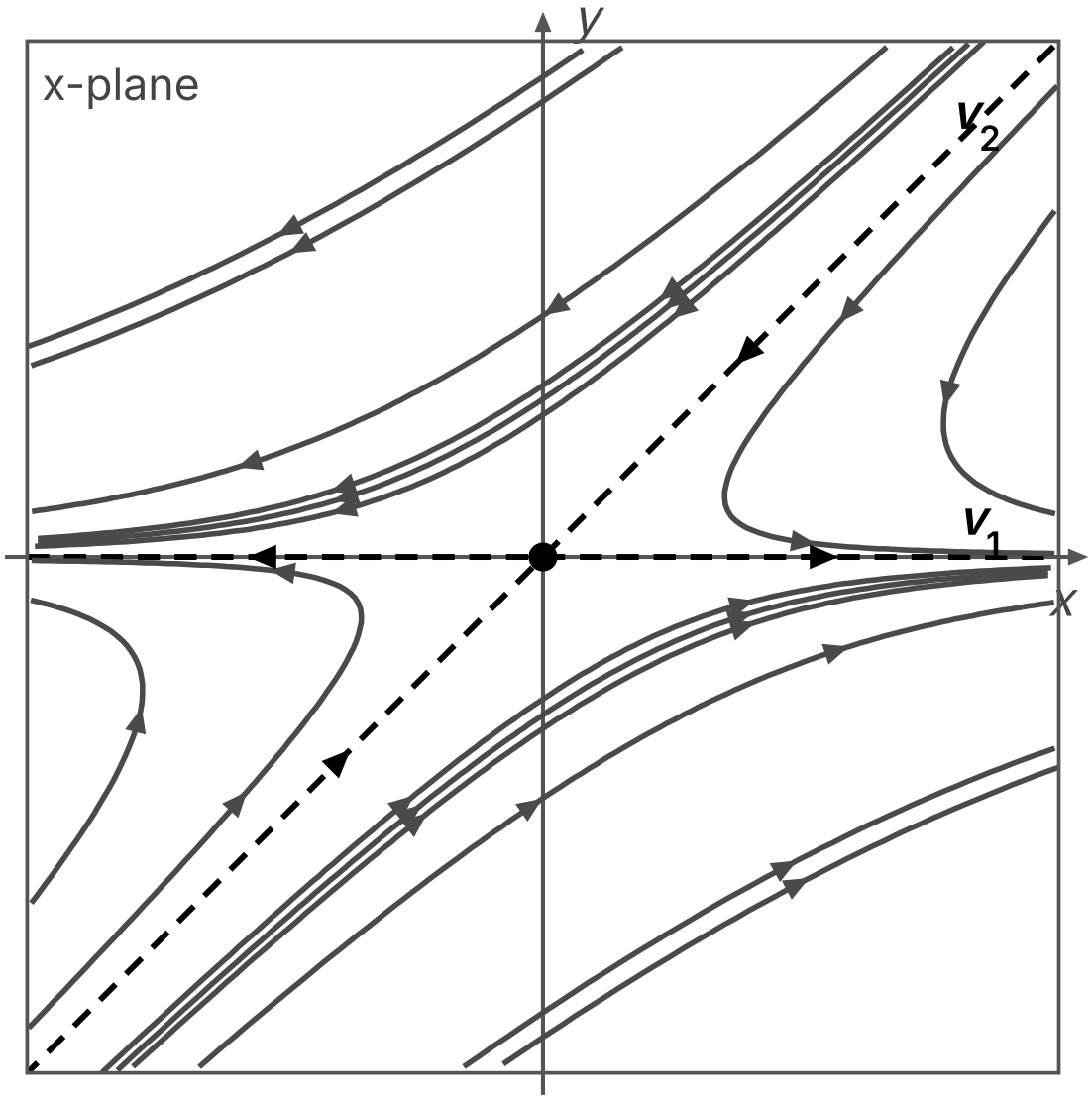
In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{1t}$ and $z_2 = c_2 e^{-2t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^{t} = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2|\,(|z_1|/|c_1|)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K\,|z_1|^{\lambda_2/\lambda_1} = K\,|z_1|^{-2}$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

The exponent $\lambda_2/\lambda_1 = -2$ is negative, so the orbits are hyperbola-like. Only solutions starting on the eigenline of the negative eigenvalue, the line through $(1, 1)$ (the **stable manifold**), reach the origin as $t \to +\infty$; the line through $(1, 0)$ (positive eigenvalue) is the **unstable manifold**. All other orbits come in close to the stable line, then leave along the unstable one.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (1, 0)$ and the z_2-axis to the line through $\mathbf{v}_2 = (1, 1)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = 1\mathbf{v}_1$ points outward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = -2\mathbf{v}_2$ points inward.



Canonical $\mathbf{z}$-plane (saddle): invariant axes and orbits $z_2 = K|z_1|^{-2}$.



Original (x, y)-plane (saddle): dashed lines are the eigenlines $\mathbf{v}_1 = (1, 0)$ and $\mathbf{v}_2 = (1, 1)$.

(iii) $\lambda_1 = -2$, $\mathbf{v}_1 = (1, 2)^T$; $\lambda_2 = -3$, $\mathbf{v}_2 = (2, -3)^T$

The statement gives the eigen-data, not the matrix. Since $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$ for $j = 1, 2$ means $AQ = Q\Lambda$ with $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, the matrix is $A = Q\Lambda Q^{-1}$ (it is needed to draw the portrait and to check the data):

$$Q = \begin{pmatrix} 1 & 2 \\ 2 & -3 \end{pmatrix}, \qquad \Lambda = \begin{pmatrix} -2 & 0 \\ 0 & -3 \end{pmatrix}, \qquad \det Q = -7, \qquad Q^{-1} = \begin{pmatrix} \frac{3}{7} & \frac{2}{7} \\ \frac{2}{7} & -\frac{1}{7} \end{pmatrix}$$

$$
\begin{aligned}
(Q\Lambda)_{11} &= 1\cdot(-2) + 2\cdot 0 = -2 \\
(Q\Lambda)_{12} &= 1\cdot 0 + 2\cdot(-3) = -6 \\
(Q\Lambda)_{21} &= 2\cdot(-2) + (-3)\cdot 0 = -4 \\
(Q\Lambda)_{22} &= 2\cdot 0 + (-3)\cdot(-3) = 9
\end{aligned}
$$

$$
Q\Lambda = \begin{pmatrix} -2 & -6 \\ -4 & 9 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= (-2)\cdot\frac{3}{7} + (-6)\cdot\frac{2}{7} = -\frac{18}{7} \\
(A)_{12} &= (-2)\cdot\frac{2}{7} + (-6)\cdot(-\frac{1}{7}) = \frac{2}{7} \\
(A)_{21} &= (-4)\cdot\frac{3}{7} + 9\cdot\frac{2}{7} = \frac{6}{7} \\
(A)_{22} &= (-4)\cdot\frac{2}{7} + 9\cdot(-\frac{1}{7}) = -\frac{17}{7}
\end{aligned}
$$

$$
A = Q\Lambda Q^{-1} = \begin{pmatrix} -\frac{18}{7} & \frac{2}{7} \\ \frac{6}{7} & -\frac{17}{7} \end{pmatrix}
$$

Change of coordinates

The eigenvectors of distinct eigenvalues are linearly independent (Topic 4, Appendix B argument), so they form a basis. Put them as columns, $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, and define the new coordinates by $\mathbf{z} = Q^{-1}\mathbf{x}$, i.e. $\mathbf{x} = Q\mathbf{z}$:

$$
Q = \begin{pmatrix} 1 & 2 \\ 2 & -3 \end{pmatrix}
$$

$\det Q = (1)(-3) - (2)(2) = -7$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$
Q^{-1} = \frac{1}{-7}\begin{pmatrix} -3 & -2 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} \frac{3}{7} & \frac{2}{7} \\ \frac{2}{7} & -\frac{1}{7} \end{pmatrix}
$$

Since Q is constant, $\dot{\mathbf{z}} = Q^{-1}\dot{\mathbf{x}} = Q^{-1}A\mathbf{x} = (Q^{-1}AQ)\mathbf{z}$. Compute AQ one column at a time, with $A\mathbf{v}_j = \lambda_j\mathbf{v}_j$:

$$A\mathbf{v}_1 = \begin{pmatrix} -\frac{18}{7} & \frac{2}{7} \\ \frac{6}{7} & -\frac{17}{7} \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ -4 \end{pmatrix} = (-2)\mathbf{v}_1, \qquad A\mathbf{v}_2 = \begin{pmatrix} -\frac{18}{7} & \frac{2}{7} \\ \frac{6}{7} & -\frac{17}{7} \end{pmatrix}\begin{pmatrix} 2 \\ -3 \end{pmatrix} = \begin{pmatrix} -6 \\ 9 \end{pmatrix} = (-3)\mathbf{v}_2$$

$$AQ = [A\mathbf{v}_1 \mid A\mathbf{v}_2] = \begin{pmatrix} -2 & -6 \\ -4 & 9 \end{pmatrix}$$

Now multiply by Q^{-1} on the left, entry by entry:

$$\begin{aligned} (Q^{-1}AQ)_{11} &= \frac{3}{7}\cdot(-2) + \frac{2}{7}\cdot(-4) = -2 \\ (Q^{-1}AQ)_{12} &= \frac{3}{7}\cdot(-6) + \frac{2}{7}\cdot 9 = 0 \\ (Q^{-1}AQ)_{21} &= \frac{2}{7}\cdot(-2) + (-\frac{1}{7})\cdot(-4) = 0 \\ (Q^{-1}AQ)_{22} &= \frac{2}{7}\cdot(-6) + (-\frac{1}{7})\cdot 9 = -3 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} -2 & 0 \\ 0 & -3 \end{pmatrix} = \Lambda = \begin{pmatrix} -2 & 0 \\ 0 & -3 \end{pmatrix}$$

The transformation has decoupled the system (the matrix is diagonal), as claimed.

Solution in the canonical plane

$$\dot{z}_1 = -2z_1, \qquad \dot{z}_2 = -3z_2$$

Each equation is separable. If $z_i \neq 0$, $\frac{dz_i}{z_i} = \lambda_i\,dt$ and integrating, $\ln|z_i| = \lambda_i t + C_i$, so $z_i = c_i e^{\lambda_i t}$ with $c_i = \pm e^{C_i}$ (the solution $z_i \equiv 0$ corresponds to $c_i = 0$). Equivalently, $\frac{d}{dt}(e^{-\lambda_i t}z_i) = 0$.

$$z_1(t) = c_1 e^{-2t}, \qquad z_2(t) = c_2 e^{-3t}$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{v}_1 + z_2\mathbf{v}_2$, so the general solution is

$$\mathbf{x}(t) = c_1 e^{-2t}\begin{pmatrix}1\\2\end{pmatrix} + c_2 e^{-3t}\begin{pmatrix}2\\-3\end{pmatrix}$$

$$x(t) = c_1 e^{-2t} + 2c_2 e^{-3t}, \qquad y(t) = 2c_1 e^{-2t} - 3c_2 e^{-3t}$$

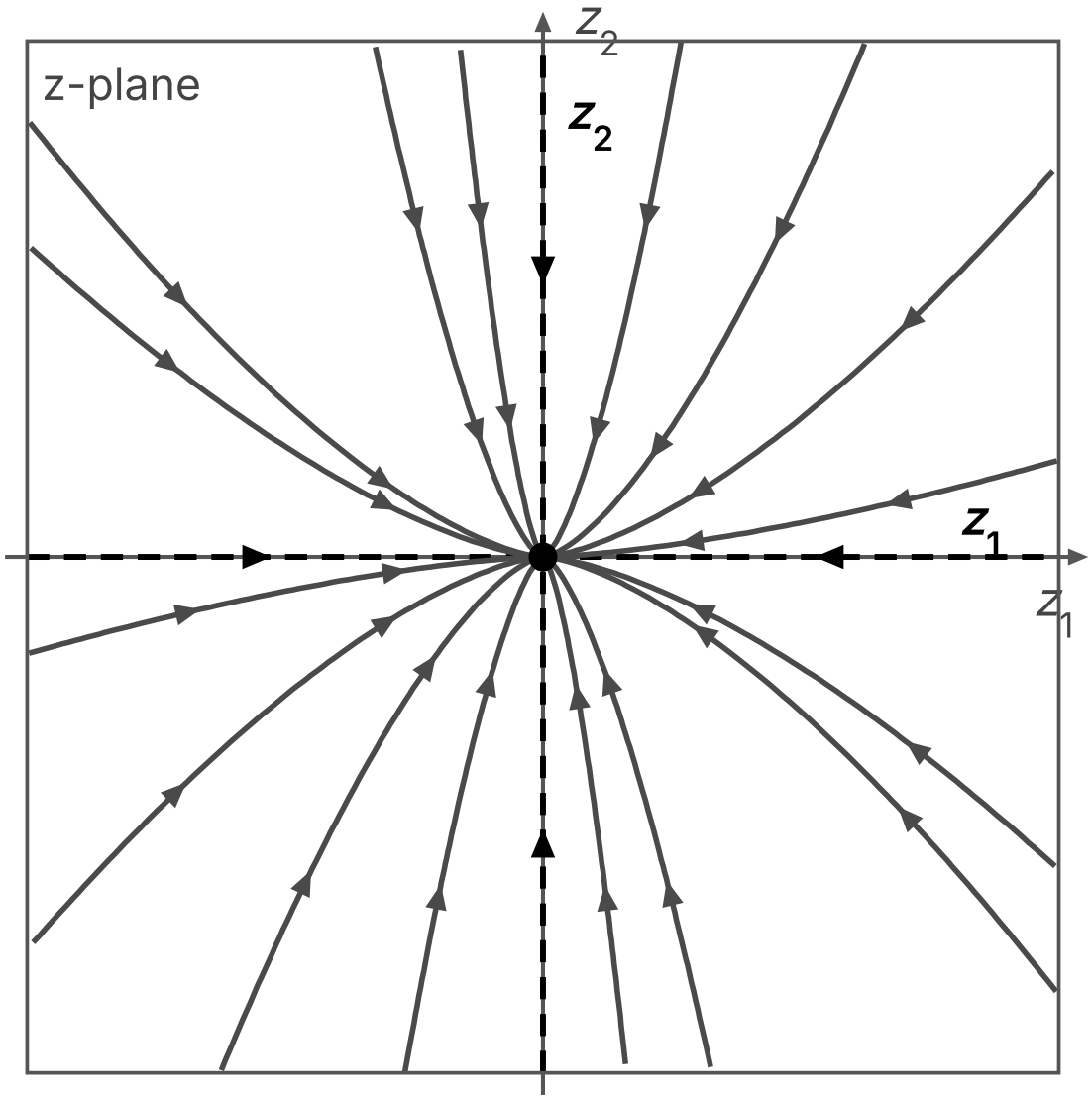
Phase portrait

$\lambda_1 = -2$ and $\lambda_2 = -3$ are both negative: the origin is a **stable node** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

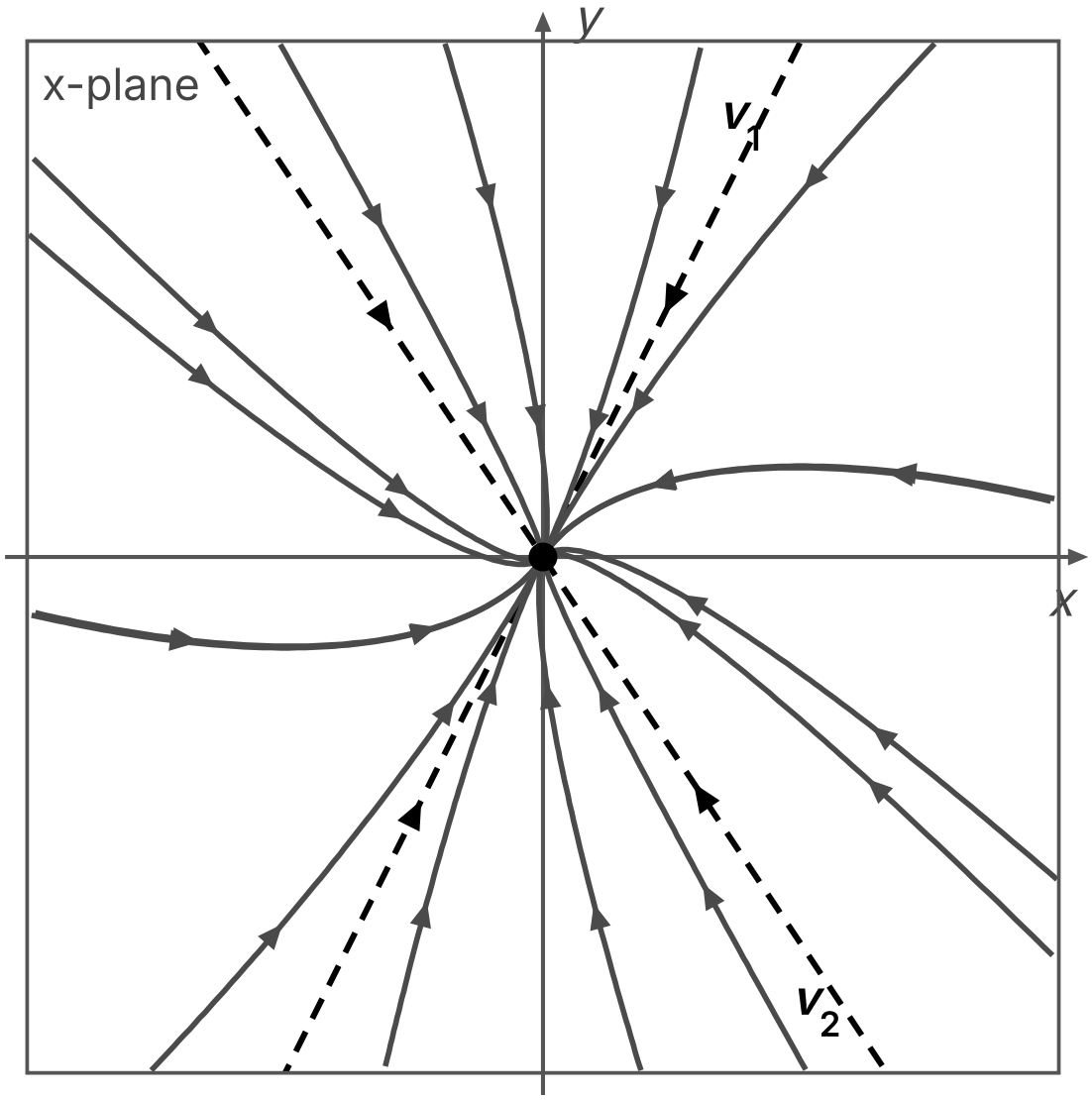
In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{-2t}$ and $z_2 = c_2 e^{-3t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^t = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2|\,(|z_1|/|c_1|)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K\,|z_1|^{\lambda_2/\lambda_1} = K\,|z_1|^{\frac{3}{2}}$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

The exponent $\lambda_2/\lambda_1 = \frac{3}{2}$ is positive, so the orbits are parabola-like curves through the origin. At the origin they are tangent to the eigenvector of the eigenvalue of **smaller modulus**, $|\lambda| = 2$, i.e. to the line through $(1, 2)$ (they arrive there as $t \to +\infty$), and far from the origin they become parallel to the other eigenvector (the fast direction). The arrows point towards the origin on both eigenlines.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (1, 2)$ and the z_2-axis to the line through $\mathbf{v}_2 = (2, -3)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = -2\mathbf{v}_1$ points inward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = -3\mathbf{v}_2$ points inward.



Canonical $\mathbf{z}$-plane (stable node): invariant axes and orbits $z_2 = K|z_1|^{\frac{3}{2}}$.



Original (x, y)-plane (stable node): dashed lines are the eigenlines $\mathbf{v}_1 = (1, 2)$ and $\mathbf{v}_2 = (2, -3)$.

(iv) $\lambda_1 = 3,\ \mathbf{v}_1 = (2, 3)^T;\ \lambda_2 = -5,\ \mathbf{v}_2 = (0, 1)^T$

The statement gives the eigen-data, not the matrix. Since $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$ for $j = 1, 2$ means $AQ = Q\Lambda$ with $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, the matrix is $A = Q\Lambda Q^{-1}$ (it is needed to draw the portrait and to check the data):

$$Q = \begin{pmatrix} 2 & 0 \\ 3 & 1 \end{pmatrix}, \qquad \Lambda = \begin{pmatrix} 3 & 0 \\ 0 & -5 \end{pmatrix}, \qquad \det Q = 2, \qquad Q^{-1} = \begin{pmatrix} \frac{1}{2} & 0 \\ -\frac{3}{2} & 1 \end{pmatrix}$$

$$
\begin{aligned}
(Q\Lambda)_{11} &= 2\cdot 3 + 0\cdot 0 = 6\\
(Q\Lambda)_{12} &= 2\cdot 0 + 0\cdot(-5) = 0\\
(Q\Lambda)_{21} &= 3\cdot 3 + 1\cdot 0 = 9\\
(Q\Lambda)_{22} &= 3\cdot 0 + 1\cdot(-5) = -5
\end{aligned}
$$

$$
Q\Lambda = \begin{pmatrix} 6 & 0 \\ 9 & -5 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= 6\cdot\frac{1}{2} + 0\cdot(-\frac{3}{2}) = 3\\
(A)_{12} &= 6\cdot 0 + 0\cdot 1 = 0\\
(A)_{21} &= 9\cdot\frac{1}{2} + (-5)\cdot(-\frac{3}{2}) = 12\\
(A)_{22} &= 9\cdot 0 + (-5)\cdot 1 = -5
\end{aligned}
$$

$$
A = Q\Lambda Q^{-1} = \begin{pmatrix} 3 & 0 \\ 12 & -5 \end{pmatrix}
$$

Change of coordinates

The eigenvectors of distinct eigenvalues are linearly independent (Topic 4, Appendix B argument), so they form a basis. Put them as columns, $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, and define the new coordinates by $\mathbf{z} = Q^{-1}\mathbf{x}$, i.e. $\mathbf{x} = Q\mathbf{z}$:

$$
Q = \begin{pmatrix} 2 & 0 \\ 3 & 1 \end{pmatrix}
$$

$\det Q = (2)(1) - (0)(3) = 2$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$
Q^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ -\frac{3}{2} & 1 \end{pmatrix}
$$

Since Q is constant, $\dot{\mathbf{z}} = Q^{-1}\dot{\mathbf{x}} = Q^{-1}A\mathbf{x} = (Q^{-1}AQ)\mathbf{z}$. Compute AQ one column at a time, with $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$:

$$A\mathbf{v}_1 = \begin{pmatrix} 3 & 0 \\ 12 & -5 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ 9 \end{pmatrix} = 3\mathbf{v}_1, \qquad A\mathbf{v}_2 = \begin{pmatrix} 3 & 0 \\ 12 & -5 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -5 \end{pmatrix} = (-5)\mathbf{v}_2$$

$$AQ = [A\mathbf{v}_1 \mid A\mathbf{v}_2] = \begin{pmatrix} 6 & 0 \\ 9 & -5 \end{pmatrix}$$

Now multiply by Q^{-1} on the left, entry by entry:

$$\begin{aligned} (Q^{-1}AQ)_{11} &= \frac{1}{2}\cdot 6 + 0 \cdot 9 = 3 \\ (Q^{-1}AQ)_{12} &= \frac{1}{2}\cdot 0 + 0 \cdot (-5) = 0 \\ (Q^{-1}AQ)_{21} &= (-\frac{3}{2})\cdot 6 + 1 \cdot 9 = 0 \\ (Q^{-1}AQ)_{22} &= (-\frac{3}{2})\cdot 0 + 1 \cdot (-5) = -5 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 3 & 0 \\ 0 & -5 \end{pmatrix} = \Lambda = \begin{pmatrix} 3 & 0 \\ 0 & -5 \end{pmatrix}$$

The transformation has decoupled the system (the matrix is diagonal), as claimed.

Solution in the canonical plane

$$\dot{z}_1 = 3z_1, \qquad \dot{z}_2 = -5z_2$$

Each equation is separable. If $z_i \neq 0$, $\frac{dz_i}{z_i} = \lambda_i\, dt$ and integrating, $\ln|z_i| = \lambda_i t + C_i$, so $z_i = c_i e^{\lambda_i t}$ with $c_i = \pm e^{C_i}$ (the solution $z_i \equiv 0$ corresponds to $c_i = 0$). Equivalently, $\frac{d}{dt}(e^{-\lambda_i t} z_i) = 0$.

$$z_1(t) = c_1 e^{3t}, \qquad z_2(t) = c_2 e^{-5t}$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{v}_1 + z_2\mathbf{v}_2$, so the general solution is

$$\mathbf{x}(t) = c_1 e^{3t}\begin{pmatrix}2\\3\end{pmatrix} + c_2 e^{-5t}\begin{pmatrix}0\\1\end{pmatrix}$$

$$x(t) = 2c_1 e^{3t}, \qquad y(t) = 3c_1 e^{3t} + c_2 e^{-5t}$$

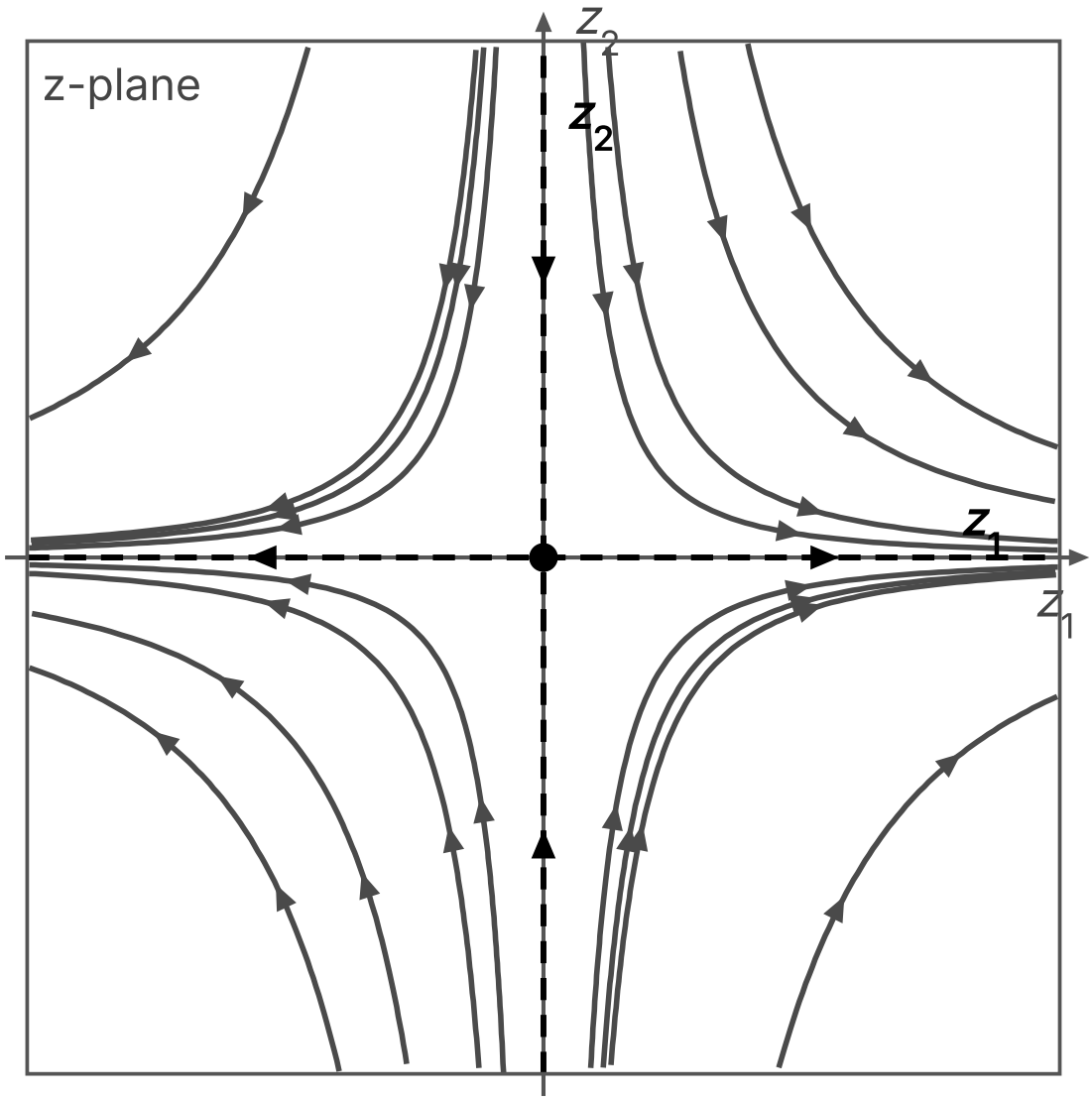
Phase portrait

$\lambda_1 = 3$ and $\lambda_2 = -5$ are of opposite signs: the origin is a **saddle** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

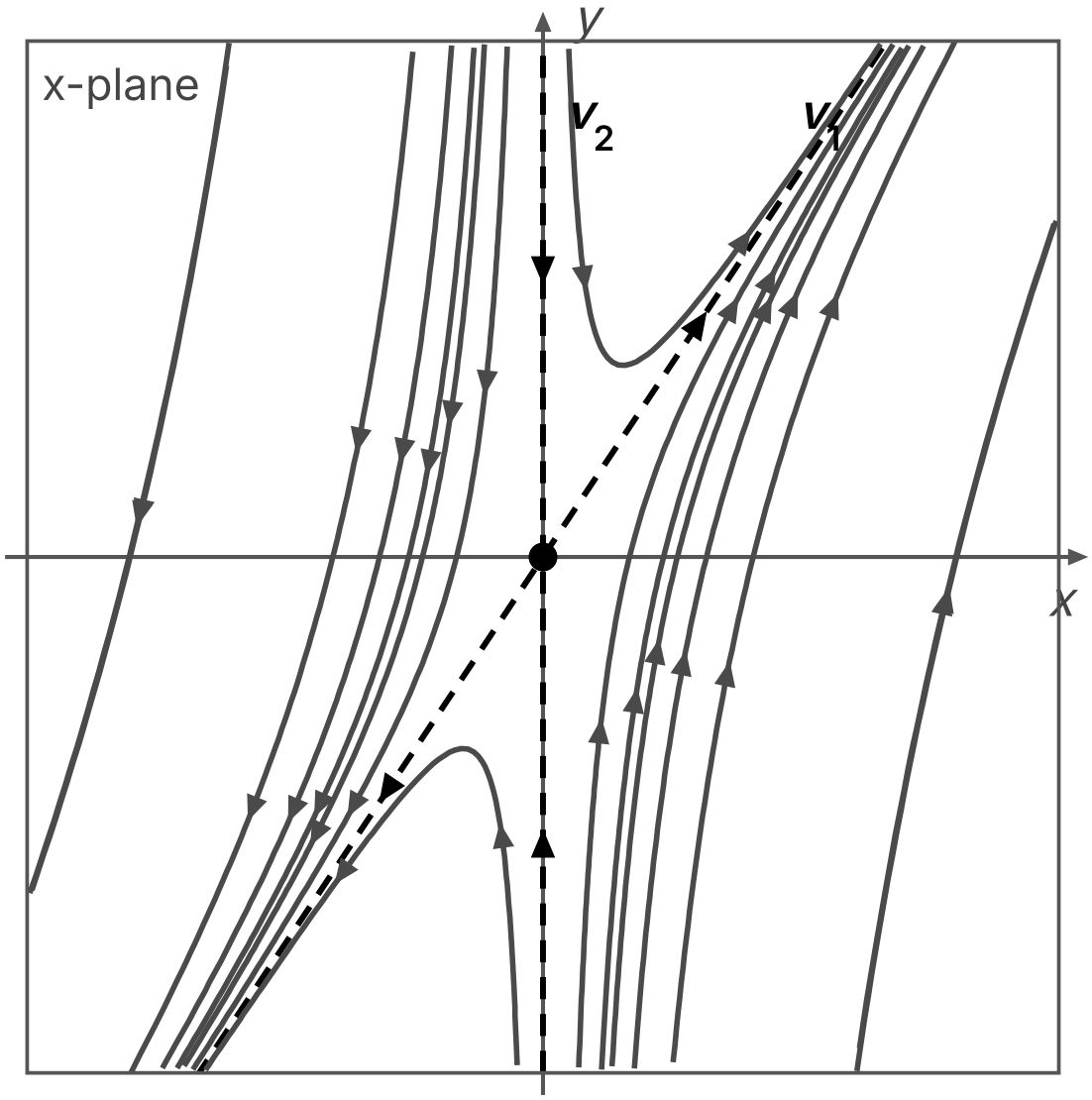
In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{3t}$ and $z_2 = c_2 e^{-5t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^t = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2|\,(|z_1|/|c_1|)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K\,|z_1|^{\lambda_2/\lambda_1} = K\,|z_1|^{-\frac{5}{3}}$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

The exponent $\lambda_2/\lambda_1 = -\frac{5}{3}$ is negative, so the orbits are hyperbola-like. Only solutions starting on the eigenline of the negative eigenvalue, the line through $(0,1)$ (the **stable manifold**), reach the origin as $t \to +\infty$; the line through $(2,3)$ (positive eigenvalue) is the **unstable manifold**. All other orbits come in close to the stable line, then leave along the unstable one.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (2,3)$ and the z_2-axis to the line through $\mathbf{v}_2 = (0,1)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = 3\mathbf{v}_1$ points outward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = -5\mathbf{v}_2$ points inward.



Canonical $\mathbf{z}$-plane (saddle): invariant axes and orbits $z_2 = K|z_1|^{-\frac{5}{3}}$.



Original (x, y)-plane (saddle): dashed lines are the eigenlines $\mathbf{v}_1 = (2, 3)$ and $\mathbf{v}_2 = (0, 1)$.

(v) $\lambda_1 = 3,\ \mathbf{v}_1 = (1, 2)^T;\ \lambda_2 = 1,\ \mathbf{v}_2 = (1, -3)^T$

The statement gives the eigen-data, not the matrix. Since $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$ for $j = 1, 2$ means $AQ = Q\Lambda$ with $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, the matrix is $A = Q\Lambda Q^{-1}$ (it is needed to draw the portrait and to check the data):

$$Q = \begin{pmatrix} 1 & 1 \\ 2 & -3 \end{pmatrix}, \qquad \Lambda = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \det Q = -5, \qquad Q^{-1} = \begin{pmatrix} \frac{3}{5} & \frac{1}{5} \\ \frac{2}{5} & -\frac{1}{5} \end{pmatrix}$$

$$
\begin{aligned}
(Q\Lambda)_{11} &= 1\cdot 3 + 1\cdot 0 = 3\\
(Q\Lambda)_{12} &= 1\cdot 0 + 1\cdot 1 = 1\\
(Q\Lambda)_{21} &= 2\cdot 3 + (-3)\cdot 0 = 6\\
(Q\Lambda)_{22} &= 2\cdot 0 + (-3)\cdot 1 = -3
\end{aligned}
$$

$$
Q\Lambda = \begin{pmatrix} 3 & 1 \\ 6 & -3 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= 3\cdot\frac{3}{5} + 1\cdot\frac{2}{5} = \frac{11}{5}\\
(A)_{12} &= 3\cdot\frac{1}{5} + 1\cdot(-\frac{1}{5}) = \frac{2}{5}\\
(A)_{21} &= 6\cdot\frac{3}{5} + (-3)\cdot\frac{2}{5} = \frac{12}{5}\\
(A)_{22} &= 6\cdot\frac{1}{5} + (-3)\cdot(-\frac{1}{5}) = \frac{9}{5}
\end{aligned}
$$

$$
A = Q\Lambda Q^{-1} = \begin{pmatrix} \frac{11}{5} & \frac{2}{5} \\ \frac{12}{5} & \frac{9}{5} \end{pmatrix}
$$

Change of coordinates

The eigenvectors of distinct eigenvalues are linearly independent (Topic 4, Appendix B argument), so they form a basis. Put them as columns, $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, and define the new coordinates by $\mathbf{z} = Q^{-1}\mathbf{x}$, i.e. $\mathbf{x} = Q\mathbf{z}$:

$$
Q = \begin{pmatrix} 1 & 1 \\ 2 & -3 \end{pmatrix}
$$

$\det Q = (1)(-3) - (1)(2) = -5$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$
Q^{-1} = \frac{1}{-5}\begin{pmatrix} -3 & -1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} \frac{3}{5} & \frac{1}{5} \\ \frac{2}{5} & -\frac{1}{5} \end{pmatrix}
$$

Since Q is constant, $\dot{\mathbf{z}} = Q^{-1}\dot{\mathbf{x}} = Q^{-1}A\mathbf{x} = (Q^{-1}AQ)\mathbf{z}$. Compute AQ one column at a time, with $A\mathbf{v}_j = \lambda_j\mathbf{v}_j$:

$$A\mathbf{v}_1 = \begin{pmatrix} \frac{11}{5} & \frac{2}{5} \\ \frac{12}{5} & \frac{8}{5} \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix} = 3\mathbf{v}_1, \qquad A\mathbf{v}_2 = \begin{pmatrix} \frac{11}{5} & \frac{2}{5} \\ \frac{12}{5} & \frac{8}{5} \end{pmatrix}\begin{pmatrix} 1 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \end{pmatrix} = 1\mathbf{v}_2$$

$$AQ = [A\mathbf{v}_1 \mid A\mathbf{v}_2] = \begin{pmatrix} 3 & 1 \\ 6 & -3 \end{pmatrix}$$

Now multiply by Q^{-1} on the left, entry by entry:

$$\begin{aligned} (Q^{-1}AQ)_{11} &= \frac{3}{5}\cdot 3 + \frac{1}{5}\cdot 6 = 3 \\ (Q^{-1}AQ)_{12} &= \frac{3}{5}\cdot 1 + \frac{1}{5}\cdot (-3) = 0 \\ (Q^{-1}AQ)_{21} &= \frac{2}{5}\cdot 3 + (-\frac{1}{5})\cdot 6 = 0 \\ (Q^{-1}AQ)_{22} &= \frac{2}{5}\cdot 1 + (-\frac{1}{5})\cdot (-3) = 1 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} = \Lambda = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$$

The transformation has decoupled the system (the matrix is diagonal), as claimed.

Solution in the canonical plane

$$\dot{z}_1 = 3z_1, \qquad \dot{z}_2 = z_2$$

Each equation is separable. If $z_i \neq 0$, $\frac{dz_i}{z_i} = \lambda_i\,dt$ and integrating, $\ln|z_i| = \lambda_i t + C_i$, so $z_i = c_i e^{\lambda_i t}$ with $c_i = \pm e^{C_i}$ (the solution $z_i \equiv 0$ corresponds to $c_i = 0$). Equivalently, $\frac{d}{dt}(e^{-\lambda_i t} z_i) = 0$.

$$z_1(t) = c_1 e^{3t}, \qquad z_2(t) = c_2 e^{1t}$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{v}_1 + z_2\mathbf{v}_2$, so the general solution is

$$\mathbf{x}(t) = c_1 e^{3t}\begin{pmatrix}1\\2\end{pmatrix} + c_2 e^{1t}\begin{pmatrix}1\\-3\end{pmatrix}$$

$$x(t) = c_1 e^{3t} + c_2 e^{t}, \qquad y(t) = 2c_1 e^{3t} - 3c_2 e^{t}$$

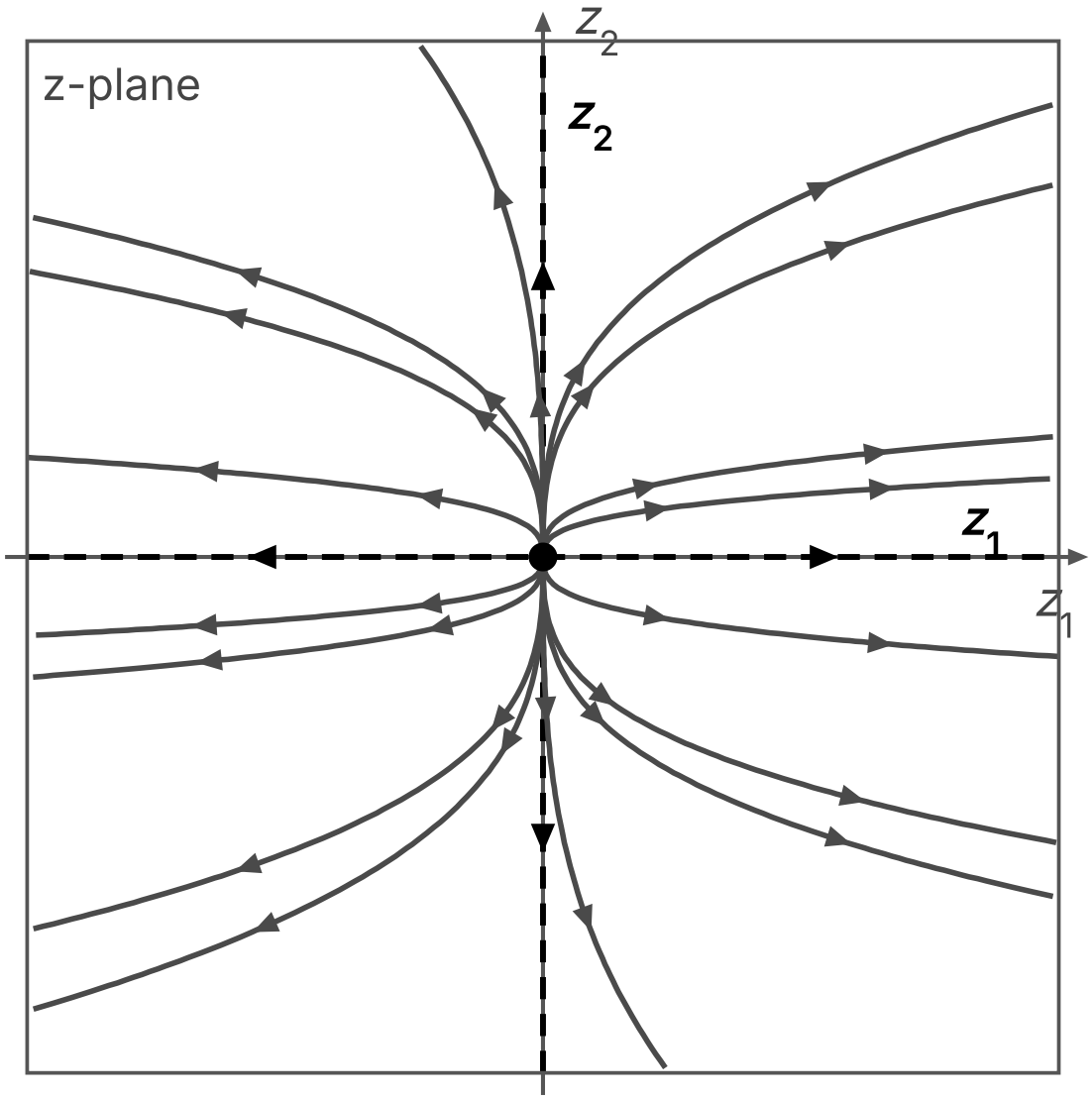
Phase portrait

$\lambda_1 = 3$ and $\lambda_2 = 1$ are both positive: the origin is a **unstable node** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

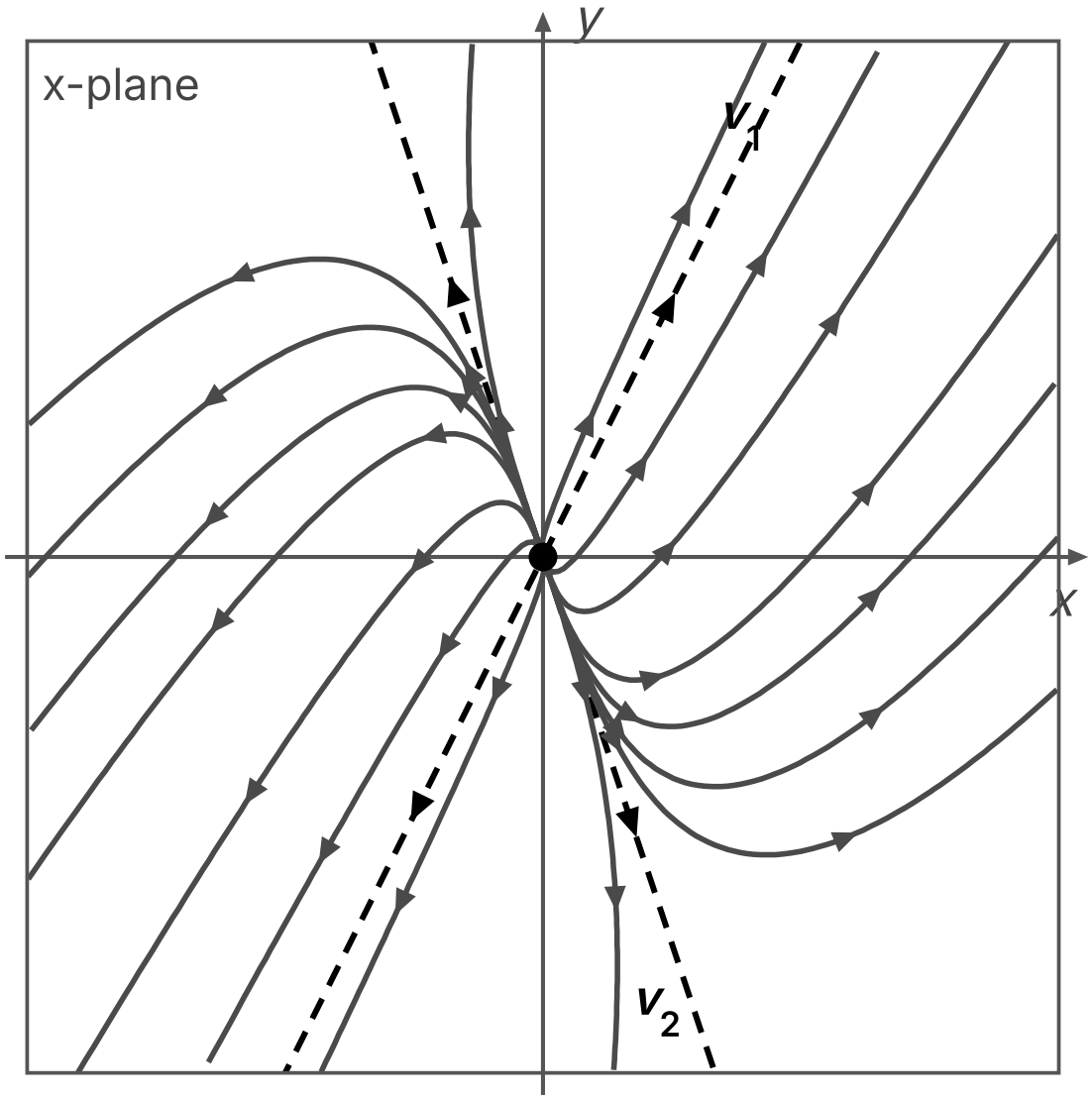
In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{3t}$ and $z_2 = c_2 e^{1t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^t = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2|\,(|z_1|/|c_1|)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K\,|z_1|^{\lambda_2/\lambda_1} = K\,|z_1|^{\frac{1}{3}}$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

The exponent $\lambda_2/\lambda_1 = \frac{1}{3}$ is positive, so the orbits are parabola-like curves through the origin. At the origin they are tangent to the eigenvector of the eigenvalue of **smaller modulus**, $|\lambda| = 1$, i.e. to the line through $(1, -3)$ (they leave from there (the limit $t \to -\infty$)), and far from the origin they become parallel to the other eigenvector (the fast direction). The arrows point away from the origin on both eigenlines.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (1, 2)$ and the z_2-axis to the line through $\mathbf{v}_2 = (1, -3)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = 3\mathbf{v}_1$ points outward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = 1\mathbf{v}_2$ points outward.



Canonical $\mathbf{z}$-plane (unstable node): invariant axes and orbits $z_2 = K|z_1|^{\frac{1}{3}}$.



Original (x, y)-plane (unstable node): dashed lines are the eigenlines $\mathbf{v}_1 = (1, 2)$ and $\mathbf{v}_2 = (1, -3)$.

(vi) $\lambda_1 = 2$, $\mathbf{v}_1 = (0, 1)^T$; $\lambda_2 = -3$, $\mathbf{v}_2 = (1, 5)^T$

The statement gives the eigen-data, not the matrix. Since $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$ for $j = 1, 2$ means $AQ = Q\Lambda$ with $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, the matrix is $A = Q\Lambda Q^{-1}$ (it is needed to draw the portrait and to check the data):

$$Q = \begin{pmatrix} 0 & 1 \\ 1 & 5 \end{pmatrix}, \qquad \Lambda = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}, \qquad \det Q = -1, \qquad Q^{-1} = \begin{pmatrix} -5 & 1 \\ 1 & 0 \end{pmatrix}$$

$$
\begin{aligned}
(Q\Lambda)_{11} &= 0 \cdot 2 + 1 \cdot 0 = 0 \\
(Q\Lambda)_{12} &= 0 \cdot 0 + 1 \cdot (-3) = -3 \\
(Q\Lambda)_{21} &= 1 \cdot 2 + 5 \cdot 0 = 2 \\
(Q\Lambda)_{22} &= 1 \cdot 0 + 5 \cdot (-3) = -15
\end{aligned}
$$

$$
Q\Lambda = \begin{pmatrix} 0 & -3 \\ 2 & -15 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= 0 \cdot (-5) + (-3) \cdot 1 = -3 \\
(A)_{12} &= 0 \cdot 1 + (-3) \cdot 0 = 0 \\
(A)_{21} &= 2 \cdot (-5) + (-15) \cdot 1 = -25 \\
(A)_{22} &= 2 \cdot 1 + (-15) \cdot 0 = 2
\end{aligned}
$$

$$
A = Q\Lambda Q^{-1} = \begin{pmatrix} -3 & 0 \\ -25 & 2 \end{pmatrix}
$$

Change of coordinates

The eigenvectors of distinct eigenvalues are linearly independent (Topic 4, Appendix B argument), so they form a basis. Put them as columns, $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, and define the new coordinates by $\mathbf{z} = Q^{-1}\mathbf{x}$, i.e. $\mathbf{x} = Q\mathbf{z}$:

$$
Q = \begin{pmatrix} 0 & 1 \\ 1 & 5 \end{pmatrix}
$$

$\det Q = (0)(5) - (1)(1) = -1$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$
Q^{-1} = \frac{1}{-1}\begin{pmatrix} 5 & -1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -5 & 1 \\ 1 & 0 \end{pmatrix}
$$

Since Q is constant, $\dot{\mathbf{z}} = Q^{-1}\dot{\mathbf{x}} = Q^{-1}A\mathbf{x} = (Q^{-1}AQ)\mathbf{z}$. Compute AQ one column at a time, with $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$:

$$A\mathbf{v}_1 = \begin{pmatrix} -3 & 0 \\ -25 & 2 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} = 2\mathbf{v}_1, \qquad A\mathbf{v}_2 = \begin{pmatrix} -3 & 0 \\ -25 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ 5 \end{pmatrix} = \begin{pmatrix} -3 \\ -15 \end{pmatrix} = (-3)\mathbf{v}_2$$

$$AQ = [A\mathbf{v}_1 \mid A\mathbf{v}_2] = \begin{pmatrix} 0 & -3 \\ 2 & -15 \end{pmatrix}$$

Now multiply by Q^{-1} on the left, entry by entry:

$$\begin{aligned} (Q^{-1}AQ)_{11} &= (-5)\cdot 0 + 1 \cdot 2 = 2 \\ (Q^{-1}AQ)_{12} &= (-5)\cdot(-3) + 1 \cdot (-15) = 0 \\ (Q^{-1}AQ)_{21} &= 1 \cdot 0 + 0 \cdot 2 = 0 \\ (Q^{-1}AQ)_{22} &= 1 \cdot (-3) + 0 \cdot (-15) = -3 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix} = \Lambda = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}$$

The transformation has decoupled the system (the matrix is diagonal), as claimed.

Solution in the canonical plane

$$\dot{z}_1 = 2z_1, \qquad \dot{z}_2 = -3z_2$$

Each equation is separable. If $z_i \neq 0$, $\frac{dz_i}{z_i} = \lambda_i\,dt$ and integrating, $\ln|z_i| = \lambda_i t + C_i$, so $z_i = c_i e^{\lambda_i t}$ with $c_i = \pm e^{C_i}$ (the solution $z_i \equiv 0$ corresponds to $c_i = 0$). Equivalently, $\frac{d}{dt}(e^{-\lambda_i t} z_i) = 0$.

$$z_1(t) = c_1 e^{2t}, \qquad z_2(t) = c_2 e^{-3t}$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{v}_1 + z_2\mathbf{v}_2$, so the general solution is

$$\mathbf{x}(t) = c_1 e^{2t}\begin{pmatrix}0\\1\end{pmatrix} + c_2 e^{-3t}\begin{pmatrix}1\\5\end{pmatrix}$$

$$x(t) = c_2 e^{-3t}, \qquad y(t) = c_1 e^{2t} + 5c_2 e^{-3t}$$

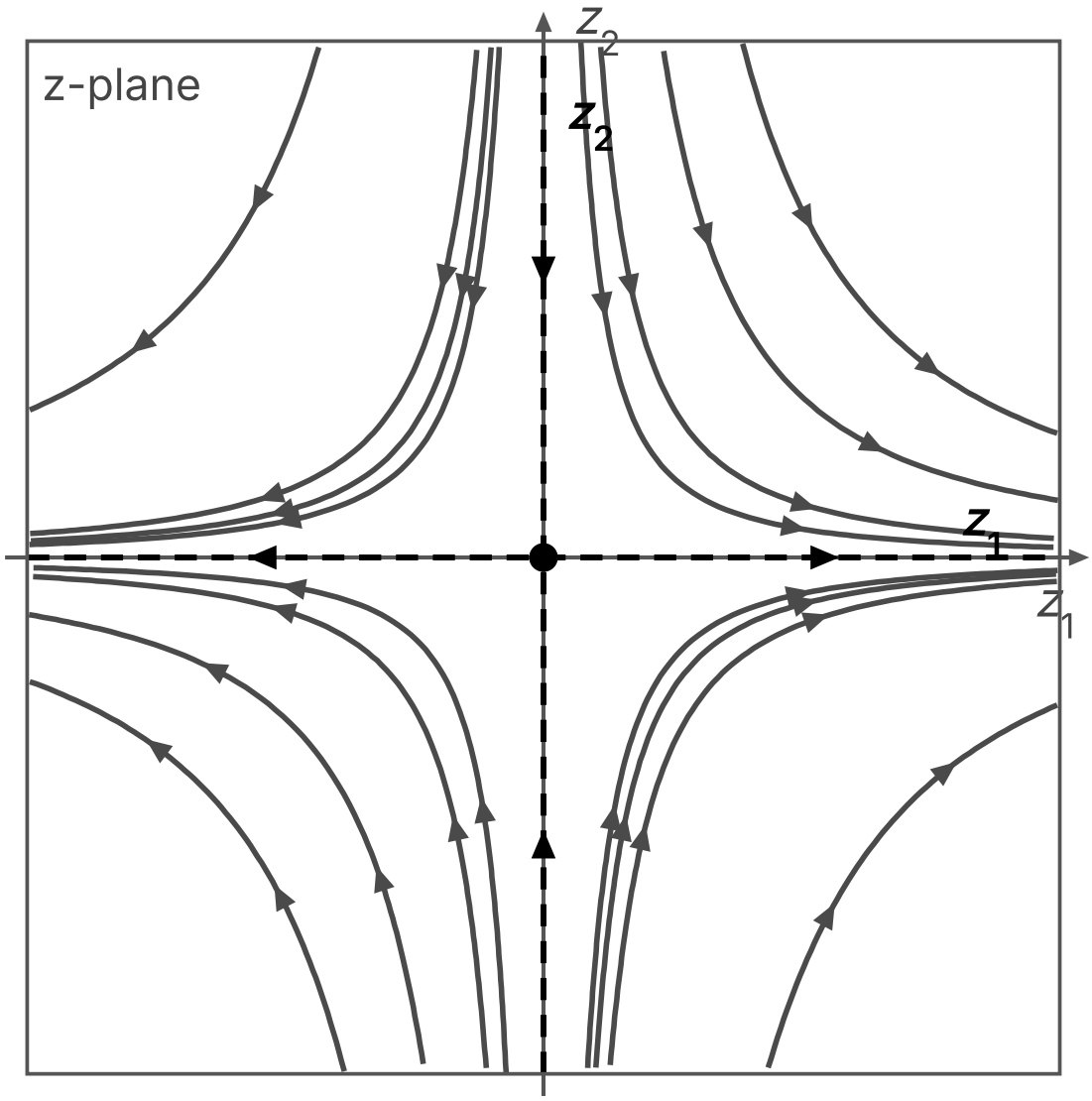
Phase portrait

$\lambda_1 = 2$ and $\lambda_2 = -3$ are of opposite signs: the origin is a **saddle** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

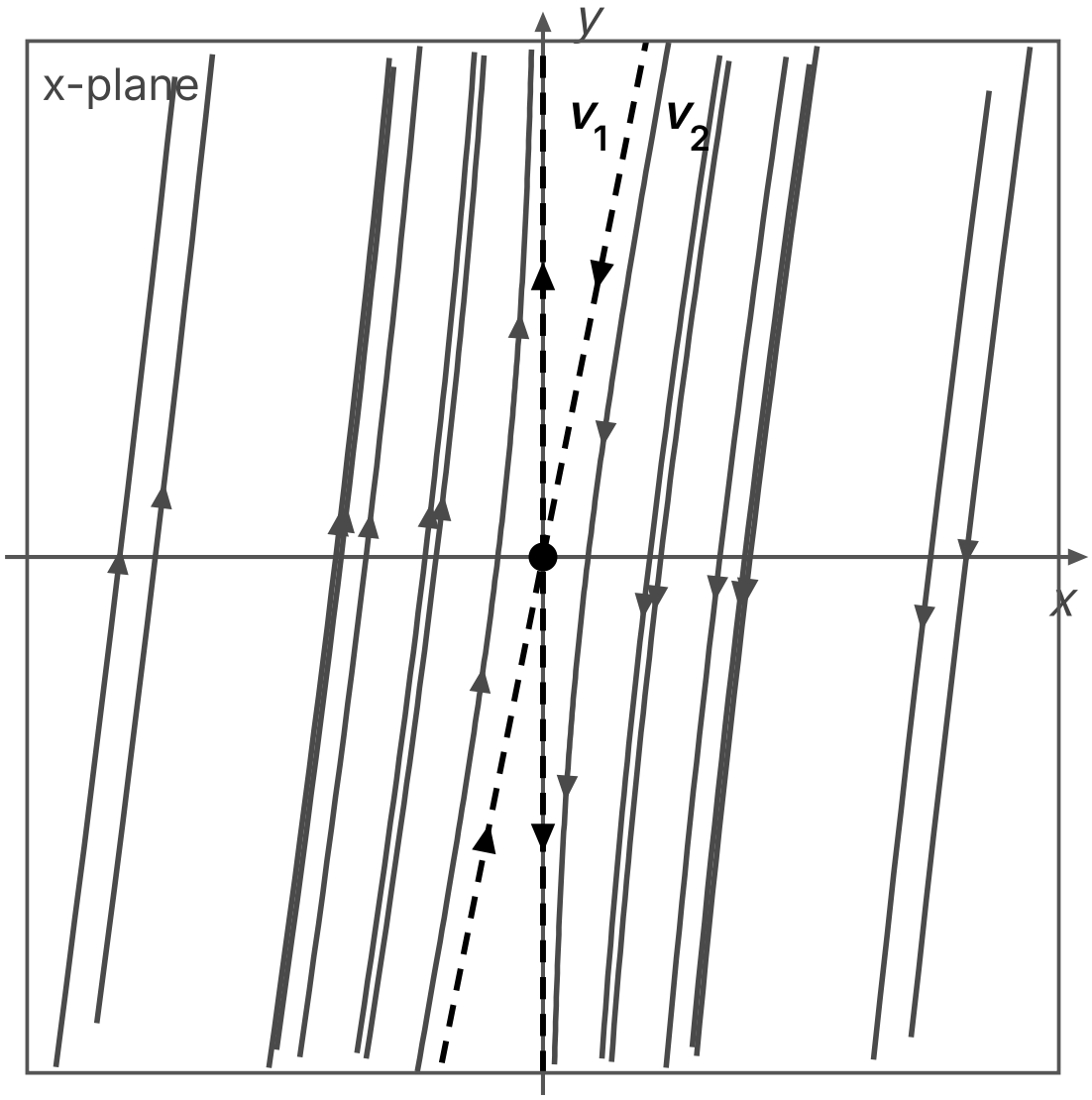
In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{2t}$ and $z_2 = c_2 e^{-3t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^t = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2|\,(|z_1|/|c_1|)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K\,|z_1|^{\lambda_2/\lambda_1} = K\,|z_1|^{-\frac{3}{2}}$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

The exponent $\lambda_2/\lambda_1 = -\frac{3}{2}$ is negative, so the orbits are hyperbola-like. Only solutions starting on the eigenline of the negative eigenvalue, the line through $(1, 5)$ (the **stable manifold**), reach the origin as $t \to +\infty$; the line through $(0, 1)$ (positive eigenvalue) is the **unstable manifold**. All other orbits come in close to the stable line, then leave along the unstable one.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (0, 1)$ and the z_2-axis to the line through $\mathbf{v}_2 = (1, 5)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = 2\mathbf{v}_1$ points outward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = -3\mathbf{v}_2$ points inward.



Canonical $\mathbf{z}$-plane (saddle): invariant axes and orbits $z_2 = K|z_1|^{-\frac{3}{2}}$.



Original (x, y)-plane (saddle): dashed lines are the eigenlines $\mathbf{v}_1 = (0, 1)$ and $\mathbf{v}_2 = (1, 5)$.

(vii) $\lambda_1 = 1,\ \mathbf{v}_1 = (1, 1)^T;\ \lambda_2 = 2,\ \mathbf{v}_2 = (2, 1)^T$

The statement gives the eigen-data, not the matrix. Since $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$ for $j = 1, 2$ means $AQ = Q\Lambda$ with $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, the matrix is $A = Q\Lambda Q^{-1}$ (it is needed to draw the portrait and to check the data):

$$Q = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}, \qquad \Lambda = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad \det Q = -1, \qquad Q^{-1} = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix}$$

$$
\begin{aligned}
(Q\Lambda)_{11} &= 1\cdot 1 + 2\cdot 0 = 1\\
(Q\Lambda)_{12} &= 1\cdot 0 + 2\cdot 2 = 4\\
(Q\Lambda)_{21} &= 1\cdot 1 + 1\cdot 0 = 1\\
(Q\Lambda)_{22} &= 1\cdot 0 + 1\cdot 2 = 2
\end{aligned}
$$

$$
Q\Lambda = \begin{pmatrix} 1 & 4 \\ 1 & 2 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= 1\cdot(-1) + 4\cdot 1 = 3\\
(A)_{12} &= 1\cdot 2 + 4\cdot(-1) = -2\\
(A)_{21} &= 1\cdot(-1) + 2\cdot 1 = 1\\
(A)_{22} &= 1\cdot 2 + 2\cdot(-1) = 0
\end{aligned}
$$

$$
A = Q\Lambda Q^{-1} = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}
$$

Change of coordinates

The eigenvectors of distinct eigenvalues are linearly independent (Topic 4, Appendix B argument), so they form a basis. Put them as columns, $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, and define the new coordinates by $\mathbf{z} = Q^{-1}\mathbf{x}$, i.e. $\mathbf{x} = Q\mathbf{z}$:

$$
Q = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}
$$

$\det Q = (1)(1) - (2)(1) = -1$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$
Q^{-1} = \frac{1}{-1}\begin{pmatrix} 1 & -2 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix}
$$

Since Q is constant, $\dot{\mathbf{z}} = Q^{-1}\dot{\mathbf{x}} = Q^{-1}A\mathbf{x} = (Q^{-1}AQ)\mathbf{z}$. Compute AQ one column at a time, with $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$:

$$A\mathbf{v}_1 = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1\mathbf{v}_1, \qquad A\mathbf{v}_2 = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} = 2\mathbf{v}_2$$

$$AQ = [A\mathbf{v}_1 \mid A\mathbf{v}_2] = \begin{pmatrix} 1 & 4 \\ 1 & 2 \end{pmatrix}$$

Now multiply by Q^{-1} on the left, entry by entry:

$$\begin{aligned} (Q^{-1}AQ)_{11} &= (-1)\cdot 1 + 2 \cdot 1 = 1 \\ (Q^{-1}AQ)_{12} &= (-1)\cdot 4 + 2 \cdot 2 = 0 \\ (Q^{-1}AQ)_{21} &= 1\cdot 1 + (-1) \cdot 1 = 0 \\ (Q^{-1}AQ)_{22} &= 1\cdot 4 + (-1) \cdot 2 = 2 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = \Lambda = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

The transformation has decoupled the system (the matrix is diagonal), as claimed.

Solution in the canonical plane

$$\dot{z}_1 = z_1, \qquad \dot{z}_2 = 2z_2$$

Each equation is separable. If $z_i \neq 0$, $\frac{dz_i}{z_i} = \lambda_i\, dt$ and integrating, $\ln|z_i| = \lambda_i t + C_i$, so $z_i = c_i e^{\lambda_i t}$ with $c_i = \pm e^{C_i}$ (the solution $z_i \equiv 0$ corresponds to $c_i = 0$). Equivalently, $\frac{d}{dt}(e^{-\lambda_i t} z_i) = 0$.

$$z_1(t) = c_1 e^{1t}, \qquad z_2(t) = c_2 e^{2t}$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{v}_1 + z_2\mathbf{v}_2$, so the general solution is

$$\mathbf{x}(t) = c_1 e^{1t}\begin{pmatrix}1\\1\end{pmatrix} + c_2 e^{2t}\begin{pmatrix}2\\1\end{pmatrix}$$

$$x(t) = c_1 e^{t} + 2c_2 e^{2t}, \qquad y(t) = c_1 e^{t} + c_2 e^{2t}$$

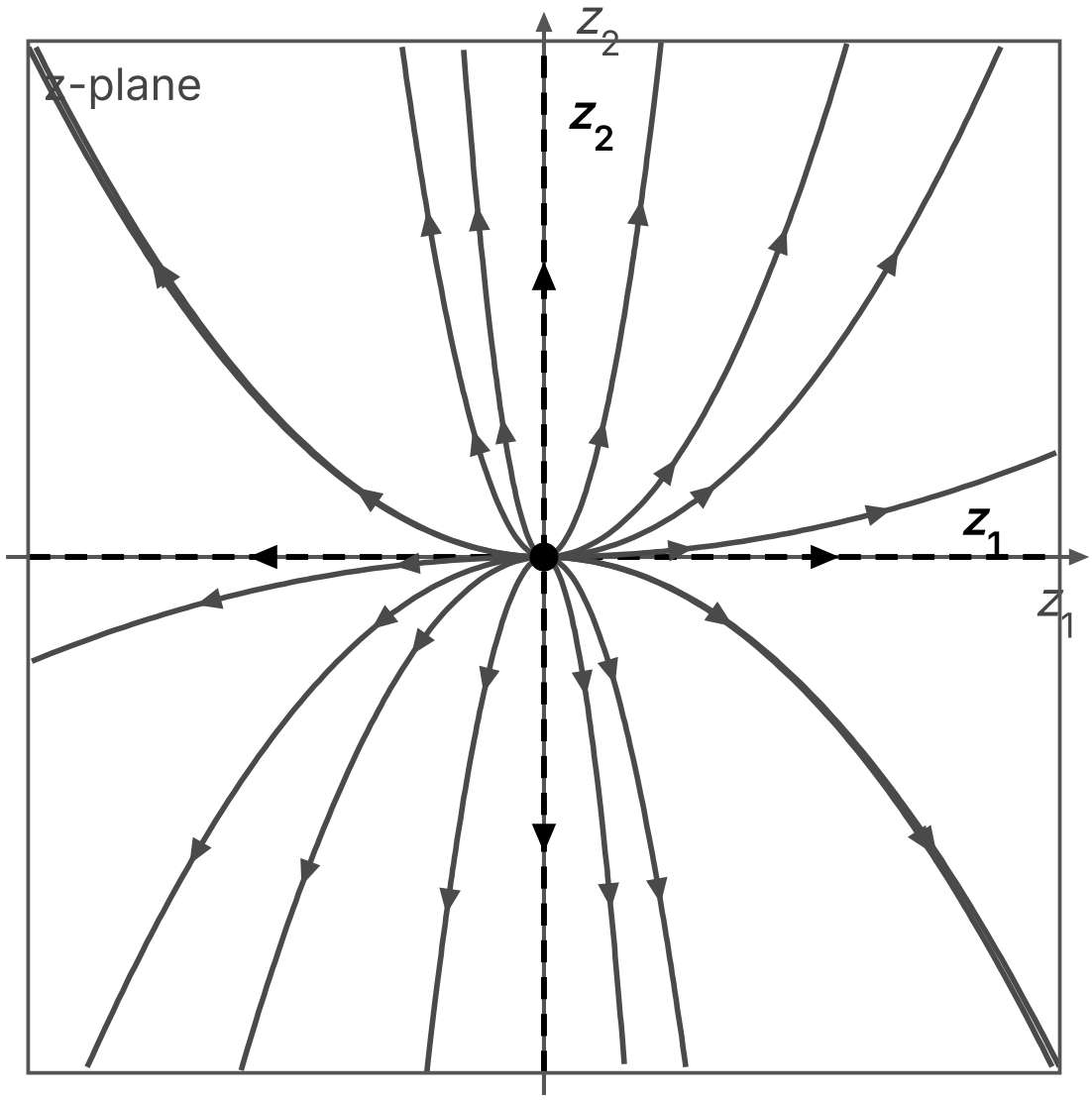
Phase portrait

$\lambda_1 = 1$ and $\lambda_2 = 2$ are both positive: the origin is a **unstable node** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

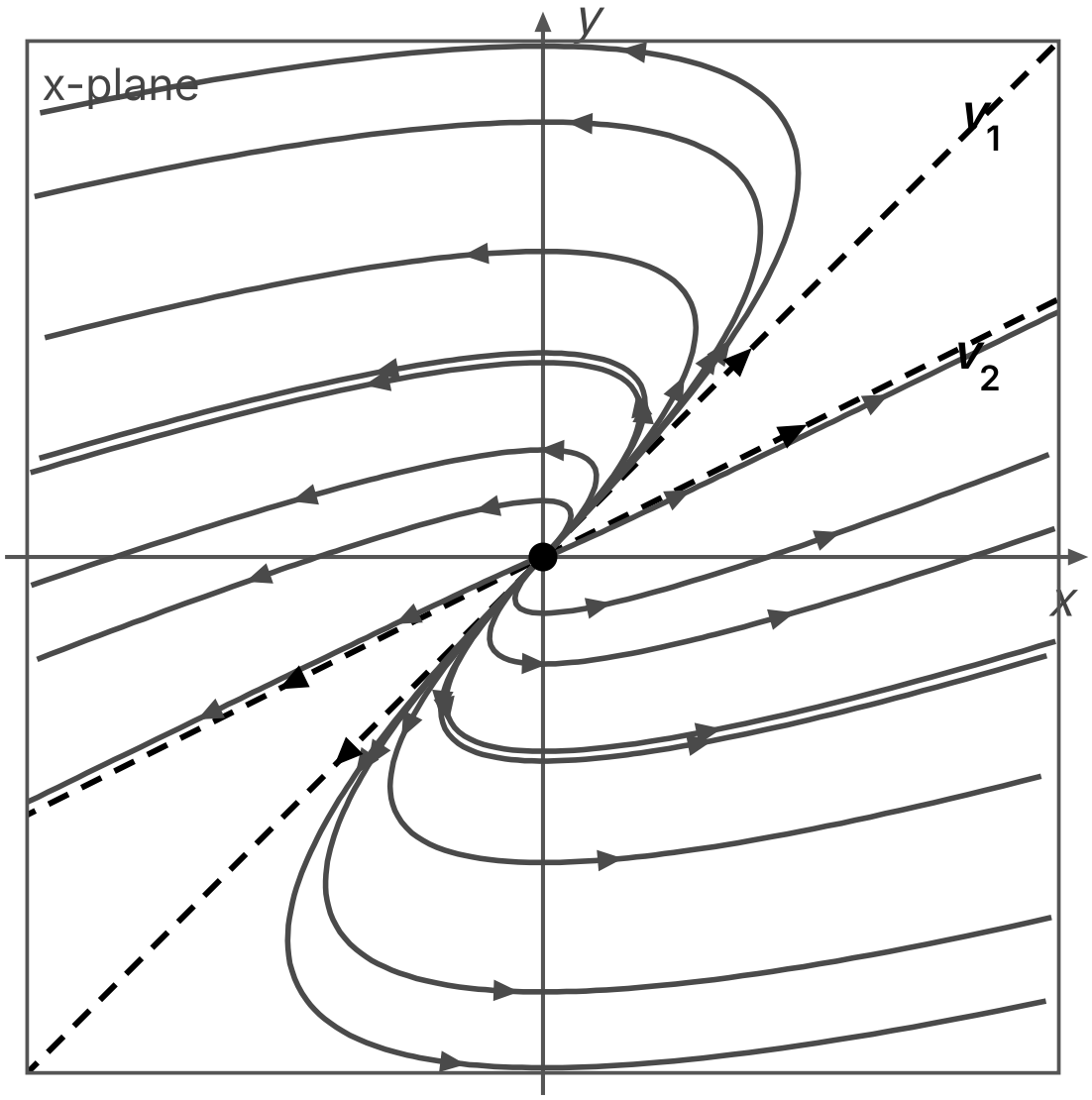
In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{1t}$ and $z_2 = c_2 e^{2t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^t = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2|\,(|z_1|/|c_1|)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K\,|z_1|^{\lambda_2/\lambda_1} = K\,|z_1|^2$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

The exponent $\lambda_2/\lambda_1 = 2$ is positive, so the orbits are parabola-like curves through the origin. At the origin they are tangent to the eigenvector of the eigenvalue of **smaller modulus**, $|\lambda| = 1$, i.e. to the line through $(1, 1)$ (they leave from there (the limit $t \to -\infty$)), and far from the origin they become parallel to the other eigenvector (the fast direction). The arrows point away from the origin on both eigenlines.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (1, 1)$ and the z_2-axis to the line through $\mathbf{v}_2 = (2, 1)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = 1\mathbf{v}_1$ points outward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = 2\mathbf{v}_2$ points outward.



Canonical $\mathbf{z}$-plane (unstable node): invariant axes and orbits $z_2 = K|z_1|^2$.



Original (x, y)-plane (unstable node): dashed lines are the eigenlines $\mathbf{v}_1 = (1, 1)$ and $\mathbf{v}_2 = (2, 1)$.

(viii) $\lambda_1 = -3,\ \mathbf{v}_1 = (1, 3)^T;\ \lambda_2 = -1,\ \mathbf{v}_2 = (-3, 2)^T$

The statement gives the eigen-data, not the matrix. Since $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$ for $j = 1, 2$ means $AQ = Q\Lambda$ with $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, the matrix is $A = Q\Lambda Q^{-1}$ (it is needed to draw the portrait and to check the data):

$$Q = \begin{pmatrix} 1 & -3 \\ 3 & 2 \end{pmatrix}, \qquad \Lambda = \begin{pmatrix} -3 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \det Q = 11, \qquad Q^{-1} = \begin{pmatrix} \frac{2}{11} & \frac{3}{11} \\ -\frac{3}{11} & \frac{1}{11} \end{pmatrix}$$

$$
\begin{aligned}
(Q\Lambda)_{11} &= 1\cdot(-3) + (-3)\cdot 0 = -3 \\
(Q\Lambda)_{12} &= 1\cdot 0 + (-3)\cdot(-1) = 3 \\
(Q\Lambda)_{21} &= 3\cdot(-3) + 2\cdot 0 = -9 \\
(Q\Lambda)_{22} &= 3\cdot 0 + 2\cdot(-1) = -2
\end{aligned}
$$

$$
Q\Lambda = \begin{pmatrix} -3 & 3 \\ -9 & -2 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= (-3)\cdot\frac{2}{11} + 3\cdot(-\frac{3}{11}) = -\frac{15}{11} \\
(A)_{12} &= (-3)\cdot\frac{3}{11} + 3\cdot\frac{1}{11} = -\frac{6}{11} \\
(A)_{21} &= (-9)\cdot\frac{2}{11} + (-2)\cdot(-\frac{3}{11}) = -\frac{12}{11} \\
(A)_{22} &= (-9)\cdot\frac{3}{11} + (-2)\cdot\frac{1}{11} = -\frac{29}{11}
\end{aligned}
$$

$$
A = Q\Lambda Q^{-1} = \begin{pmatrix} -\frac{15}{11} & -\frac{6}{11} \\ -\frac{12}{11} & -\frac{29}{11} \end{pmatrix}
$$

Change of coordinates

The eigenvectors of distinct eigenvalues are linearly independent (Topic 4, Appendix B argument), so they form a basis. Put them as columns, $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, and define the new coordinates by $\mathbf{z} = Q^{-1}\mathbf{x}$, i.e. $\mathbf{x} = Q\mathbf{z}$:

$$
Q = \begin{pmatrix} 1 & -3 \\ 3 & 2 \end{pmatrix}
$$

$\det Q = (1)(2) - (-3)(3) = 11$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$
Q^{-1} = \frac{1}{11}\begin{pmatrix} 2 & 3 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} \frac{2}{11} & \frac{3}{11} \\ -\frac{3}{11} & \frac{1}{11} \end{pmatrix}
$$

Since Q is constant, $\dot{\mathbf{z}} = Q^{-1}\dot{\mathbf{x}} = Q^{-1}A\mathbf{x} = (Q^{-1}AQ)\mathbf{z}$. Compute AQ one column at a time, with $A\mathbf{v}_j = \lambda_j\mathbf{v}_j$:

$$A\mathbf{v}_1 = \begin{pmatrix} -\frac{15}{11} & -\frac{6}{11} \\ -\frac{12}{11} & -\frac{29}{11} \end{pmatrix}\begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} -3 \\ -9 \end{pmatrix} = (-3)\mathbf{v}_1, \qquad A\mathbf{v}_2 = \begin{pmatrix} -\frac{15}{11} & -\frac{6}{11} \\ -\frac{12}{11} & -\frac{29}{11} \end{pmatrix}\begin{pmatrix} -3 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} = (-1)\mathbf{v}_2$$

$$AQ = [A\mathbf{v}_1 \mid A\mathbf{v}_2] = \begin{pmatrix} -3 & 3 \\ -9 & -2 \end{pmatrix}$$

Now multiply by Q^{-1} on the left, entry by entry:

$$\begin{aligned} (Q^{-1}AQ)_{11} &= \frac{2}{11}\cdot(-3) + \frac{3}{11}\cdot(-9) = -3 \\ (Q^{-1}AQ)_{12} &= \frac{2}{11}\cdot 3 + \frac{3}{11}\cdot(-2) = 0 \\ (Q^{-1}AQ)_{21} &= (-\frac{3}{11})\cdot(-3) + \frac{1}{11}\cdot(-9) = 0 \\ (Q^{-1}AQ)_{22} &= (-\frac{3}{11})\cdot 3 + \frac{1}{11}\cdot(-2) = -1 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} -3 & 0 \\ 0 & -1 \end{pmatrix} = \Lambda = \begin{pmatrix} -3 & 0 \\ 0 & -1 \end{pmatrix}$$

The transformation has decoupled the system (the matrix is diagonal), as claimed.

Solution in the canonical plane

$$\dot{z}_1 = -3z_1, \qquad \dot{z}_2 = -z_2$$

Each equation is separable. If $z_i \neq 0$, $\frac{dz_i}{z_i} = \lambda_i\,dt$ and integrating, $\ln|z_i| = \lambda_i t + C_i$, so $z_i = c_i e^{\lambda_i t}$ with $c_i = \pm e^{C_i}$ (the solution $z_i \equiv 0$ corresponds to $c_i = 0$). Equivalently, $\frac{d}{dt}(e^{-\lambda_i t} z_i) = 0$.

$$z_1(t) = c_1 e^{-3t}, \qquad z_2(t) = c_2 e^{-1t}$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{v}_1 + z_2\mathbf{v}_2$, so the general solution is

$$\mathbf{x}(t) = c_1 e^{-3t}\begin{pmatrix}1\\3\end{pmatrix} + c_2 e^{-1t}\begin{pmatrix}-3\\2\end{pmatrix}$$

$$x(t) = c_1 e^{-3t} - 3c_2 e^{-t}, \qquad y(t) = 3c_1 e^{-3t} + 2c_2 e^{-t}$$

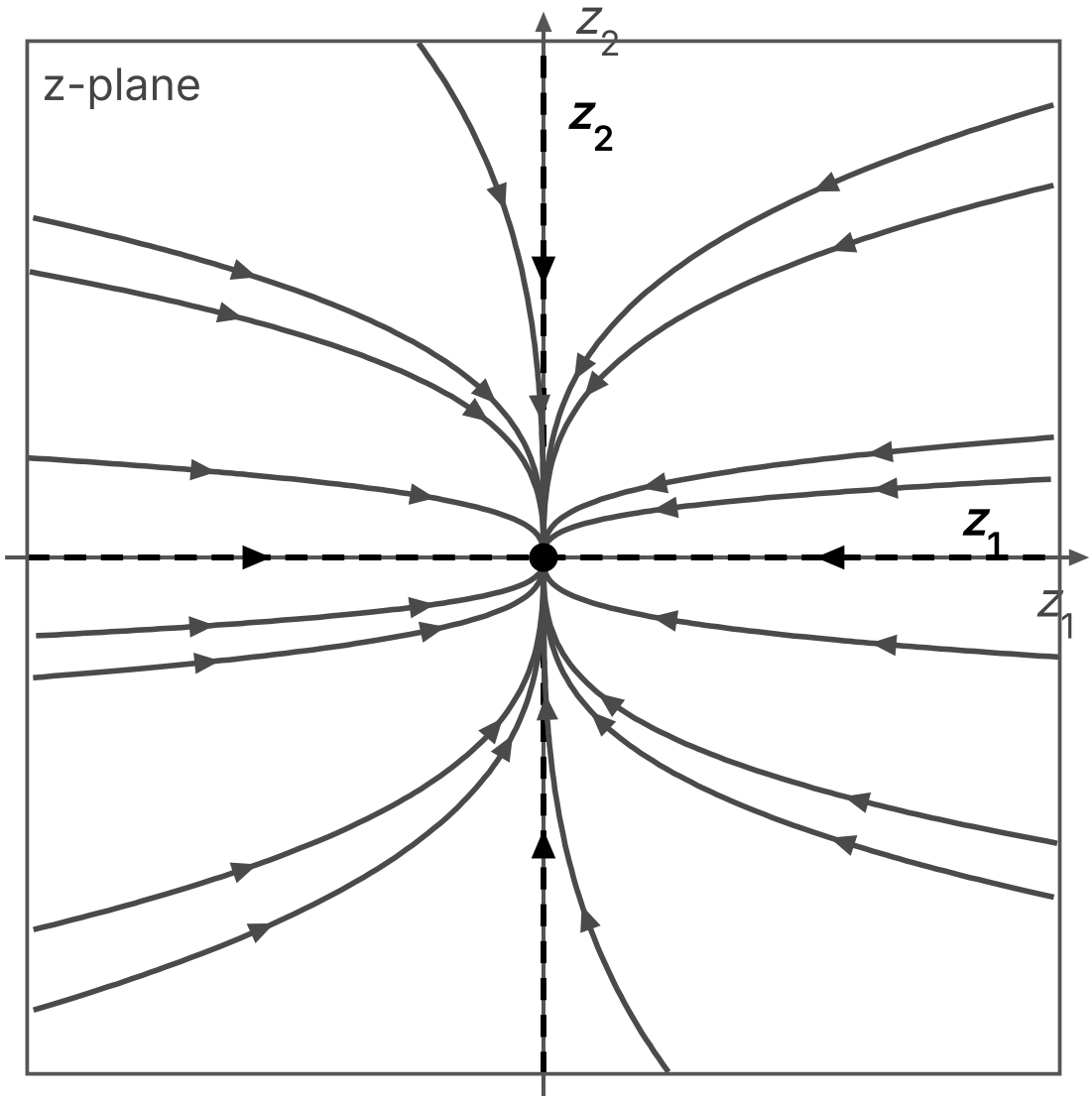
Phase portrait

$\lambda_1 = -3$ and $\lambda_2 = -1$ are both negative: the origin is a **stable node** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

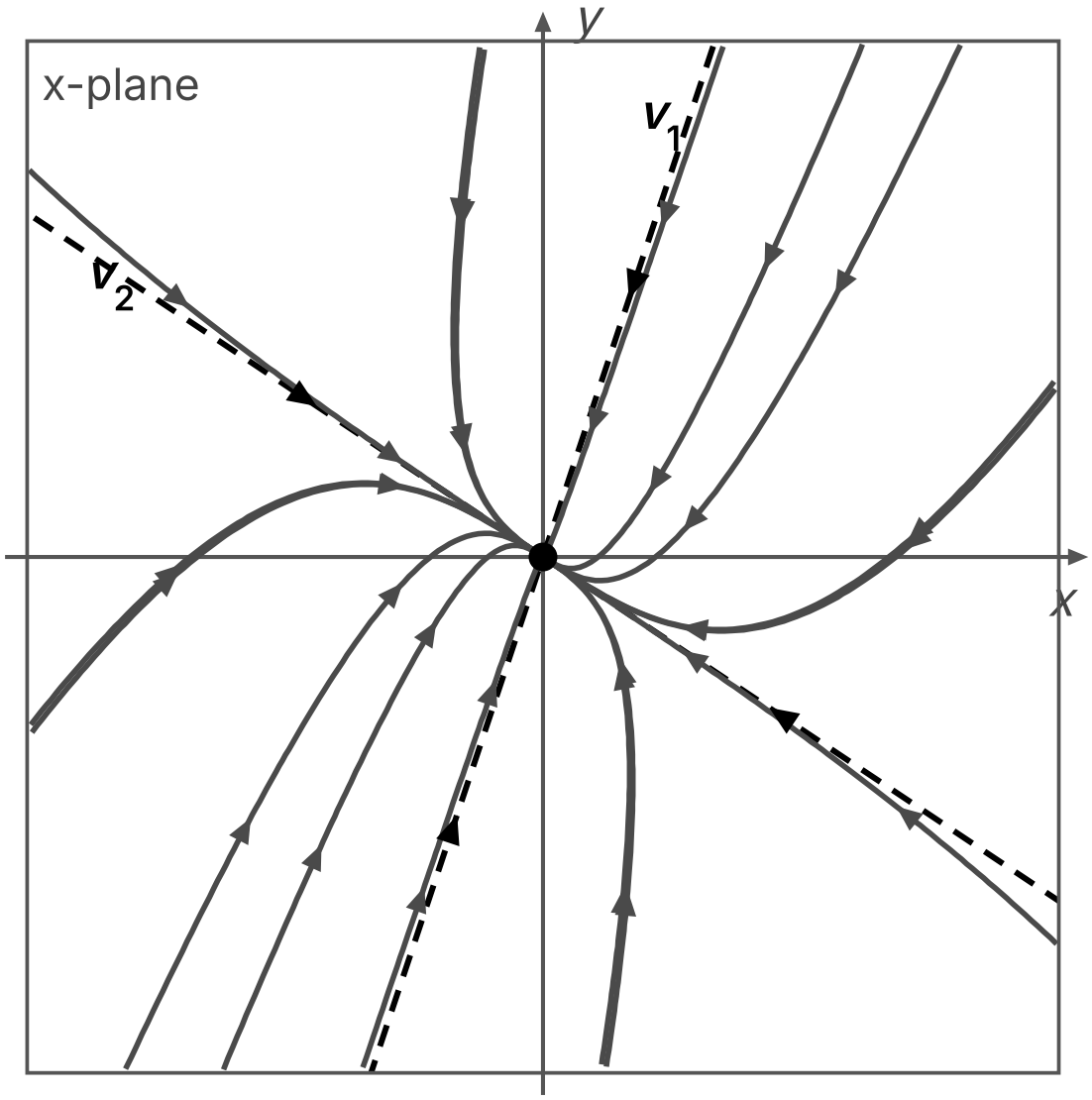
In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{-3t}$ and $z_2 = c_2 e^{-1t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^t = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2|\,(|z_1|/|c_1|)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K\,|z_1|^{\lambda_2/\lambda_1} = K\,|z_1|^{\frac{1}{3}}$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

The exponent $\lambda_2/\lambda_1 = \frac{1}{3}$ is positive, so the orbits are parabola-like curves through the origin. At the origin they are tangent to the eigenvector of the eigenvalue of **smaller modulus**, $|\lambda| = 1$, i.e. to the line through $(-3, 2)$ (they arrive there as $t \to +\infty$), and far from the origin they become parallel to the other eigenvector (the fast direction). The arrows point towards the origin on both eigenlines.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (1, 3)$ and the z_2-axis to the line through $\mathbf{v}_2 = (-3, 2)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = -3\mathbf{v}_1$ points inward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = -1\mathbf{v}_2$ points inward.



Canonical $\mathbf{z}$-plane (stable node): invariant axes and orbits $z_2 = K|z_1|^{\frac{1}{3}}$.



Original (x, y)-plane (stable node): dashed lines are the eigenlines $\mathbf{v}_1 = (1, 3)$ and $\mathbf{v}_2 = (-3, 2)$.

INTERPRETATION AND CHECKS

Item	General solution
(i)	$\mathbf{x} = c_1 e^{1t}\begin{pmatrix}1\\1\end{pmatrix} + c_2 e^{2t}\begin{pmatrix}1\\-1\end{pmatrix}$
(ii)	$\mathbf{x} = c_1 e^{1t}\begin{pmatrix}1\\0\end{pmatrix} + c_2 e^{-2t}\begin{pmatrix}1\\1\end{pmatrix}$
(iii)	$\mathbf{x} = c_1 e^{-2t}\begin{pmatrix}1\\2\end{pmatrix} + c_2 e^{-3t}\begin{pmatrix}2\\-3\end{pmatrix}$
(iv)	$\mathbf{x} = c_1 e^{3t}\begin{pmatrix}2\\3\end{pmatrix} + c_2 e^{-5t}\begin{pmatrix}0\\1\end{pmatrix}$

Item	General solution
(v)	$\mathbf{x} = c_1 e^{3t}\begin{pmatrix}1\\2\end{pmatrix} + c_2 e^{1t}\begin{pmatrix}1\\-3\end{pmatrix}$
(vi)	$\mathbf{x} = c_1 e^{2t}\begin{pmatrix}0\\1\end{pmatrix} + c_2 e^{-3t}\begin{pmatrix}1\\5\end{pmatrix}$
(vii)	$\mathbf{x} = c_1 e^{1t}\begin{pmatrix}1\\1\end{pmatrix} + c_2 e^{2t}\begin{pmatrix}2\\1\end{pmatrix}$
(viii)	$\mathbf{x} = c_1 e^{-3t}\begin{pmatrix}1\\3\end{pmatrix} + c_2 e^{-1t}\begin{pmatrix}-3\\2\end{pmatrix}$

- In every item the reconstructed A satisfies $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$ (checked in the text) and $\det A = \lambda_1\lambda_2$, $\operatorname{tr} A = \lambda_1 + \lambda_2$.
- Nodes: the tangent direction at the origin is the eigenvector of the **smaller** $|\lambda|$ because $|z_2|/|z_1|^{\lambda_2/\lambda_1}$ is constant: if $|\lambda_2| > |\lambda_1|$ the exponent $r = \lambda_2/\lambda_1$ exceeds 1, so $z_2 = K|z_1|^r$ tends to 0 faster than z_1 and the curve hugs the z_1-axis (the slower eigendirection) near the origin; if $r < 1$ it hugs the z_2-axis, which is then the slower one. The phase portraits drawn above show this numerically (curves integrated from the matrix A itself, not from the formula, so the tangency is a genuine check).
- Saddles (ii), (iv), (vi): the stable manifold is the eigenline of the negative eigenvalue.
- Items (ii), (iv) and (vi) have an eigenvector with a zero entry ((ii) $\mathbf{v}_1 = (1,0)$, (iv) $\mathbf{v}_2 = (0,1)$, (vi) $\mathbf{v}_1 = (0,1)$): that eigenline is an axis of the (x,y)-plane.

EXERCISE 9

Decoupling Four Systems with Real Eigenvalues

Original statement · probls_ch4-i_2627_v2.pdf, p. 3

Diagonalisation • Decoupling • Four Matrices

STATEMENT

Find the eigenvalues and eigenvectors of the matrix on the right-hand side in each case, and hence find the coordinate transformation that will decouple the corresponding equations. Show that this transformation has the desired effect. You can also write down the general solution and draw the phase portrait for the equation.

- (i) $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} 8 & 14 \\ 7 & 1 \end{pmatrix} \mathbf{x}$;
- (ii) $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} 2 & 0 \\ -5 & -3 \end{pmatrix} \mathbf{x}$;
- (iii) $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} 11 & -2 \\ 3 & 4 \end{pmatrix} \mathbf{x}$;
- (iv) $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} 1 & 20 \\ 40 & -19 \end{pmatrix} \mathbf{x}$.

SETUP

Each system is $\dot{\mathbf{x}} = A\mathbf{x}$ with a constant 2×2 matrix. Plan (Topic 4 Sec. 4 and 5): (a) characteristic equation $\det(A - \lambda I) = 0$; (b) eigenvectors from $(A - \lambda I)\mathbf{v} = \mathbf{0}$; (c) if the eigenvalues are real and distinct the eigenvectors are independent and $Q = [\mathbf{v}_1\ \mathbf{v}_2]$ defines $\mathbf{z} = Q^{-1}\mathbf{x}$ with $\dot{\mathbf{z}} = Q^{-1}AQ\,\mathbf{z}$; (d) show $Q^{-1}AQ = \Lambda$ by explicit multiplication.

FORMULATION

Since $\mathbf{x} = Q\mathbf{z}$ with Q constant, $\dot{\mathbf{x}} = Q\dot{\mathbf{z}}$ and $A\mathbf{x} = AQ\mathbf{z}$, so $\dot{\mathbf{z}} = Q^{-1}AQ\,\mathbf{z}$. If $Q^{-1}AQ = \Lambda = \operatorname{diag}(\lambda_1, \lambda_2)$ the equations $\dot{z}_1 = \lambda_1 z_1$, $\dot{z}_2 = \lambda_2 z_2$ are uncoupled.

METHOD OF THE CLASS NOTES [NOTES, P. 9]

The diagonalisation of the notes: $A = PDP^{-1}$ with $P = (\mathbf{v}_1, \mathbf{v}_2)$ and $D = \operatorname{diag}(\lambda_1, \lambda_2)$; with $\mathbf{z} = P^{-1}\mathbf{x}$, $\dot{\mathbf{z}} = P^{-1}A\mathbf{x} = DP^{-1}\mathbf{x} = D\mathbf{z}$, so $z_1 = C_1 e^{\lambda_1 t}$ and $z_2 = C_2 e^{\lambda_2 t}$ are uncoupled. The Q and Λ of this page are the notes' P and D. Item (i) is the matrix of Exercise 7 (vii): elimination and eigenvalues give the same characteristic polynomial, $\lambda^2 - 9\lambda - 90$.

DERIVATION

(i) $A = \begin{pmatrix} 8 & 14 \\ 7 & 1 \end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix} 8-\lambda & 14 \\ 7 & 1-\lambda \end{vmatrix} = (8-\lambda)(1-\lambda) - (14)(7)$$

$$= \lambda^2 - (8+1)\lambda + (8\cdot 1 - 14\cdot 7) = \lambda^2 - (9)\lambda + (-90) = \lambda^2 - 9\lambda - 90$$

$\operatorname{tr} A = 9$, $\det A = -90$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4\det A = 81 - (-360) = 441$.

$$\lambda = \frac{9 \pm \sqrt{441}}{2} = \frac{9 \pm 21}{2} \quad \Longrightarrow \quad \lambda_1 = 15,\ \lambda_2 = -6$$

Check: $\lambda_1 + \lambda_2 = 9 = \operatorname{tr} A$ and $\lambda_1\lambda_2 = -90 = \det A$.

Eigenvectors

For $\lambda = 15$: $(A - (15)I)\mathbf{v_1} = \mathbf{0}$, that is

$$\begin{pmatrix} -7 & 14 \\ 7 & -14 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} -7v_1 + 14v_2 = 0 \quad &\text{(row 1)} \\ 7v_1 - 14v_2 = 0 \quad &\text{(row 2)} \end{aligned}$$

The coefficient matrix is singular ($\det(A - \lambda I) = 0$), so the rows are proportional (row 2 equals -1 times row 1) and express a single condition. Row 1 reads $-7v_1 + 14v_2 = 0$, so $v_1 = 2\,v_2$: every eigenvector is a multiple of $(2, 1)^T$. The smallest integer representative is

$$\mathbf{v_1} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

Check: $A\mathbf{v_1} = \begin{pmatrix} 30 \\ 15 \end{pmatrix} = 15\mathbf{v_1}$.

For $\lambda = -6$: $(A - (-6)I)\mathbf{v_2} = \mathbf{0}$, that is

$$\begin{pmatrix} 14 & 14 \\ 7 & 7 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} 14v_1 + 14v_2 = 0 \quad &\text{(row 1)} \\ 7v_1 + 7v_2 = 0 \quad &\text{(row 2)} \end{aligned}$$

The coefficient matrix is singular ($\det(A - \lambda I) = 0$), so the rows are proportional (row 2 equals $\frac{1}{2}$ times row 1) and express a single condition. Row 1 reads $14v_1 + 14v_2 = 0$, so $v_1 = -v_2$: every eigenvector is a multiple of $(-1, 1)^T$. The smallest integer representative is

$$\mathbf{v_2} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Check: $A\mathbf{v_2} = \begin{pmatrix} -6 \\ 6 \end{pmatrix} = -6\mathbf{v_2}$.

Change of coordinates

The eigenvectors of distinct eigenvalues are linearly independent (Topic 4, Appendix B argument), so they form a basis. Put them as columns, $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, and define the new coordinates by $\mathbf{z} = Q^{-1}\mathbf{x}$, i.e. $\mathbf{x} = Q\mathbf{z}$:

$$Q = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$$

$\det Q = (2)(-1) - (1)(1) = -3$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-3}\begin{pmatrix} -1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{pmatrix}$$

Since Q is constant, $\dot{\mathbf{z}} = Q^{-1}\dot{\mathbf{x}} = Q^{-1}A\mathbf{x} = (Q^{-1}AQ)\mathbf{z}$. Compute AQ one column at a time, with $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$:

$$A\mathbf{v}_1 = \begin{pmatrix} 8 & 14 \\ 7 & 1 \end{pmatrix}\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 30 \\ 15 \end{pmatrix} = 15\mathbf{v}_1, \qquad A\mathbf{v}_2 = \begin{pmatrix} 8 & 14 \\ 7 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -6 \\ 6 \end{pmatrix} = (-6)\mathbf{v}_2$$

$$AQ = [A\mathbf{v}_1 \mid A\mathbf{v}_2] = \begin{pmatrix} 30 & -6 \\ 15 & 6 \end{pmatrix}$$

Now multiply by Q^{-1} on the left, entry by entry:

$$\begin{aligned} (Q^{-1}AQ)_{11} &= \frac{1}{3}\cdot 30 + \frac{1}{3}\cdot 15 = 15 \\ (Q^{-1}AQ)_{12} &= \frac{1}{3}\cdot (-6) + \frac{1}{3}\cdot 6 = 0 \\ (Q^{-1}AQ)_{21} &= \frac{1}{3}\cdot 30 + (-\frac{2}{3})\cdot 15 = 0 \\ (Q^{-1}AQ)_{22} &= \frac{1}{3}\cdot (-6) + (-\frac{2}{3})\cdot 6 = -6 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 15 & 0 \\ 0 & -6 \end{pmatrix} = \Lambda = \begin{pmatrix} 15 & 0 \\ 0 & -6 \end{pmatrix}$$

The transformation has decoupled the system (the matrix is diagonal), as claimed.

Solution in the canonical plane

$$\dot{z}_1 = 15 z_1, \qquad \dot{z}_2 = -6 z_2$$

Each equation is separable. If $z_i \neq 0$, $\frac{dz_i}{z_i} = \lambda_i \, dt$ and integrating, $\ln|z_i| = \lambda_i t + C_i$, so $z_i = c_i e^{\lambda_i t}$ with $c_i = \pm e^{C_i}$ (the solution $z_i \equiv 0$ corresponds to $c_i = 0$). Equivalently, $\frac{d}{dt}(e^{-\lambda_i t} z_i) = 0$.

$$z_1(t) = c_1 e^{15t}, \qquad z_2(t) = c_2 e^{-6t}$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1 \mathbf{v}_1 + z_2 \mathbf{v}_2$, so the general solution is

$$\mathbf{x}(t) = c_1 e^{15t} \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 e^{-6t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$x(t) = 2c_1 e^{15t} + c_2 e^{-6t}, \qquad y(t) = c_1 e^{15t} - c_2 e^{-6t}$$

The decoupling transformation

In the original variables $\mathbf{x} = (x, y)^T$ the new coordinates are $\mathbf{z} = Q^{-1}\mathbf{x}$:

$$z_1 = \frac{x}{3} + \frac{y}{3}, \qquad z_2 = \frac{x}{3} - \frac{2y}{3}$$

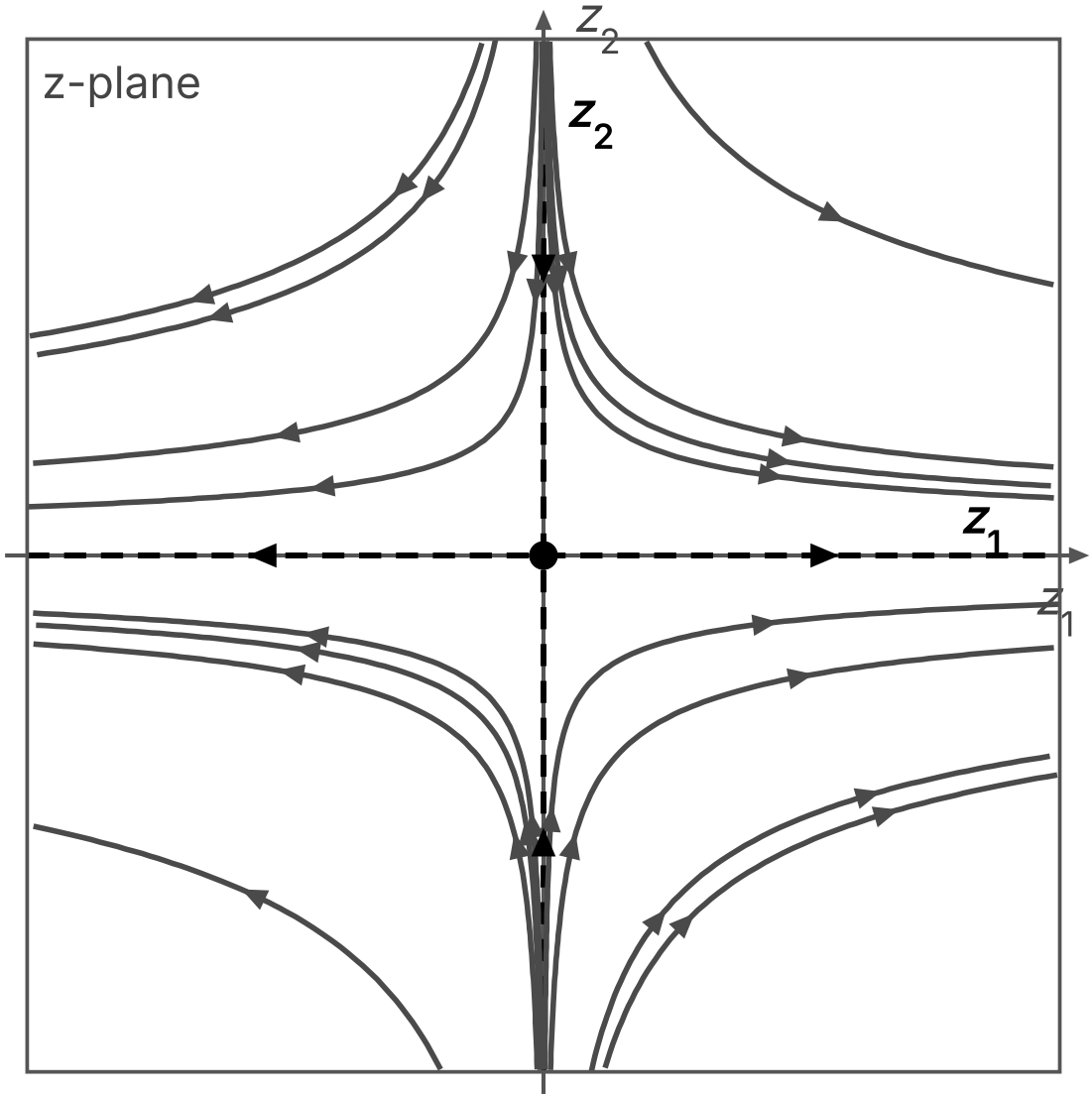
Phase portrait

$\lambda_1 = 15$ and $\lambda_2 = -6$ are of opposite signs: the origin is a **saddle** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

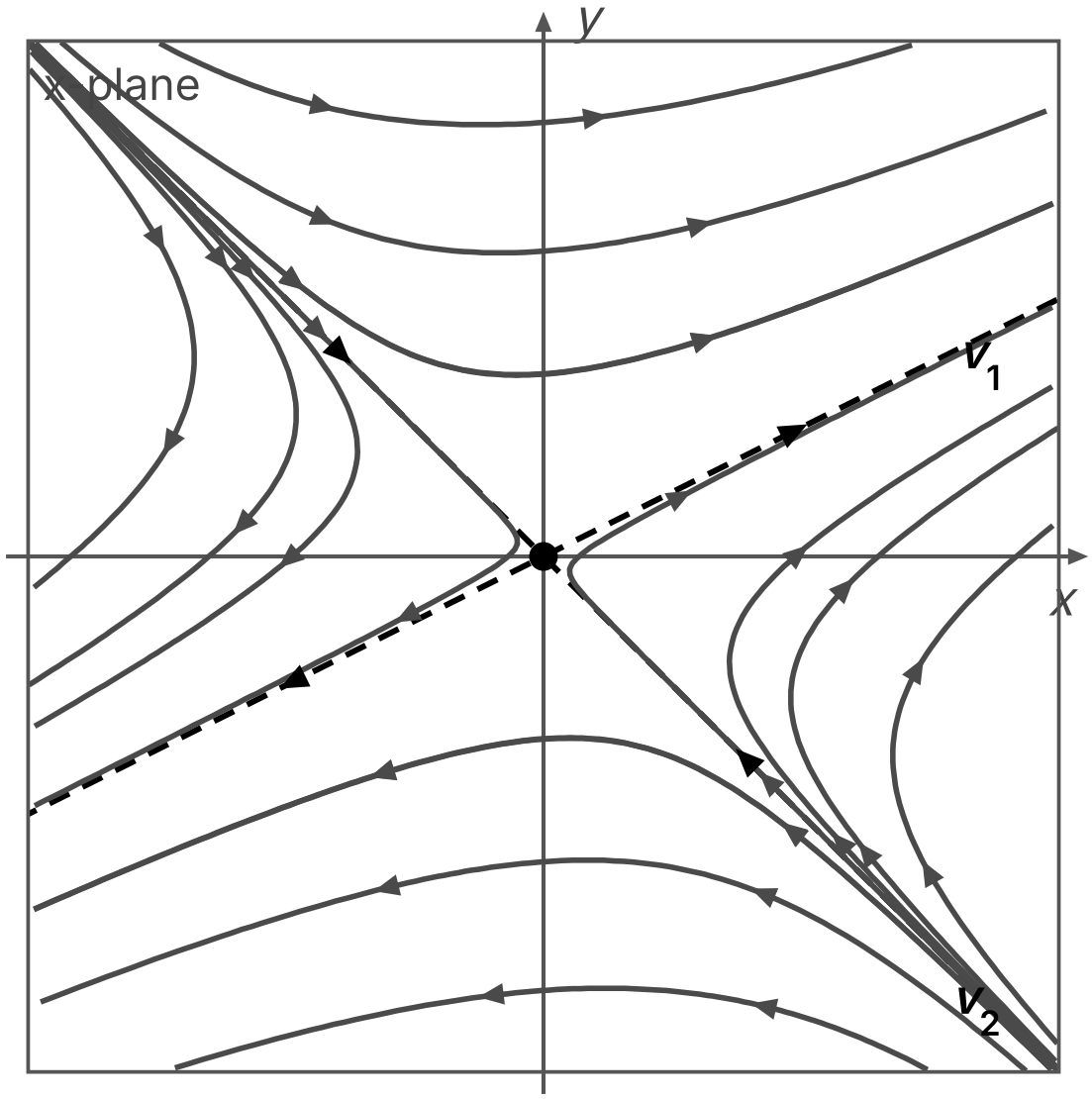
In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{15t}$ and $z_2 = c_2 e^{-6t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^t = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2| \, (|z_1|/|c_1|)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K \, |z_1|^{\lambda_2/\lambda_1} = K \, |z_1|^{-\frac{2}{5}}$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

The exponent $\lambda_2/\lambda_1 = -\frac{2}{5}$ is negative, so the orbits are hyperbola-like. Only solutions starting on the eigenline of the negative eigenvalue, the line through $(1, -1)$ (the **stable manifold**), reach the origin as $t \to +\infty$; the line through $(2, 1)$ (positive eigenvalue) is the **unstable manifold**. All other orbits come in close to the stable line, then leave along the unstable one.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (2, 1)$ and the z_2-axis to the line through $\mathbf{v}_2 = (1, -1)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = 15\mathbf{v}_1$ points outward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = -6\mathbf{v}_2$ points inward.



Canonical $\mathbf{z}$-plane (saddle): invariant axes and orbits $z_2 = K|z_1|^{-\frac{2}{5}}$.



Original (x, y)-plane (saddle): dashed lines are the eigenlines $\mathbf{v}_1 = (2, 1)$ and $\mathbf{v}_2 = (1, -1)$.

(ii) $A = \begin{pmatrix} 2 & 0 \\ -5 & -3 \end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix} 2 - \lambda & 0 \\ -5 & -3 - \lambda \end{vmatrix} = (2 - \lambda)(-3 - \lambda) - (0)(-5)$$

$$= \lambda^2 - (2 + (-3))\lambda + (2 \cdot (-3) - 0 \cdot (-5)) = \lambda^2 - (-1)\lambda + (-6) = \lambda^2 + \lambda - 6$$

$\operatorname{tr} A = -1$, $\det A = -6$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4 \det A = 1 - (-24) = 25$.

$$\lambda = \frac{-1 \pm \sqrt{25}}{2} = \frac{-1 \pm 5}{2} \quad \Longrightarrow \quad \lambda_1 = 2,\ \lambda_2 = -3$$

Check: $\lambda_1 + \lambda_2 = -1 = \operatorname{tr} A$ and $\lambda_1 \lambda_2 = -6 = \det A$.

Eigenvectors

For $\lambda = 2$: $(A - (2)I)\mathbf{v_1} = \mathbf{0}$, that is

$$\begin{pmatrix} 0 & 0 \\ -5 & -5 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} 0v_1 + 0v_2 &= 0 \quad \text{(row 1)} \\ -5v_1 - 5v_2 &= 0 \quad \text{(row 2)} \end{aligned}$$

The coefficient matrix is singular ($\det(A - \lambda I) = 0$), so the rows are proportional (row 1 is identically zero) and express a single condition. Row 2 reads $-5v_1 + (-5)v_2 = 0$, so $v_1 = -v_2$: every eigenvector is a multiple of $(-1, 1)^T$. The smallest integer representative is

$$\mathbf{v_1} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Check: $A\mathbf{v_1} = \begin{pmatrix} 2 \\ -2 \end{pmatrix} = 2\mathbf{v_1}$.

For $\lambda = -3$: $(A - (-3)I)\mathbf{v_2} = \mathbf{0}$, that is

$$\begin{pmatrix} 5 & 0 \\ -5 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} 5v_1 + 0v_2 &= 0 \quad \text{(row 1)} \\ -5v_1 + 0v_2 &= 0 \quad \text{(row 2)} \end{aligned}$$

The coefficient matrix is singular ($\det(A - \lambda I) = 0$), so the rows are proportional (row 2 equals -1 times row 1) and express a single condition. Row 1 reads $5v_1 + 0v_2 = 0$, so $v_1 = 0$ and v_2 is free; the simplest choice is

$$\mathbf{v_2} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Check: $A\mathbf{v_2} = \begin{pmatrix} 0 \\ -3 \end{pmatrix} = -3\mathbf{v_2}$.

Change of coordinates

The eigenvectors of distinct eigenvalues are linearly independent (Topic 4, Appendix B argument), so they form a basis. Put them as columns, $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, and define the new coordinates by $\mathbf{z} = Q^{-1}\mathbf{x}$, i.e. $\mathbf{x} = Q\mathbf{z}$:

$$Q = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$$

$\det Q = (1)(1) - (0)(-1) = 1$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{1}\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

Since Q is constant, $\dot{\mathbf{z}} = Q^{-1}\dot{\mathbf{x}} = Q^{-1}A\mathbf{x} = (Q^{-1}AQ)\mathbf{z}$. Compute AQ one column at a time, with $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$:

$$A\mathbf{v}_1 = \begin{pmatrix} 2 & 0 \\ -5 & -3 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix} = 2\mathbf{v}_1, \qquad A\mathbf{v}_2 = \begin{pmatrix} 2 & 0 \\ -5 & -3 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -3 \end{pmatrix} = (-3)\mathbf{v}_2$$

$$AQ = [A\mathbf{v}_1 \mid A\mathbf{v}_2] = \begin{pmatrix} 2 & 0 \\ -2 & -3 \end{pmatrix}$$

Now multiply by Q^{-1} on the left, entry by entry:

$$\begin{aligned} (Q^{-1}AQ)_{11} &= 1 \cdot 2 + 0 \cdot (-2) = 2 \\ (Q^{-1}AQ)_{12} &= 1 \cdot 0 + 0 \cdot (-3) = 0 \\ (Q^{-1}AQ)_{21} &= 1 \cdot 2 + 1 \cdot (-2) = 0 \\ (Q^{-1}AQ)_{22} &= 1 \cdot 0 + 1 \cdot (-3) = -3 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix} = \Lambda = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}$$

The transformation has decoupled the system (the matrix is diagonal), as claimed.

Solution in the canonical plane

$$\dot{z}_1 = 2z_1, \qquad \dot{z}_2 = -3z_2$$

Each equation is separable. If $z_i \neq 0$, $\frac{dz_i}{z_i} = \lambda_i \, dt$ and integrating, $\ln|z_i| = \lambda_i t + C_i$, so $z_i = c_i e^{\lambda_i t}$ with $c_i = \pm e^{C_i}$ (the solution $z_i \equiv 0$ corresponds to $c_i = 0$). Equivalently, $\frac{d}{dt}(e^{-\lambda_i t} z_i) = 0$.

$$z_1(t) = c_1 e^{2t}, \qquad z_2(t) = c_2 e^{-3t}$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{v}_1 + z_2\mathbf{v}_2$, so the general solution is

$$\mathbf{x}(t) = c_1 e^{2t}\begin{pmatrix} 1 \\ -1 \end{pmatrix} + c_2 e^{-3t}\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$x(t) = c_1 e^{2t}, \qquad y(t) = -c_1 e^{2t} + c_2 e^{-3t}$$

The decoupling transformation

In the original variables $\mathbf{x} = (x, y)^T$ the new coordinates are $\mathbf{z} = Q^{-1}\mathbf{x}$:

$$z_1 = x, \qquad z_2 = x + y$$

Phase portrait

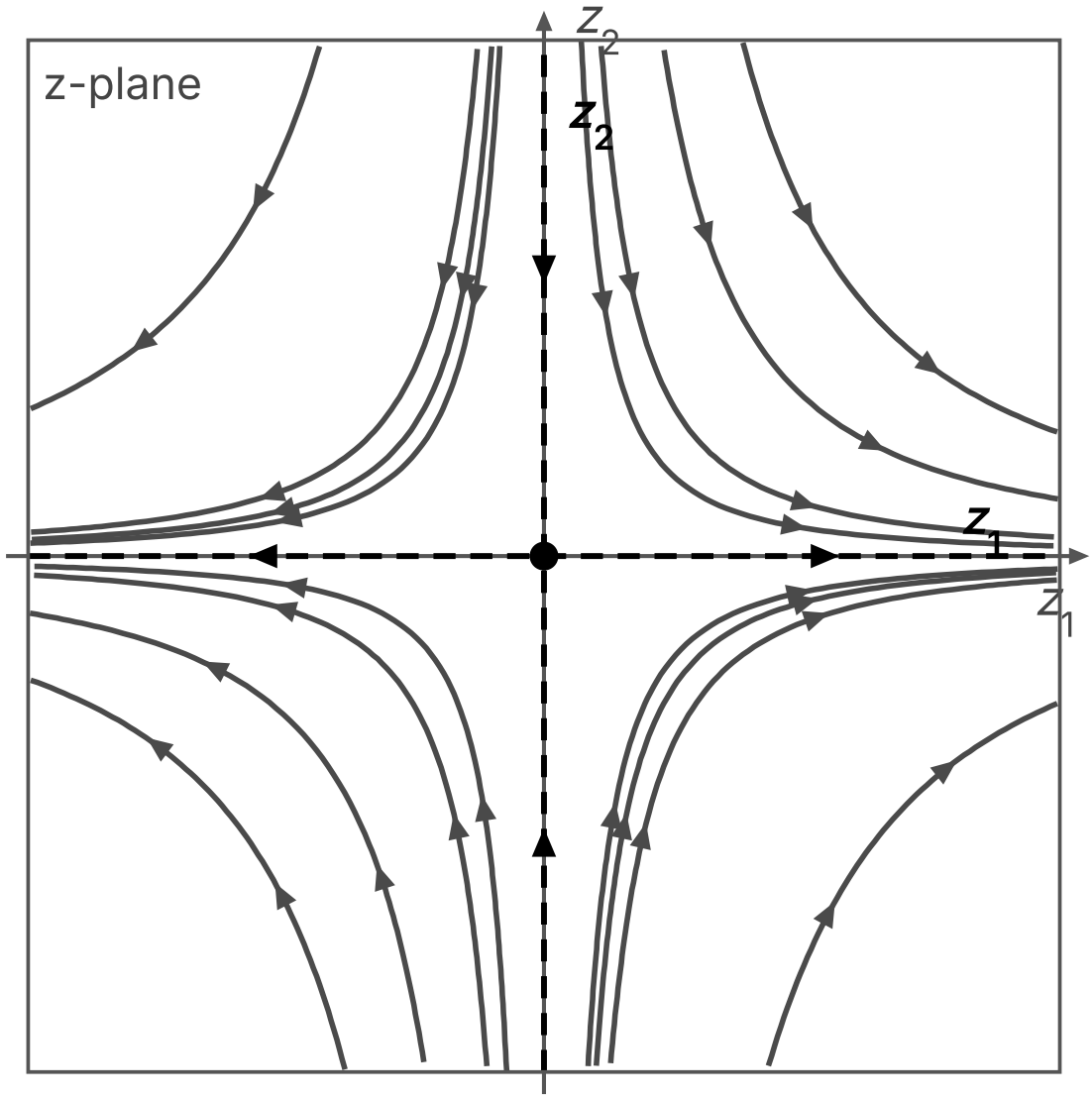
$\lambda_1 = 2$ and $\lambda_2 = -3$ are of opposite signs: the origin is a **saddle** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{2t}$ and $z_2 = c_2 e^{-3t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^t = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2| \, (|z_1|/|c_1|)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K\,|z_1|^{\lambda_2/\lambda_1} = K\,|z_1|^{-\frac{3}{2}}$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

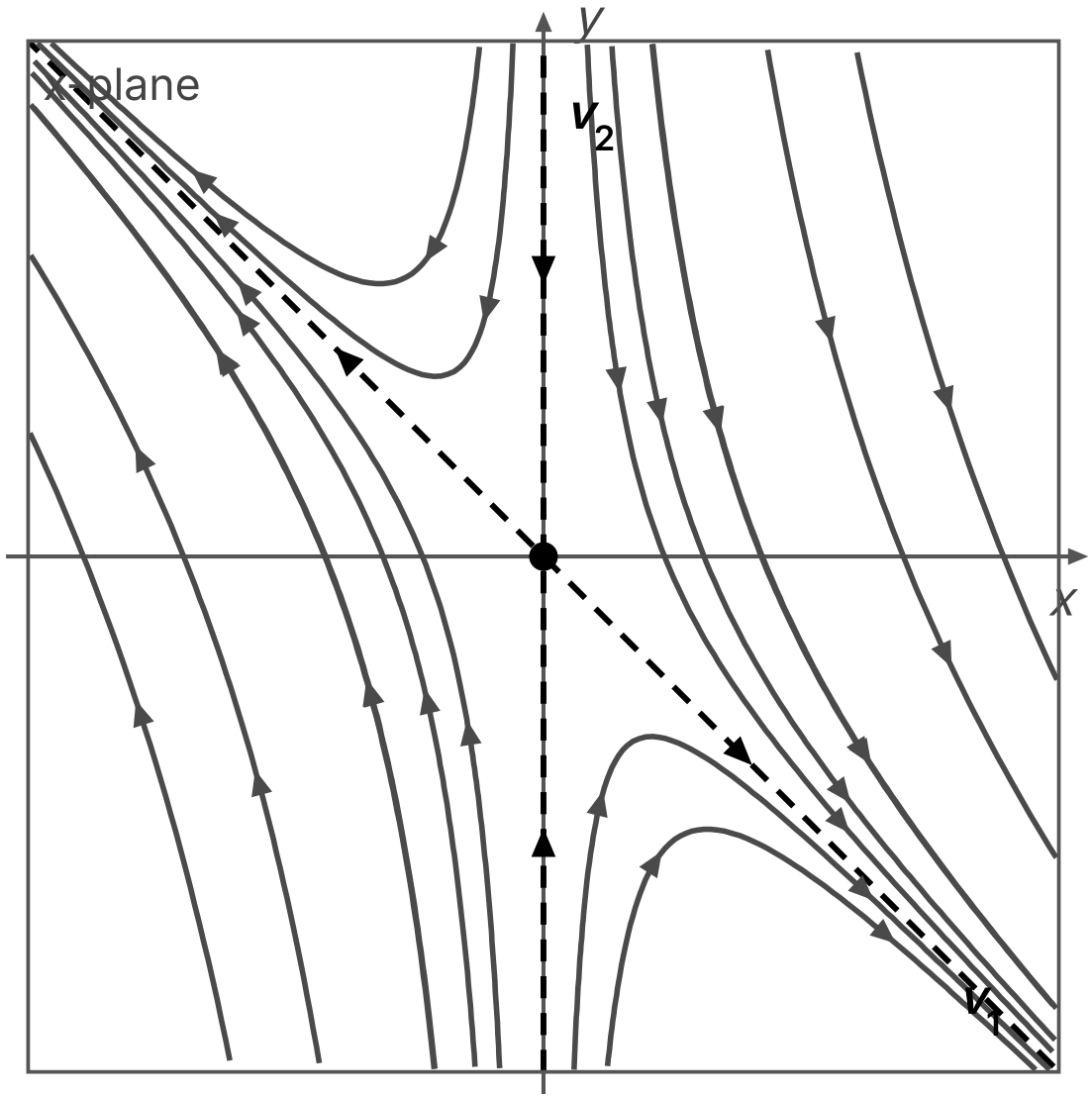
The exponent $\lambda_2/\lambda_1 = -\frac{3}{2}$ is negative, so the orbits are hyperbola-like. Only solutions starting on the eigenline of the negative eigenvalue, the line through $(0, 1)$ (the **stable manifold**), reach the origin as $t \to +\infty$; the line through $(1, -1)$ (positive eigenvalue) is the **unstable manifold**. All other

orbits come in close to the stable line, then leave along the unstable one.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (1, -1)$ and the z_2-axis to the line through $\mathbf{v}_2 = (0, 1)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = 2\mathbf{v}_1$ points outward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = -3\mathbf{v}_2$ points inward.



Canonical $\mathbf{z}$-plane (saddle): invariant axes and orbits $z_2 = K|z_1|^{-\frac{3}{2}}$.



Original (x, y)-plane (saddle): dashed lines are the eigenlines $\mathbf{v}_1 = (1, -1)$ and $\mathbf{v}_2 = (0, 1)$.

(iii) $A = \begin{pmatrix} 11 & -2 \\ 3 & 4 \end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix} 11 - \lambda & -2 \\ 3 & 4 - \lambda \end{vmatrix} = (11 - \lambda)(4 - \lambda) - (-2)(3)$$

$$= \lambda^2 - (11 + 4)\lambda + (11 \cdot 4 - (-2) \cdot 3) = \lambda^2 - (15)\lambda + (50) = \lambda^2 - 15\lambda + 50$$

$\operatorname{tr} A = 15$, $\det A = 50$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4 \det A = 225 - 200 = 25$.

$$\lambda = \frac{15 \pm \sqrt{25}}{2} = \frac{15 \pm 5}{2} \quad \Longrightarrow \quad \lambda_1 = 10,\ \lambda_2 = 5$$

Check: $\lambda_1 + \lambda_2 = 15 = \operatorname{tr} A$ and $\lambda_1 \lambda_2 = 50 = \det A$.

Eigenvectors

For $\lambda = 10$: $(A - (10)I)\mathbf{v_1} = \mathbf{0}$, that is

$$\begin{pmatrix} 1 & -2 \\ 3 & -6 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} 1v_1 - 2v_2 &= 0 \quad \text{(row 1)} \\ 3v_1 - 6v_2 &= 0 \quad \text{(row 2)} \end{aligned}$$

The coefficient matrix is singular ($\det(A - \lambda I) = 0$), so the rows are proportional (row 2 equals 3 times row 1) and express a single condition. Row 1 reads $1v_1 + (-2)v_2 = 0$, so $v_1 = 2\,v_2$: every eigenvector is a multiple of $(2, 1)^T$. The smallest integer representative is

$$\mathbf{v_1} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

Check: $A\mathbf{v_1} = \begin{pmatrix} 20 \\ 10 \end{pmatrix} = 10\mathbf{v_1}$.

For $\lambda = 5$: $(A - (5)I)\mathbf{v_2} = \mathbf{0}$, that is

$$\begin{pmatrix} 6 & -2 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} 6v_1 - 2v_2 &= 0 \quad \text{(row 1)} \\ 3v_1 - 1v_2 &= 0 \quad \text{(row 2)} \end{aligned}$$

The coefficient matrix is singular ($\det(A - \lambda I) = 0$), so the rows are proportional (row 2 equals $\frac{1}{2}$ times row 1) and express a single condition. Row 1 reads $6v_1 + (-2)v_2 = 0$, so $v_1 = \frac{1}{3}\,v_2$: every eigenvector is a multiple of $(\frac{1}{3}, 1)^T$. The smallest integer representative is

$$\mathbf{v_2} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

Check: $A\mathbf{v_2} = \begin{pmatrix} 5 \\ 15 \end{pmatrix} = 5\mathbf{v_2}$.

Change of coordinates

The eigenvectors of distinct eigenvalues are linearly independent (Topic 4, Appendix B argument), so they form a basis. Put them as columns, $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, and define the new coordinates by $\mathbf{z} = Q^{-1}\mathbf{x}$, i.e. $\mathbf{x} = Q\mathbf{z}$:

$$Q = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$$

$\det Q = (2)(3) - (1)(1) = 5$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{5}\begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} \frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{pmatrix}$$

Since Q is constant, $\dot{\mathbf{z}} = Q^{-1}\dot{\mathbf{x}} = Q^{-1}A\mathbf{x} = (Q^{-1}AQ)\mathbf{z}$. Compute AQ one column at a time, with $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$:

$$A\mathbf{v}_1 = \begin{pmatrix} 11 & -2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 20 \\ 10 \end{pmatrix} = 10\mathbf{v}_1, \qquad A\mathbf{v}_2 = \begin{pmatrix} 11 & -2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 15 \end{pmatrix} = 5\mathbf{v}_2$$

$$AQ = [A\mathbf{v}_1 \mid A\mathbf{v}_2] = \begin{pmatrix} 20 & 5 \\ 10 & 15 \end{pmatrix}$$

Now multiply by Q^{-1} on the left, entry by entry:

$$
\begin{aligned}
(Q^{-1}AQ)_{11} &= \frac{3}{5}\cdot 20 + (-\frac{1}{5})\cdot 10 = 10\\
(Q^{-1}AQ)_{12} &= \frac{3}{5}\cdot 5 + (-\frac{1}{5})\cdot 15 = 0\\
(Q^{-1}AQ)_{21} &= (-\frac{1}{5})\cdot 20 + \frac{2}{5}\cdot 10 = 0\\
(Q^{-1}AQ)_{22} &= (-\frac{1}{5})\cdot 5 + \frac{2}{5}\cdot 15 = 5
\end{aligned}
$$

$$
Q^{-1}AQ = \begin{pmatrix} 10 & 0 \\ 0 & 5 \end{pmatrix} = \Lambda = \begin{pmatrix} 10 & 0 \\ 0 & 5 \end{pmatrix}
$$

The transformation has decoupled the system (the matrix is diagonal), as claimed.

Solution in the canonical plane

$$
\dot{z}_1 = 10 z_1, \qquad \dot{z}_2 = 5 z_2
$$

Each equation is separable. If $z_i \neq 0$, $\frac{dz_i}{z_i} = \lambda_i \, dt$ and integrating, $\ln|z_i| = \lambda_i t + C_i$, so $z_i = c_i e^{\lambda_i t}$ with $c_i = \pm e^{C_i}$ (the solution $z_i \equiv 0$ corresponds to $c_i = 0$). Equivalently, $\frac{d}{dt}(e^{-\lambda_i t} z_i) = 0$.

$$
z_1(t) = c_1 e^{10t}, \qquad z_2(t) = c_2 e^{5t}
$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1 \mathbf{v}_1 + z_2 \mathbf{v}_2$, so the general solution is

$$
\mathbf{x}(t) = c_1 e^{10t} \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 e^{5t} \begin{pmatrix} 1 \\ 3 \end{pmatrix}
$$

$$
x(t) = 2c_1 e^{10t} + c_2 e^{5t}, \qquad y(t) = c_1 e^{10t} + 3c_2 e^{5t}
$$

The decoupling transformation

In the original variables $\mathbf{x} = (x, y)^T$ the new coordinates are $\mathbf{z} = Q^{-1}\mathbf{x}$:

$$z_1 = \frac{3x}{5} - \frac{y}{5}, \qquad z_2 = -\frac{x}{5} + \frac{2y}{5}$$

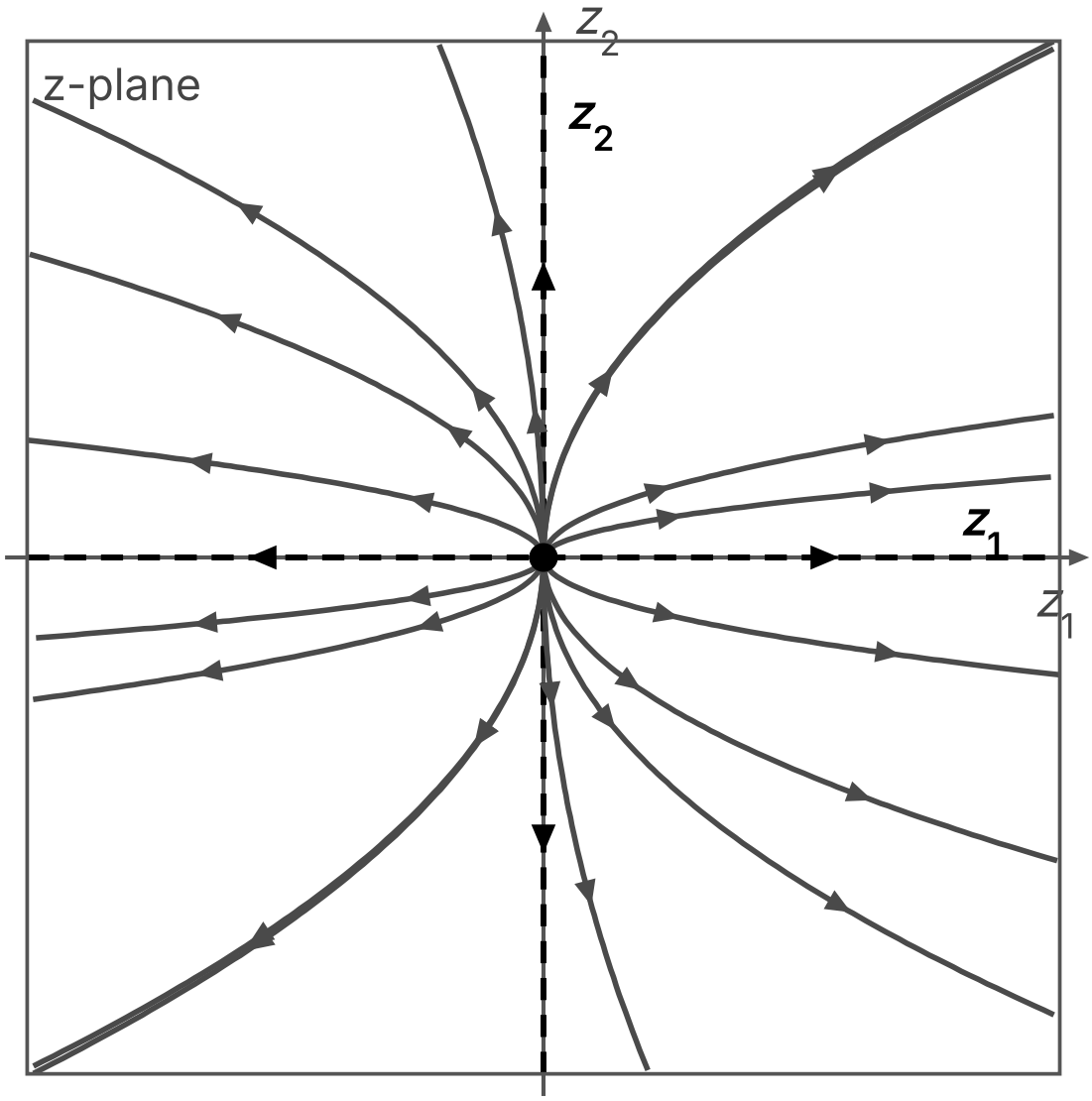
Phase portrait

$\lambda_1 = 10$ and $\lambda_2 = 5$ are both positive: the origin is a **unstable node** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

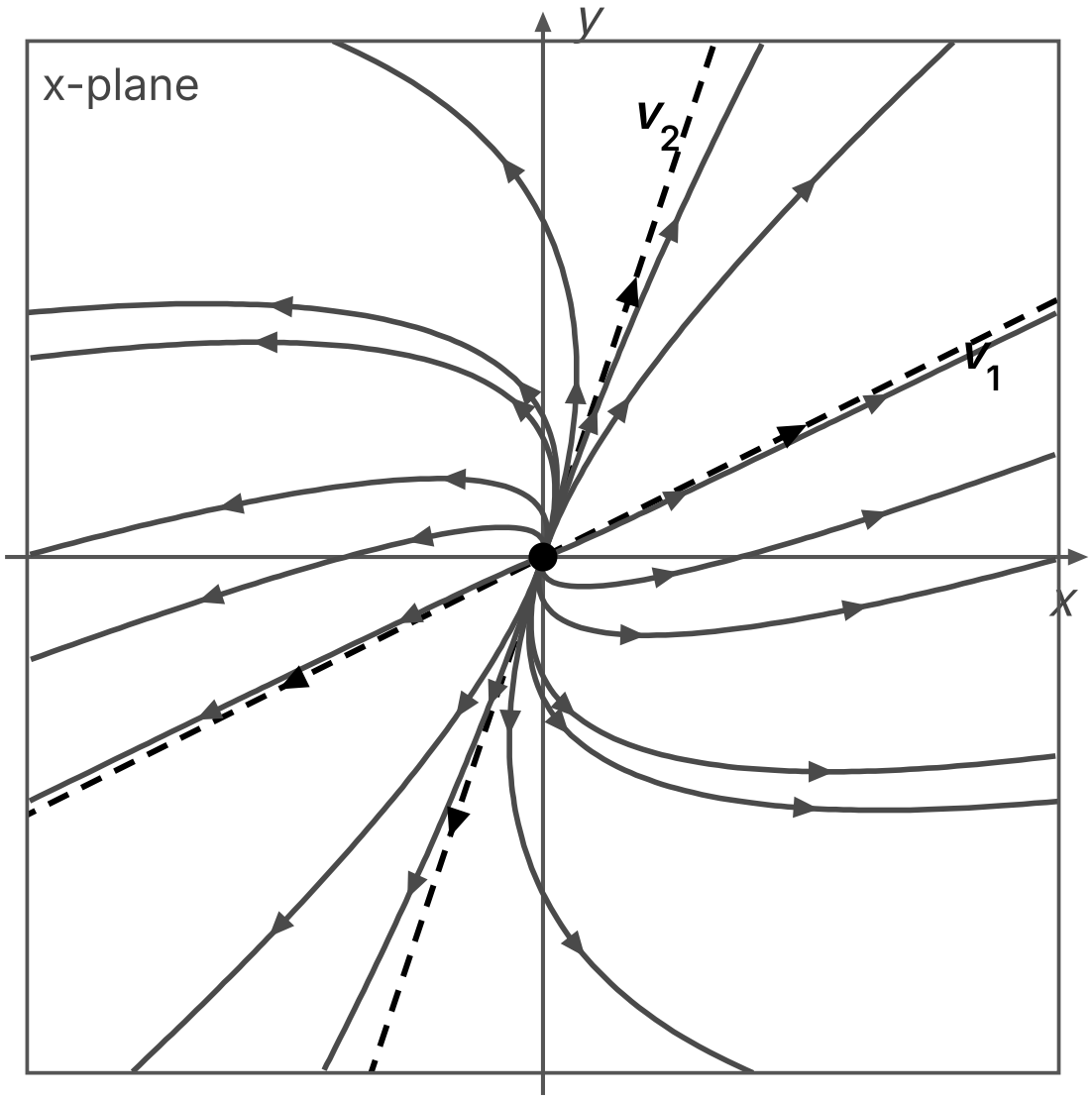
In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{10t}$ and $z_2 = c_2 e^{5t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^t = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2|\,(|z_1|/|c_1|)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K\,|z_1|^{\lambda_2/\lambda_1} = K\,|z_1|^{\frac{1}{2}}$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

The exponent $\lambda_2/\lambda_1 = \frac{1}{2}$ is positive, so the orbits are parabola-like curves through the origin. At the origin they are tangent to the eigenvector of the eigenvalue of **smaller modulus**, $|\lambda| = 5$, i.e. to the line through $(1, 3)$ (they leave from there (the limit $t \to -\infty$)), and far from the origin they become parallel to the other eigenvector (the fast direction). The arrows point away from the origin on both eigenlines.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (2, 1)$ and the z_2-axis to the line through $\mathbf{v}_2 = (1, 3)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = 10\mathbf{v}_1$ points outward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = 5\mathbf{v}_2$ points outward.



Canonical $\mathbf{z}$-plane (unstable node): invariant axes and orbits $z_2 = K|z_1|^{\frac{1}{2}}$.



Original (x, y)-plane (unstable node): dashed lines are the eigenlines $\mathbf{v}_1 = (2, 1)$ and $\mathbf{v}_2 = (1, 3)$.

(iv) $A = \begin{pmatrix} 1 & 20 \\ 40 & -19 \end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix} 1 - \lambda & 20 \\ 40 & -19 - \lambda \end{vmatrix} = (1 - \lambda)(-19 - \lambda) - (20)(40)$$

$$= \lambda^2 - (1 + (-19))\lambda + (1 \cdot (-19) - 20 \cdot 40) = \lambda^2 - (-18)\lambda + (-819) = \lambda^2 + 18\lambda - 819$$

$\operatorname{tr} A = -18$, $\det A = -819$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4 \det A = 324 - (-3276) = 3600$.

$$\lambda = \frac{-18 \pm \sqrt{3600}}{2} = \frac{-18 \pm 60}{2} \quad \Longrightarrow \quad \lambda_1 = 21,\ \lambda_2 = -39$$

Check: $\lambda_1 + \lambda_2 = -18 = \operatorname{tr} A$ and $\lambda_1 \lambda_2 = -819 = \det A$.

Eigenvectors

For $\lambda = 21$: $(A - (21)I)\mathbf{v_1} = \mathbf{0}$, that is

$$\begin{pmatrix} -20 & 20 \\ 40 & -40 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} -20v_1 + 20v_2 = 0 \quad &\text{(row 1)} \\ 40v_1 - 40v_2 = 0 \quad &\text{(row 2)} \end{aligned}$$

The coefficient matrix is singular ($\det(A - \lambda I) = 0$), so the rows are proportional (row 2 equals -2 times row 1) and express a single condition. Row 1 reads $-20v_1 + 20v_2 = 0$, so $v_1 = v_2$: every eigenvector is a multiple of $(1, 1)^T$. The smallest integer representative is

$$\mathbf{v_1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Check: $A\mathbf{v_1} = \begin{pmatrix} 21 \\ 21 \end{pmatrix} = 21\mathbf{v_1}$.

For $\lambda = -39$: $(A - (-39)I)\mathbf{v_2} = \mathbf{0}$, that is

$$\begin{pmatrix} 40 & 20 \\ 40 & 20 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} 40v_1 + 20v_2 = 0 \quad &\text{(row 1)} \\ 40v_1 + 20v_2 = 0 \quad &\text{(row 2)} \end{aligned}$$

The coefficient matrix is singular ($\det(A - \lambda I) = 0$), so the rows are proportional (row 2 is identical to row 1) and express a single condition. Row 1 reads $40v_1 + 20v_2 = 0$, so $v_1 = -\frac{1}{2}\, v_2$: every eigenvector is a multiple of $(-\frac{1}{2}, 1)^T$. The smallest integer representative is

$$\mathbf{v_2} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

Check: $A\mathbf{v_2} = \begin{pmatrix} -39 \\ 78 \end{pmatrix} = -39\mathbf{v_2}$.

Change of coordinates

The eigenvectors of distinct eigenvalues are linearly independent (Topic 4, Appendix B argument), so they form a basis. Put them as columns, $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$, and define the new coordinates by $\mathbf{z} = Q^{-1}\mathbf{x}$, i.e. $\mathbf{x} = Q\mathbf{z}$:

$$Q = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}$$

$\det Q = (1)(-2) - (1)(1) = -3$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-3}\begin{pmatrix} -2 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{pmatrix}$$

Since Q is constant, $\dot{\mathbf{z}} = Q^{-1}\dot{\mathbf{x}} = Q^{-1}A\mathbf{x} = (Q^{-1}AQ)\mathbf{z}$. Compute AQ one column at a time, with $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$:

$$A\mathbf{v}_1 = \begin{pmatrix} 1 & 20 \\ 40 & -19 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 21 \\ 21 \end{pmatrix} = 21\mathbf{v}_1, \qquad A\mathbf{v}_2 = \begin{pmatrix} 1 & 20 \\ 40 & -19 \end{pmatrix}\begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} -39 \\ 78 \end{pmatrix} = (-39)\mathbf{v}_2$$

$$AQ = [A\mathbf{v}_1 \mid A\mathbf{v}_2] = \begin{pmatrix} 21 & -39 \\ 21 & 78 \end{pmatrix}$$

Now multiply by Q^{-1} on the left, entry by entry:

$$
\begin{aligned}
(Q^{-1}AQ)_{11} &= \frac{2}{3}\cdot 21 + \frac{1}{3}\cdot 21 = 21\\
(Q^{-1}AQ)_{12} &= \frac{2}{3}\cdot (-39) + \frac{1}{3}\cdot 78 = 0\\
(Q^{-1}AQ)_{21} &= \frac{1}{3}\cdot 21 + (-\frac{1}{3})\cdot 21 = 0\\
(Q^{-1}AQ)_{22} &= \frac{1}{3}\cdot (-39) + (-\frac{1}{3})\cdot 78 = -39
\end{aligned}
$$

$$
Q^{-1}AQ = \begin{pmatrix} 21 & 0 \\ 0 & -39 \end{pmatrix} = \Lambda = \begin{pmatrix} 21 & 0 \\ 0 & -39 \end{pmatrix}
$$

The transformation has decoupled the system (the matrix is diagonal), as claimed.

Solution in the canonical plane

$$
\dot{z}_1 = 21 z_1, \qquad \dot{z}_2 = -39 z_2
$$

Each equation is separable. If $z_i \neq 0$, $\frac{dz_i}{z_i} = \lambda_i\, dt$ and integrating, $\ln|z_i| = \lambda_i t + C_i$, so $z_i = c_i e^{\lambda_i t}$ with $c_i = \pm e^{C_i}$ (the solution $z_i \equiv 0$ corresponds to $c_i = 0$). Equivalently, $\frac{d}{dt}(e^{-\lambda_i t} z_i) = 0$.

$$
z_1(t) = c_1 e^{21t}, \qquad z_2(t) = c_2 e^{-39t}
$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1 \mathbf{v}_1 + z_2 \mathbf{v}_2$, so the general solution is

$$
\mathbf{x}(t) = c_1 e^{21t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-39t} \begin{pmatrix} 1 \\ -2 \end{pmatrix}
$$

$$
x(t) = c_1 e^{21t} + c_2 e^{-39t}, \qquad y(t) = c_1 e^{21t} - 2c_2 e^{-39t}
$$

The decoupling transformation

In the original variables $\mathbf{x} = (x, y)^T$ the new coordinates are $\mathbf{z} = Q^{-1}\mathbf{x}$:

$$z_1 = \frac{2x}{3} + \frac{y}{3}, \qquad z_2 = \frac{x}{3} - \frac{y}{3}$$

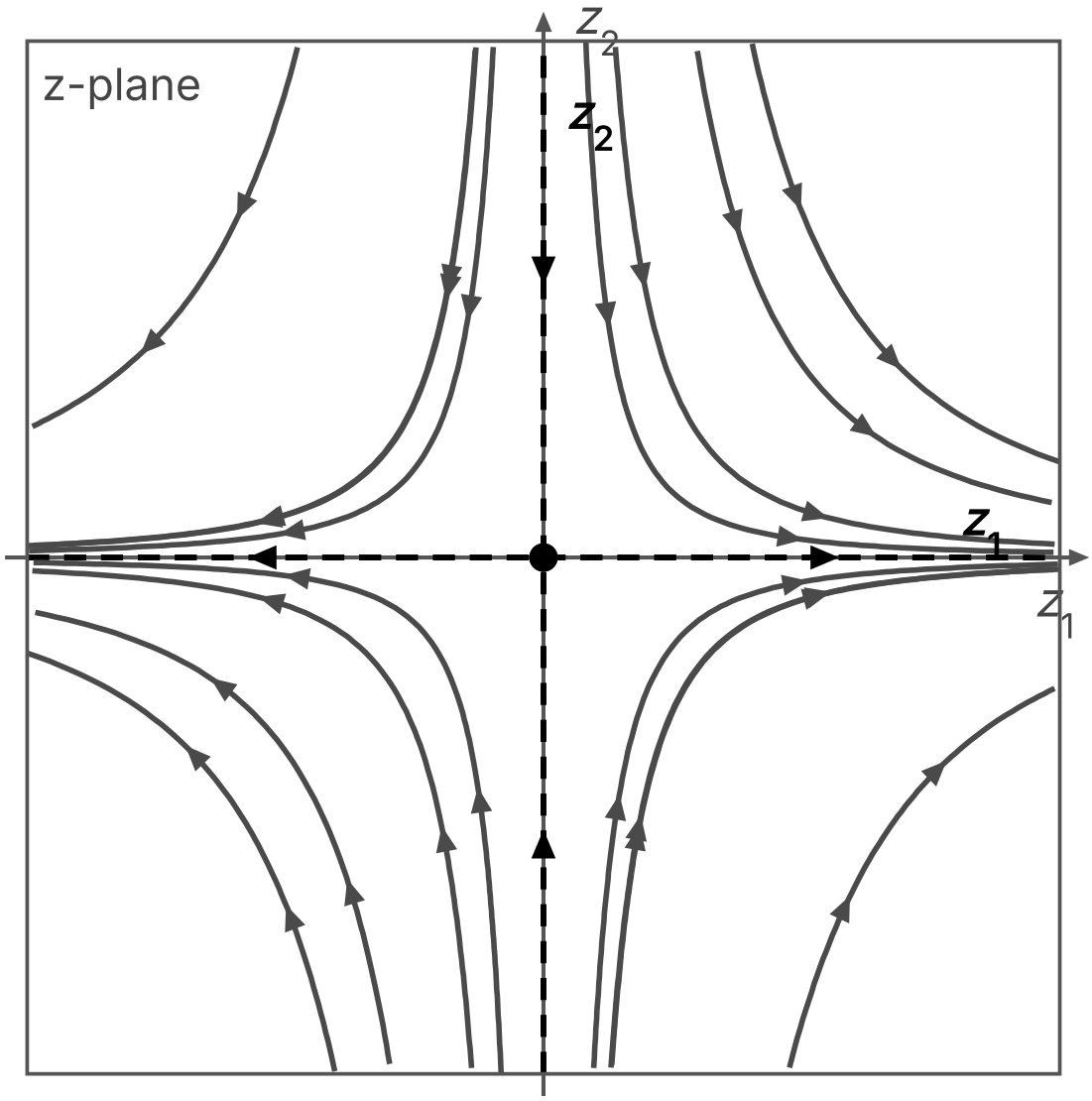
Phase portrait

$\lambda_1 = 21$ and $\lambda_2 = -39$ are of opposite signs: the origin is a **saddle** (Robinson Ch. 28, Topic 4 Sec. 6 and 9).

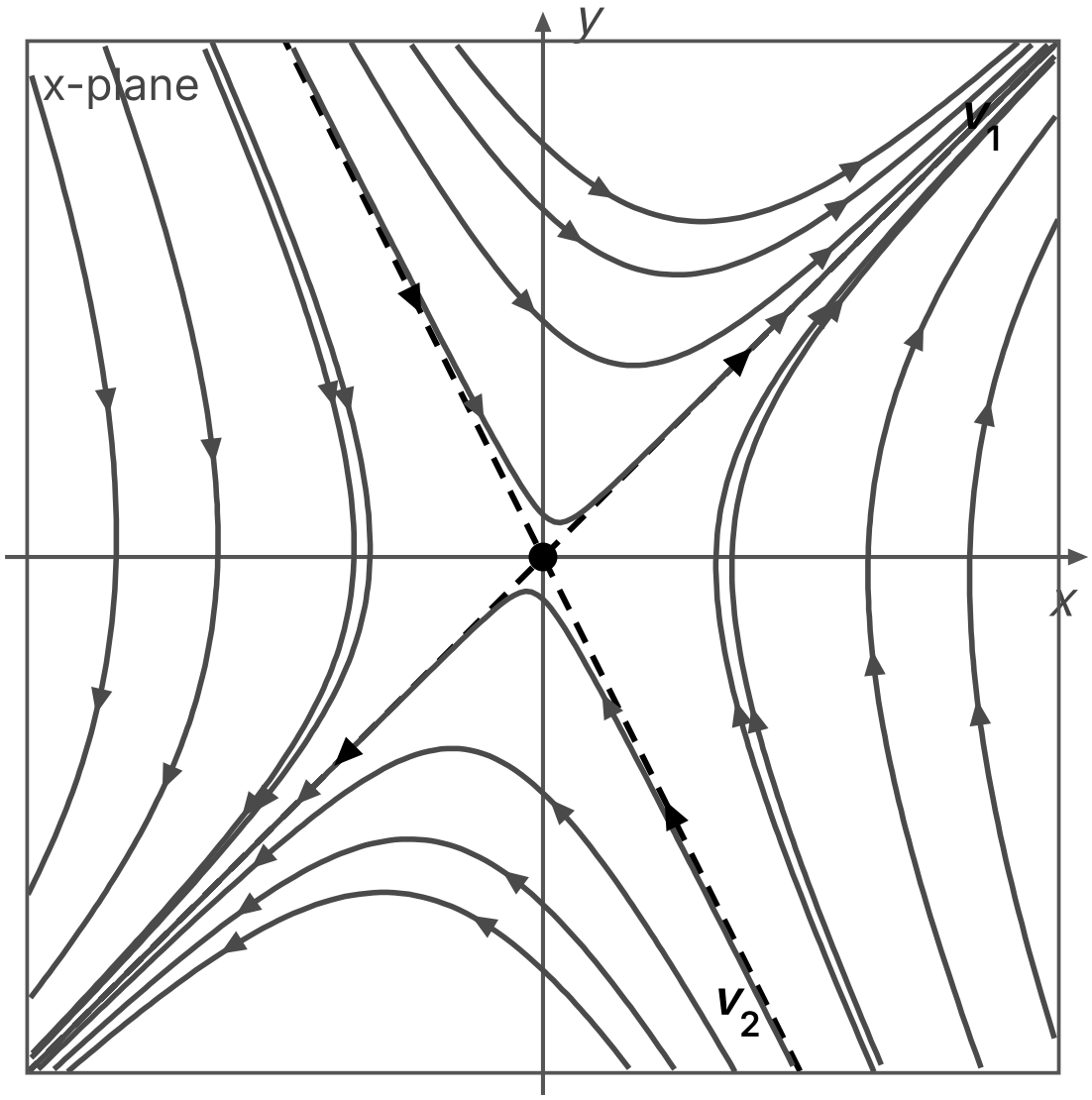
In the canonical $\mathbf{z}$-plane, eliminating t from $z_1 = c_1 e^{21t}$ and $z_2 = c_2 e^{-39t}$ (for $c_1 \neq 0$): $|z_1|/|c_1| = e^{\lambda_1 t}$, so $e^t = (|z_1|/|c_1|)^{1/\lambda_1}$ and $|z_2| = |c_2| e^{\lambda_2 t} = |c_2|\,(|z_1|/|c_1|)^{\lambda_2/\lambda_1}$, i.e. $z_2 = K\,|z_1|^{\lambda_2/\lambda_1} = K\,|z_1|^{-\frac{13}{7}}$ with $K = c_2 |c_1|^{-\lambda_2/\lambda_1}$ (equivalently, integrate the separable equation $\frac{dz_2}{dz_1} = \frac{\lambda_2 z_2}{\lambda_1 z_1}$ by the chain rule, as in Exercise 10). The axes $z_1 = 0$ and $z_2 = 0$ are invariant: a solution that starts on an axis stays on it.

The exponent $\lambda_2/\lambda_1 = -\frac{13}{7}$ is negative, so the orbits are hyperbola-like. Only solutions starting on the eigenline of the negative eigenvalue, the line through $(1, -2)$ (the **stable manifold**), reach the origin as $t \to +\infty$; the line through $(1, 1)$ (positive eigenvalue) is the **unstable manifold**. All other orbits come in close to the stable line, then leave along the unstable one.

The map $\mathbf{x} = Q\mathbf{z}$ sends the z_1-axis to the line through $\mathbf{v}_1 = (1, 1)$ and the z_2-axis to the line through $\mathbf{v}_2 = (1, -2)$. Straight lines stay straight and the origin stays fixed, so the $\mathbf{x}$-portrait is the canonical one sheared onto the eigenlines. On $\mathbf{v}_1$: $\dot{\mathbf{x}} = A\mathbf{v}_1 = 21\mathbf{v}_1$ points outward; on $\mathbf{v}_2$: $\dot{\mathbf{x}} = -39\mathbf{v}_2$ points inward.



Canonical $\mathbf{z}$-plane (saddle): invariant axes and orbits $z_2 = K|z_1|^{-\frac{13}{7}}$.



Original (x, y)-plane (saddle): dashed lines are the eigenlines $\mathbf{v}_1 = (1, 1)$ and $\mathbf{v}_2 = (1, -2)$.

INTERPRETATION AND CHECKS

- Each $Q^{-1}AQ$ was multiplied out entry by entry above and equals Λ exactly; the eigenvalues satisfy $\lambda_1 + \lambda_2 = \operatorname{tr} A$ and $\lambda_1\lambda_2 = \det A$.
- (i) 15 and -6: saddle; (ii) 2 and -3 (lower-triangular matrix: the eigenvalues are the diagonal entries): saddle; (iii) 10 and 5: unstable node, tangent at the origin to the eigenvector of 5; (iv) 21 and -39: saddle with very different rates.
- (i) is the homogeneous part of Exercise 7(vii) and (ii) shows that for a triangular matrix the eigenvalues can be read off the diagonal (Topic 4 Examples 27.2 and 27.3).
- The change of coordinates is not the efficient way to solve the system (the general solution follows directly from the eigenpairs), but it is what justifies the portrait: it reduces every case to the uncoupled one (Robinson Sec. 28.2).

- Dimensional check: with t in seconds the eigenvalues have units s^{-1}, and Q is dimensionless (it mixes coordinates of equal units).

EXERCISE 10

Trajectories of an Uncoupled System by the Chain Rule

Original statement · probls_ch4-i_2627_v2.pdf, p. 3

Chain Rule • Separable Equation • Orbit Equation

STATEMENT

Using the chain rule for $y = y(x(t))$,

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt} \implies \frac{dy}{dx} = \frac{\dfrac{dy}{dt}}{\dfrac{dx}{dt}}.$$

Therefore, if

$$\frac{dx}{dt} = \lambda_1 x, \qquad \frac{dy}{dt} = \lambda_2 y$$

we have

$$\frac{dy}{dx} = \frac{\lambda_2 y}{\lambda_1 x}.$$

Solve the latter equation to determine the curves traced out by trajectories of the system of equations for $x(t)$ and $y(t)$.

SETUP AND HYPOTHESES

Real constants λ_1, λ_2. The chain rule is valid where $x(t)$ is a differentiable function with $\dot{x} \neq 0$, so that $t \mapsto x$ is locally invertible and y can be regarded as a function of x. Here $\dot{x} = \lambda_1 x \neq 0$ requires $\lambda_1 \neq 0$ and $x \neq 0$. The cases $x = 0$ (the y-axis), $y = 0$ (the x-axis) and $\lambda_1 = 0$ are treated separately at the end.

FORMULATION

The orbit equation $\frac{dy}{dx} = \frac{\lambda_2}{\lambda_1}\frac{y}{x}$ has no explicit t and is **separable**: all the y on the left and all the x on the right.

METHOD OF THE CLASS NOTES [NOTES, P. 9]

This is the step of the notes that turns Case 1 into a portrait. From $z_1 = C_1 e^{\lambda_1 t}$ and $z_2 = C_2 e^{\lambda_2 t}$ they eliminate t (for $C_1 > 0$):

$$z_2 = C_2\left(e^{\lambda_1 t}\right)^{\lambda_2/\lambda_1} = C_2 C_1^{-\lambda_2/\lambda_1}\left(C_1 e^{\lambda_1 t}\right)^{\lambda_2/\lambda_1} = \tilde{C}\, z_1^{\lambda_2/\lambda_1}.$$

The chain-rule route below reaches the same curves, $y = \tilde{C}\,|x|^{\lambda_2/\lambda_1}$ on each half-plane, without solving for t.

DERIVATION

Integration

Separate variables (for $y \neq 0$, $x \neq 0$) and integrate both sides with their antiderivatives, $\int \frac{dy}{y} = \ln|y|$:

$$\frac{dy}{y} = \frac{\lambda_2}{\lambda_1}\frac{dx}{x} \implies \ln|y| = \frac{\lambda_2}{\lambda_1}\ln|x| + C$$

Exponentiate and absorb e^C in a constant $K = \pm e^C$:

$$|y| = e^C |x|^{\lambda_2/\lambda_1} \implies \boxed{y = K\,|x|^{\lambda_2/\lambda_1}}, \qquad K \neq 0$$

Checking with the explicit solution

The equations are solved directly by $x = x_0 e^{\lambda_1 t}$, $y = y_0 e^{\lambda_2 t}$ (separation of variables, $\frac{d}{dt}(e^{-\lambda t}u) = 0$). For $x_0 \neq 0$, $|x|/|x_0| = e^{\lambda_1 t}$ so $e^t = (|x|/|x_0|)^{1/\lambda_1}$ and

$$y = y_0\left(e^{\lambda_1 t}\right)^{\lambda_2/\lambda_1} = y_0\left(\frac{|x|}{|x_0|}\right)^{\lambda_2/\lambda_1} = \underbrace{y_0|x_0|^{-\lambda_2/\lambda_1}}_{K}|x|^{\lambda_2/\lambda_1}$$

which is the same family, with $K = y_0|x_0|^{-\lambda_2/\lambda_1}$ (the sign of y is the sign of y_0, preserved since $y = y_0 e^{\lambda_2 t}$).

The axes

If $x_0 = 0$ then $x(t) \equiv 0$ and the orbit is on the y-axis (with $y = y_0 e^{\lambda_2 t}$); if $y_0 = 0$ then $y \equiv 0$ (that is $K = 0$) and the orbit lies on the x-axis. The origin is the equilibrium. If $\lambda_1 = 0$, x is constant and the orbits are vertical lines.

Shapes of the curves

Exponent λ_2/λ_1	**Curves** $y = K\lvert x\rvert^r$	**Origin**
$r > 1$ (same sign, $\lvert\lambda_2\rvert > \lvert\lambda_1\rvert$)	parabola-like, tangent to the x-axis at 0	node (stable if $\lambda_i < 0$, unstable if > 0)
$r = 1$ ($\lambda_1 = \lambda_2$)	straight lines $y = Kx$ (rays)	star
$0 < r < 1$	tangent to the y-axis at 0	node
$r < 0$ (opposite signs)	hyperbola-like, $y\lvert x\rvert^{-r} = K$, asymptotic to the axes	saddle
$r = -1$	rectangular hyperbolas $xy = \pm K$	saddle

INTERPRETATION AND CHECKS

- This is the derivation behind the canonical portraits of Topic 4 Sec. 6.1: replace (x, y) by (z_1, z_2) and λ_1, λ_2 by the two eigenvalues to get $z_2 = K\lvert z_1\rvert^{\lambda_2/\lambda_1}$ (Robinson Eq. 28.12).
- Examples: $\lambda_1 = 1, \lambda_2 = 2$: $y = Kx^2$ (parabolas); $\lambda_1 = 1, \lambda_2 = -1$: $y = K/x$ (hyperbolas).
- Direction along the orbits is **not** given by the orbit equation: it is read from $\dot{x} = \lambda_1 x, \dot{y} = \lambda_2 y$ (away from 0 along an axis if $\lambda > 0$, towards it if $\lambda < 0$).
- Differentiating the answer: $\frac{dy}{dx} = K\frac{\lambda_2}{\lambda_1}\lvert x\rvert^{\lambda_2/\lambda_1 - 1}\operatorname{sign}(x) = \frac{\lambda_2}{\lambda_1}\frac{y}{x}$, which recovers the ODE (using $\lvert x\rvert^r/x = \lvert x\rvert^{r-1}\operatorname{sign} x$).

EXERCISE 11

A Zero Eigenvalue: Lines of Equilibria

Original statement · probls_ch4-i_2627_v2.pdf, p. 4

Zero Eigenvalue • Line of Equilibria • Parallel Orbits

STATEMENT

Suppose that A has eigenvalues $\lambda_1 = 0$, $\lambda_2 \neq 0$, with eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$, respectively.

- (i) Write down the general solution of $\dot{\mathbf{x}} = A\mathbf{x}$.
- (ii) After changing to a coordinate system $\tilde{\mathbf{x}}$ referred to the eigenvectors, we obtain $\dfrac{d\tilde{\mathbf{x}}}{dt} = \begin{pmatrix} 0 & 0 \\ 0 & \lambda_2 \end{pmatrix} \tilde{\mathbf{x}}$. By solving these equations draw the phase portrait in the $\tilde{\mathbf{x}}$ system, and sketch the phase portrait for the original coordinates $\mathbf{x}$.
- (iii) Draw the phase portrait for the equation $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix} \mathbf{x}$.

SETUP AND HYPOTHESES

A is 2×2 real with eigenvalues 0 and $\lambda_2 \neq 0$ (both real: $\lambda_1 = 0$ is real and $\lambda_2 = \operatorname{tr} A$ is real because A is real; being distinct, they have independent eigenvectors; $\det A = 0$). $A\mathbf{v}_1 = \mathbf{0}$, $A\mathbf{v}_2 = \lambda_2 \mathbf{v}_2$. Because $\det A = 0$ the origin is **not** an isolated equilibrium: every point of $\ker A = \operatorname{span}\{\mathbf{v}_1\}$ is an equilibrium ($A\mathbf{x} = \mathbf{0}$).

FORMULATION

Use the general result for distinct eigenvalues, $\mathbf{x} = c_1 e^{\lambda_1 t}\mathbf{v}_1 + c_2 e^{\lambda_2 t}\mathbf{v}_2$ (Robinson Eq. 28.1), valid whenever the eigenvalues differ, in particular if one of them is 0; and $Q = [\mathbf{v}_1\ \mathbf{v}_2]$, $\mathbf{z} = Q^{-1}\mathbf{x}$.

METHOD OF THE CLASS NOTES [NOTES, P. 9]

Special case $\lambda_1 = 0$ of the notes: $A\mathbf{v}_1 = \mathbf{0}$, so the line spanned by $\mathbf{v}_1$ is made of equilibria, and $\mathbf{x} = C_1\mathbf{v}_1 + C_2 e^{\lambda_2 t}\mathbf{v}_2$: straight orbits parallel to $\mathbf{v}_2$.

Reading of the handwriting: the notes write $\mathbf{x}(t) = C_2\mathbf{v}_2 e^{\lambda_2 t}$, which drops the line of equilibria $C_1\mathbf{v}_1$. The general solution needs both terms, since it must match any $\mathbf{x}(0)$.

DERIVATION

(i) General solution

With $\lambda_1 = 0$, $e^{\lambda_1 t} = e^0 = 1$:

$$\mathbf{x}(t) = c_1\mathbf{v}_1 + c_2 e^{\lambda_2 t}\mathbf{v}_2$$

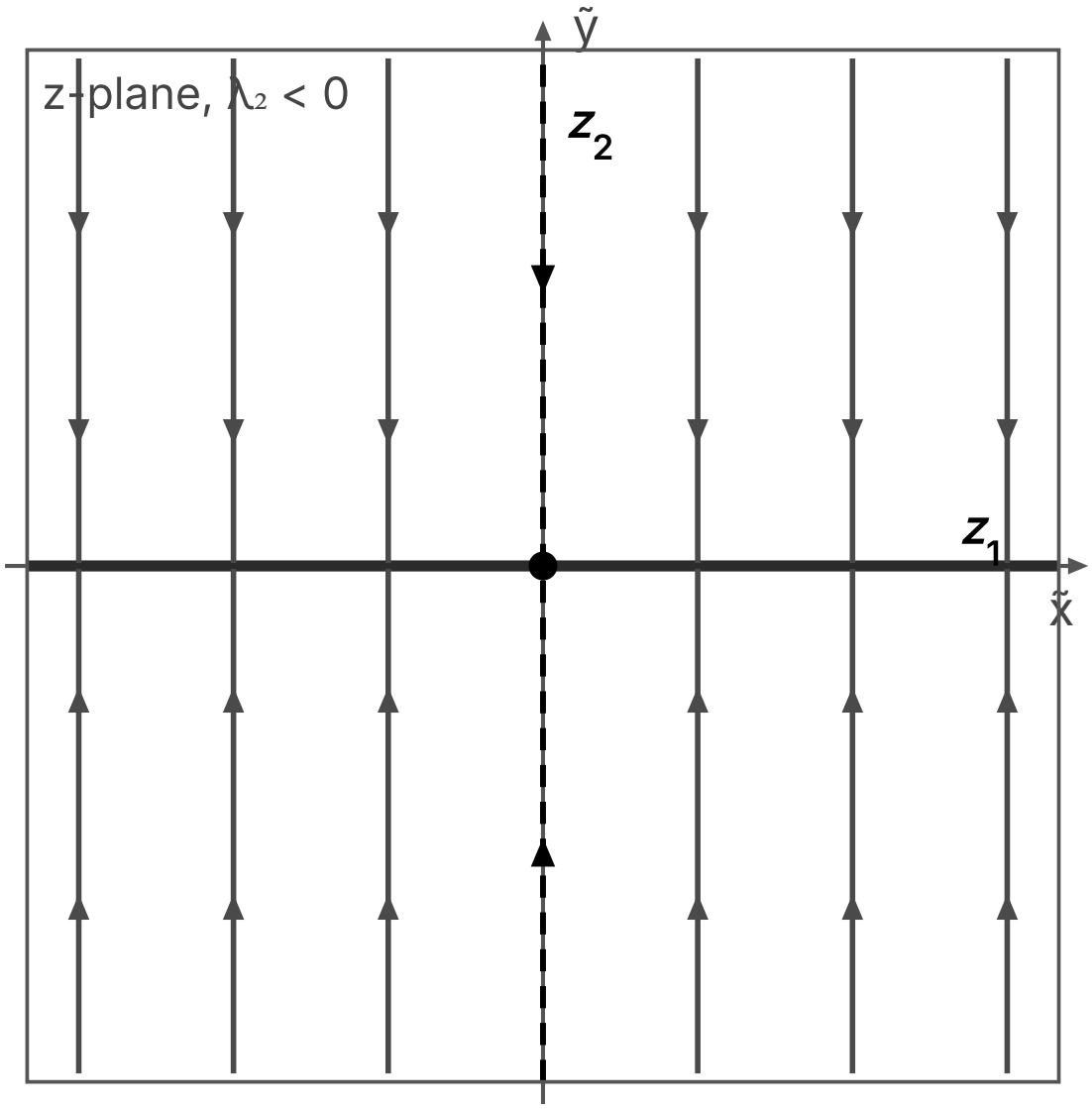
Check: $\dot{\mathbf{x}} = c_2\lambda_2 e^{\lambda_2 t}\mathbf{v}_2$ and $A\mathbf{x} = c_1 A\mathbf{v}_1 + c_2 e^{\lambda_2 t} A\mathbf{v}_2 = \mathbf{0} + c_2\lambda_2 e^{\lambda_2 t}\mathbf{v}_2$. They agree. The constant vector $c_1\mathbf{v}_1$ is an equilibrium and $c_2 e^{\lambda_2 t}\mathbf{v}_2$ the time-dependent part.

(ii) Canonical phase portrait and its image

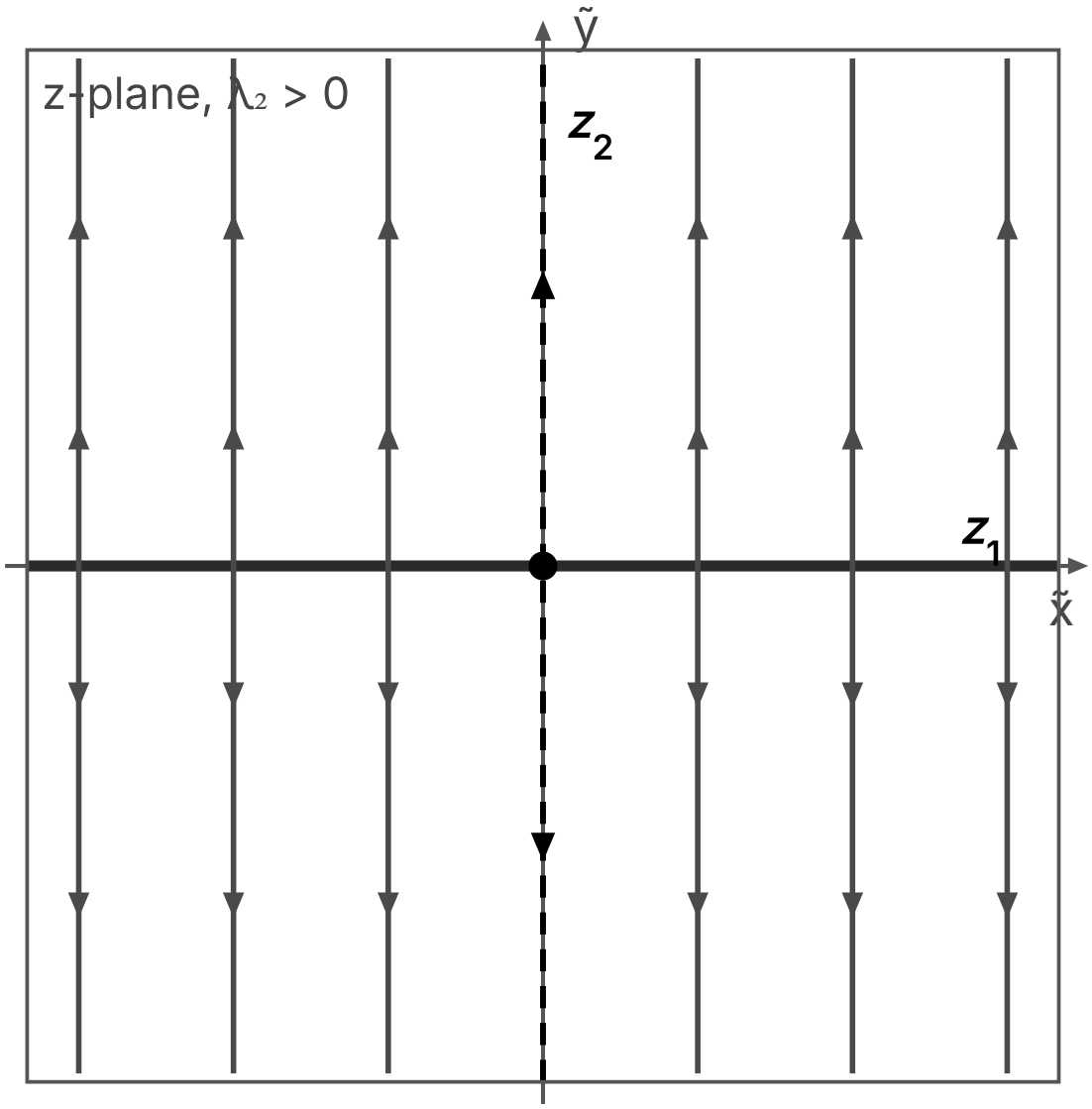
In the coordinates $\tilde{\mathbf{x}} = (\tilde{x}, \tilde{y})^T$ referred to $\mathbf{v}_1, \mathbf{v}_2$ the system is $\dot{\tilde{x}} = 0$, $\dot{\tilde{y}} = \lambda_2\tilde{y}$. The first gives $\tilde{x} = c_1$ (constant); the second is separable: $\frac{d\tilde{y}}{\tilde{y}} = \lambda_2\, dt$, $\ln|\tilde{y}| = \lambda_2 t + C$, so $\tilde{y} = c_2 e^{\lambda_2 t}$.

$$\tilde{x}(t) = c_1, \qquad \tilde{y}(t) = c_2 e^{\lambda_2 t}$$

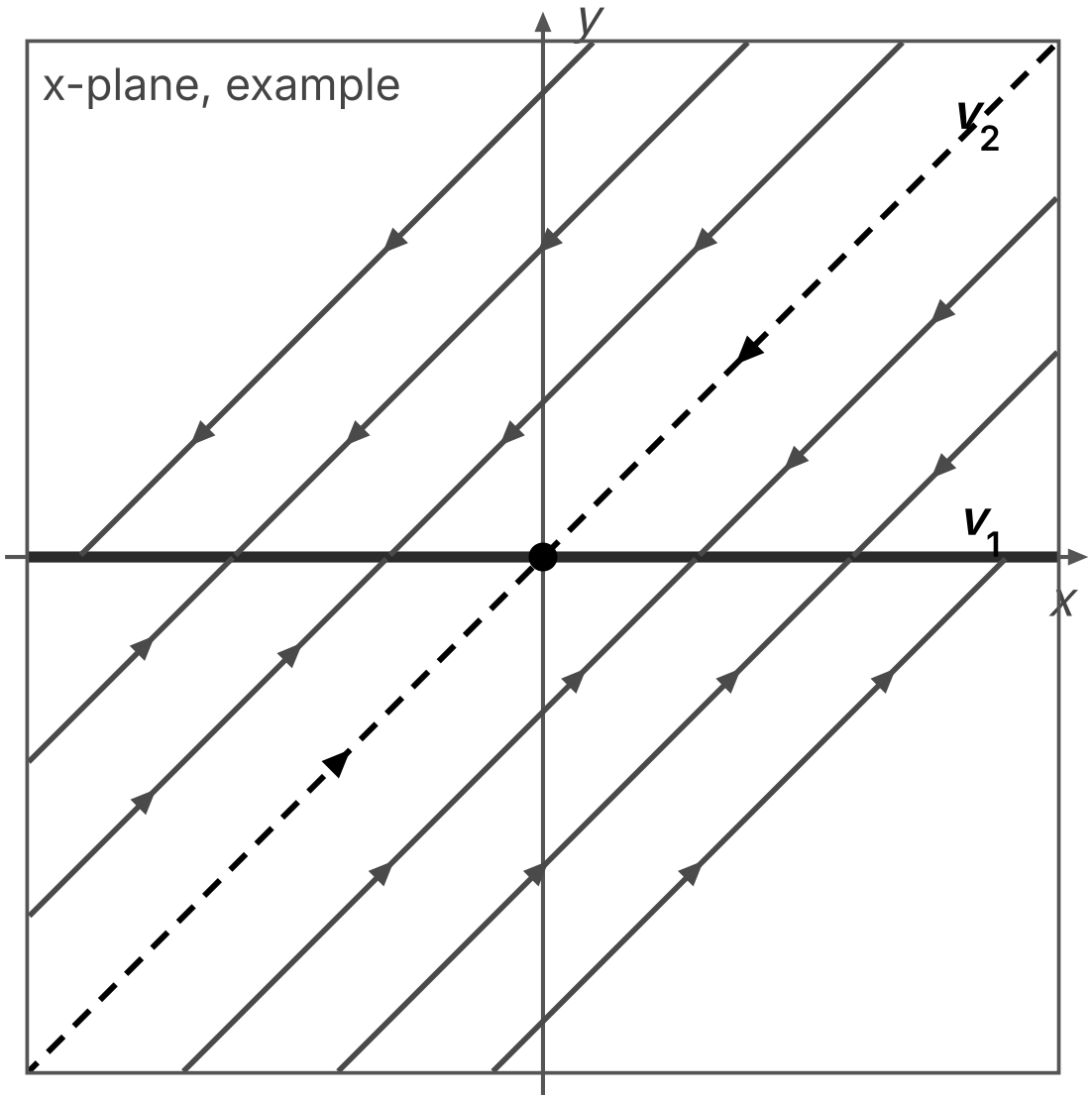
- Orbits: the **vertical lines** $\tilde{x} = c_1$, because $\tilde{x}$ never changes. Each such line is a pair of half-lines (for $\tilde{y} > 0$ and $\tilde{y} < 0$) and the point $(c_1, 0)$ is an equilibrium.
- The $\tilde{x}$-axis ($\tilde{y} = 0$) consists entirely of equilibria (it is the image of $\ker A$).
- Direction: if $\lambda_2 < 0$, $|\tilde{y}| = |c_2|e^{\lambda_2 t}$ decreases: the arrows on every vertical line point **towards** the $\tilde{x}$-axis, which they approach as $t \to +\infty$ without reaching it. If $\lambda_2 > 0$ the arrows point **away** from the axis. The sign of $\dot{\tilde{y}} = \lambda_2\tilde{y}$ gives the direction at once.
- Original coordinates: $\mathbf{x} = Q\tilde{\mathbf{x}} = \tilde{x}\mathbf{v}_1 + \tilde{y}\mathbf{v}_2$. The $\tilde{x}$-axis goes to the line $\operatorname{span}\{\mathbf{v}_1\}$ of equilibria and the vertical lines go to the lines $c_1\mathbf{v}_1 + s\mathbf{v}_2$, $s \in \mathbb{R}$, **parallel to $\mathbf{v}_2$**, drawn through the points of the line of equilibria. Straight lines remain straight: the picture is the canonical one sheared, with orbits along $\mathbf{v}_2$ towards/away from the line of equilibria.



Canonical $\tilde{\mathbf{x}}$-plane, $\lambda_2 < 0$: the $\tilde{x}$-axis (green) is a line of equilibria and the vertical orbits approach it.



Canonical $\tilde{\mathbf{x}}$-plane, $\lambda_2 > 0$: the vertical orbits leave the line of equilibria.



Original (x, y)-plane for the example $\mathbf{v}_1 = (1, 0)$, $\mathbf{v}_2 = (1, 1)$, $\lambda_2 = -2$: equilibria on the line of $\mathbf{v}_1$, orbits parallel to $\mathbf{v}_2$.

(For the sketch in the original plane, take the example $\mathbf{v}_1 = (1, 0)$, $\mathbf{v}_2 = (1, 1)$, $\lambda_2 = -2$, whose matrix is $A = Q\Lambda Q^{-1} = \begin{pmatrix} 0 & -2 \\ 0 & -2 \end{pmatrix}$; any other eigen-data give the same picture up to a linear change of the axes.)

(iii) The matrix $\begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix} -2 - \lambda & 2 \\ 1 & -1 - \lambda \end{vmatrix} = (-2 - \lambda)(-1 - \lambda) - (2)(1)$$

$$= \lambda^2 - (-2 + (-1))\lambda + (-2 \cdot (-1) - 2 \cdot 1) = \lambda^2 - (-3)\lambda + (0) = \lambda^2 + 3\lambda$$

$\operatorname{tr} A = -3$, $\det A = 0$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4 \det A = 9 - 0 = 9$.

$$\lambda = \frac{-3 \pm \sqrt{9}}{2} = \frac{-3 \pm 3}{2} \quad \Longrightarrow \quad \lambda_1 = 0,\ \lambda_2 = -3$$

Check: $\lambda_1 + \lambda_2 = -3 = \operatorname{tr} A$ and $\lambda_1 \lambda_2 = 0 = \det A$.

Eigenvectors

For $\lambda = 0$: $(A - (0)I)\mathbf{v_1} = \mathbf{0}$, that is

$$\begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} -2v_1 + 2v_2 &= 0 \quad \text{(row 1)} \\ 1v_1 - 1v_2 &= 0 \quad \text{(row 2)} \end{aligned}$$

The coefficient matrix is singular ($\det(A - \lambda I) = 0$), so the rows are proportional (row 2 equals $-\frac{1}{2}$ times row 1) and express a single condition. Row 1 reads $-2v_1 + 2v_2 = 0$, so $v_1 = v_2$: every eigenvector is a multiple of $(1, 1)^T$. The smallest integer representative is

$$\mathbf{v_1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Check: $A\mathbf{v_1} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0\mathbf{v_1}$.

For $\lambda = -3$: $(A - (-3)I)\mathbf{v_2} = \mathbf{0}$, that is

$$\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} 1v_1 + 2v_2 &= 0 \quad \text{(row 1)} \\ 1v_1 + 2v_2 &= 0 \quad \text{(row 2)} \end{aligned}$$

The coefficient matrix is singular ($\det(A-\lambda I)=0$), so the rows are proportional (row 2 is identical to row 1) and express a single condition. Row 1 reads $1v_1+2v_2=0$, so $v_1=-2\,v_2$: every eigenvector is a multiple of $(-2,1)^T$. The smallest integer representative is

$$\mathbf{v_2}=\begin{pmatrix}2\\-1\end{pmatrix}$$

Check: $A\mathbf{v_2}=\begin{pmatrix}-6\\3\end{pmatrix}=-3\mathbf{v_2}$.

Change of coordinates

The eigenvectors of distinct eigenvalues are linearly independent (Topic 4, Appendix B argument), so they form a basis. Put them as columns, $Q=[\mathbf{v}_1\mid\mathbf{v}_2]$, and define the new coordinates by $\mathbf{z}=Q^{-1}\mathbf{x}$, i.e. $\mathbf{x}=Q\mathbf{z}$:

$$Q=\begin{pmatrix}1&2\\1&-1\end{pmatrix}$$

$\det Q=(1)(-1)-(2)(1)=-3$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix}a&b\\c&d\end{pmatrix}^{-1}=\frac{1}{ad-bc}\begin{pmatrix}d&-b\\-c&a\end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1}=\frac{1}{-3}\begin{pmatrix}-1&-2\\-1&1\end{pmatrix}=\begin{pmatrix}\frac{1}{3}&\frac{2}{3}\\\frac{1}{3}&-\frac{1}{3}\end{pmatrix}$$

Since Q is constant, $\dot{\mathbf{z}}=Q^{-1}\dot{\mathbf{x}}=Q^{-1}A\mathbf{x}=(Q^{-1}AQ)\mathbf{z}$. Compute AQ one column at a time, with $A\mathbf{v}_j=\lambda_j\mathbf{v}_j$:

$$A\mathbf{v}_1=\begin{pmatrix}-2&2\\1&-1\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}=0\mathbf{v}_1,\qquad A\mathbf{v}_2=\begin{pmatrix}-2&2\\1&-1\end{pmatrix}\begin{pmatrix}2\\-1\end{pmatrix}=\begin{pmatrix}-6\\3\end{pmatrix}=(-3)\mathbf{v}_2$$

$$AQ=[A\mathbf{v}_1\mid A\mathbf{v}_2]=\begin{pmatrix}0&-6\\0&3\end{pmatrix}$$

Now multiply by Q^{-1} on the left, entry by entry:

$$
\begin{aligned}
(Q^{-1}AQ)_{11} &= \frac{1}{3}\cdot 0 + \frac{2}{3}\cdot 0 = 0 \\
(Q^{-1}AQ)_{12} &= \frac{1}{3}\cdot (-6) + \frac{2}{3}\cdot 3 = 0 \\
(Q^{-1}AQ)_{21} &= \frac{1}{3}\cdot 0 + (-\frac{1}{3})\cdot 0 = 0 \\
(Q^{-1}AQ)_{22} &= \frac{1}{3}\cdot (-6) + (-\frac{1}{3})\cdot 3 = -3
\end{aligned}
$$

$$
Q^{-1}AQ = \begin{pmatrix} 0 & 0 \\ 0 & -3 \end{pmatrix} = \Lambda = \begin{pmatrix} 0 & 0 \\ 0 & -3 \end{pmatrix}
$$

The transformation has decoupled the system (the matrix is diagonal), as claimed.

Solution in the canonical plane

$$
\dot{z}_1 = 0, \qquad \dot{z}_2 = -3z_2
$$

Each equation is separable. If $z_i \neq 0$, $\frac{dz_i}{z_i} = \lambda_i\, dt$ and integrating, $\ln|z_i| = \lambda_i t + C_i$, so $z_i = c_i e^{\lambda_i t}$ with $c_i = \pm e^{C_i}$ (the solution $z_i \equiv 0$ corresponds to $c_i = 0$). Equivalently, $\frac{d}{dt}(e^{-\lambda_i t} z_i) = 0$.

$$
z_1(t) = c_1 e^{0t}, \qquad z_2(t) = c_2 e^{-3t}
$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{v}_1 + z_2\mathbf{v}_2$, so the general solution is

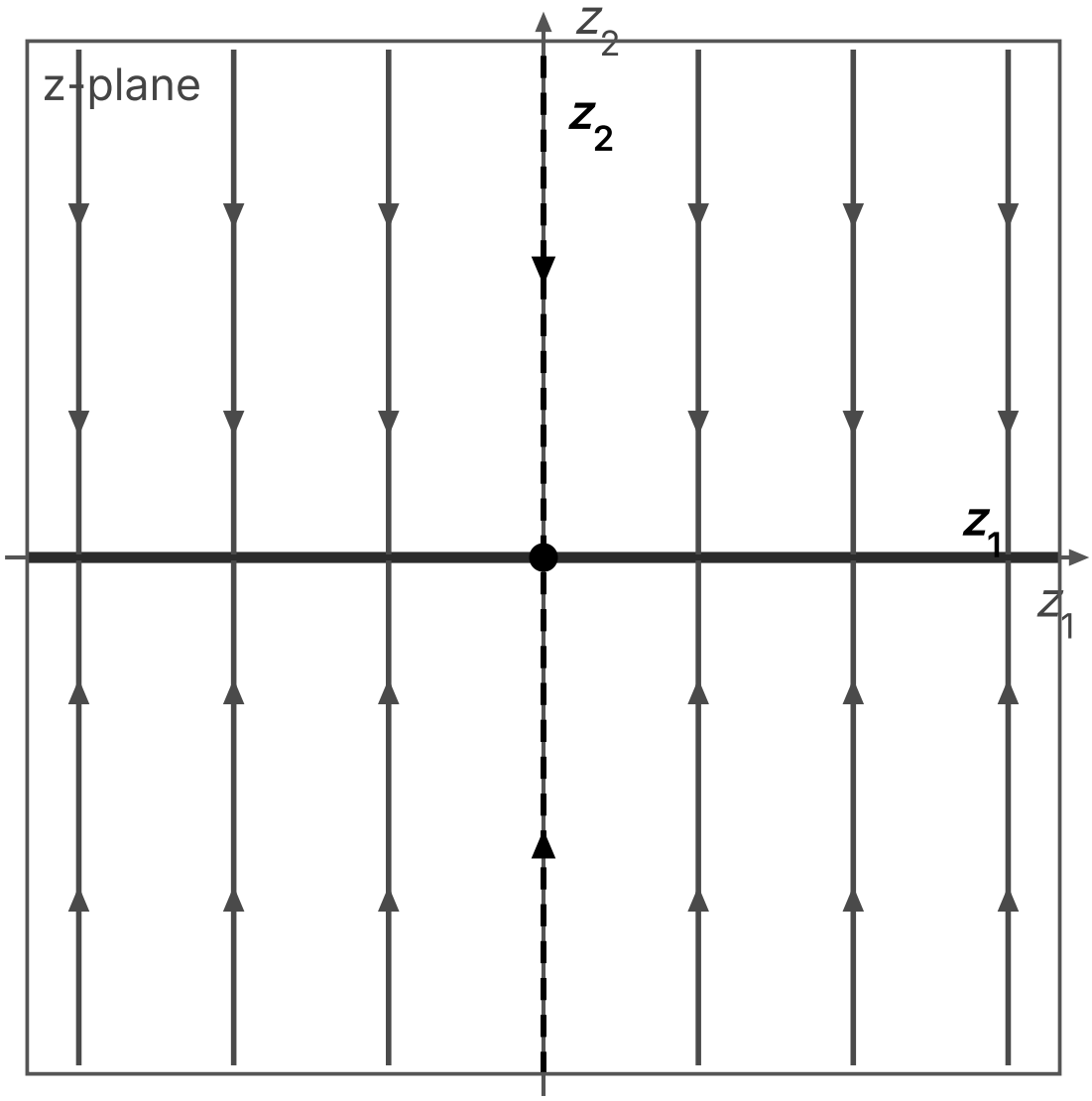
$$
\mathbf{x}(t) = c_1 e^{0t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-3t}\begin{pmatrix} 2 \\ -1 \end{pmatrix}
$$

$$
x(t) = c_1 + 2c_2 e^{-3t}, \qquad y(t) = c_1 - c_2 e^{-3t}
$$

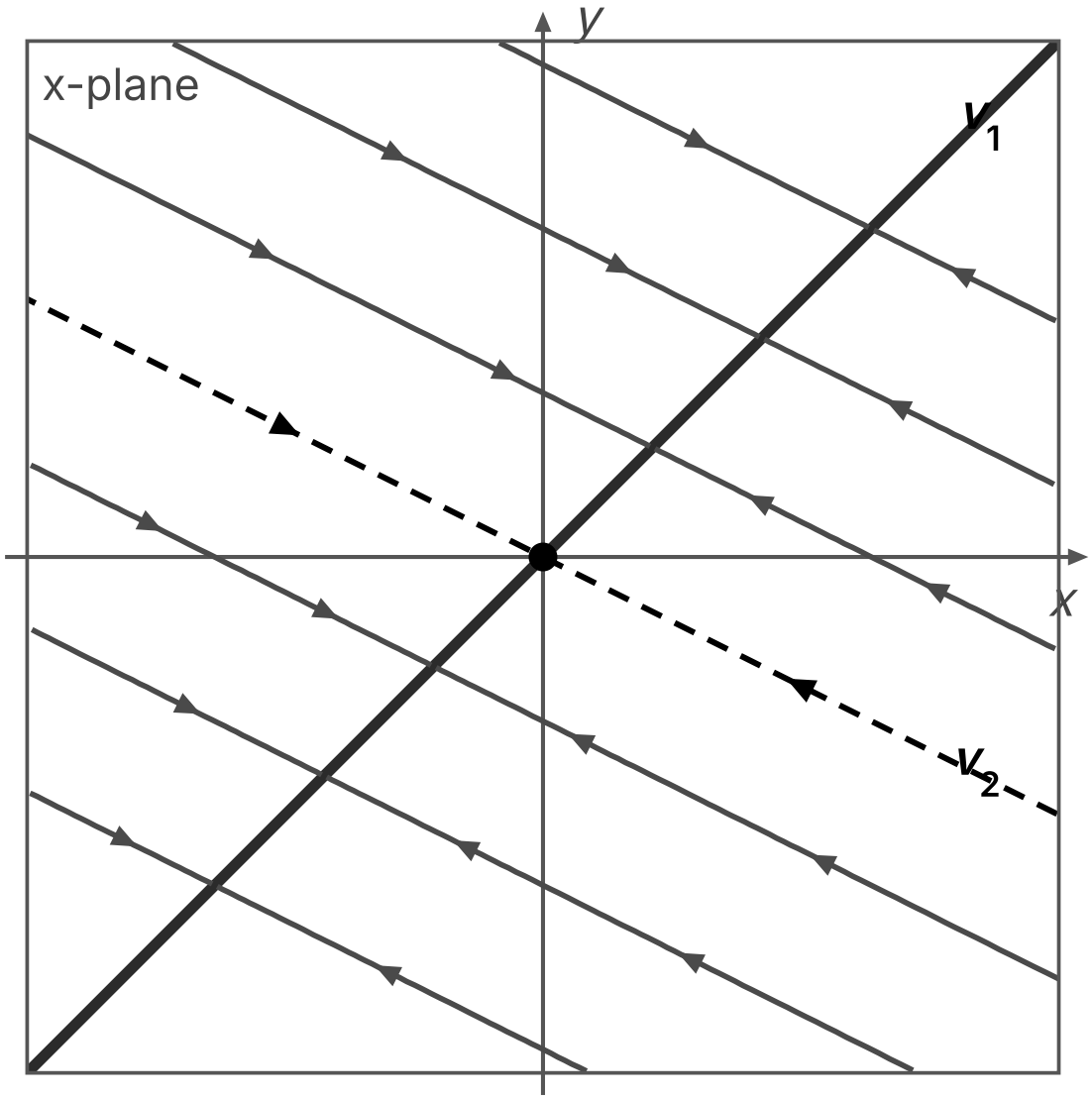
Phase portrait

One eigenvalue is zero and the other is $\lambda_2 = -3 \neq 0$. In the canonical plane $\dot{z}_1 = 0$, so z_1 is constant along every solution (Topic 4, case $\lambda = 0$).

Every point of the line through $\mathbf{v}_1 = (1, 1)$ is an equilibrium, because $A\mathbf{v}_1 = \mathbf{0}$. Every other solution moves on a straight line parallel to $\mathbf{v}_2 = (2, -1)$ with $\mathbf{x} - \mathbf{x}_{eq} \propto e^{-3t}$: it approaches the line of equilibria as $t \to +\infty$ (the equilibria are stable but each orbit ends at a different point). The arrows point towards the line of equilibria.



Canonical $\mathbf{z}$-plane: the line of equilibria (green) and the trajectories parallel to the other axis.



Original (x, y)-plane: equilibria on the line through $(1, 1)$ (green); trajectories parallel to $(2, -1)$, towards that line.

INTERPRETATION AND CHECKS

- (iii): $\operatorname{tr} A = -3$ and $\det A = (-2)(-1) - 2 \cdot 1 = 0$, so the eigenvalues are 0 and -3: the general solution is $\mathbf{x} = c_1(1, 1)^T + c_2 e^{-3t}(2, -1)^T$, every orbit converges to the equilibrium $c_1(1, 1)^T$, i.e. $\mathbf{x}(t) \to \mathbf{x}_\infty$ on the line $y = x$ along the direction $(2, -1)$.
- Explicit limit for $\mathbf{x}(0) = (x_0, y_0)$: $\mathbf{x}_\infty = c_1(1, 1)^T$ with $c_1 = (x_0 + 2y_0)/3$ (from Q^{-1} computed above, first row $(\frac{1}{3}, \frac{2}{3})$). Example: $\mathbf{x}(0) = (2, -1)$ gives $c_1 = 0$ and the orbit is the line $\mathbf{x} = e^{-3t}(2, -1)$ going to the origin.
- The origin is no longer an isolated equilibrium: the node/saddle/spiral classification of Topic 4 Sec. 9 assumes $\det A \neq 0$ (see the trace-determinant discussion), and here $\det A = 0$, which is the borderline between a stable node ($\lambda_2 < 0$) and a saddle.

- Conserved quantity: the combination $\tilde{x} = c_1$ is a first integral of the motion ($x + 2y$ in (iii), up to the factor $\frac{1}{3}$): $\frac{d}{dt}(x + 2y) = (-2x + 2y) + 2(x - y) = 0$ ✓.